Exact Second-Order Asymptotics for the Gaussian Gray–Wyner Problem

Lin Zhou and Lei Yu

Abstract

We revisit the Gray-Wyner problem of lossy compression of correlated Gaussian sources under the quadratic distortion measure and derive the exact second-order asymptotics under a joint excess-distortion probability criterion. In particular, we first classify the boundary rate tuples into two cases: with a nonzero common rate and with a zero common rate. For the case with a nonzero common rate, our main contribution is the achievability proof. Specifically, we derive a covariance shell covering lemma that is analogous to the type covering lemma for the discrete case and manage to obtain an achievability result that matches the existing converse result of Liu et al. (IEEE TIT 2020). For the case with a zero common rate, we use the same covering lemma and derive a new converse bound. We would like to emphasize that the boundary rate tuple with the zero common rate was ignored by previous second-order analyses of the Gray-Wyner problem for the discrete sources. Thus, our results in this case are of independent interest and provide some new insights.

Index Terms

Dispersion, Rate-distortion, Covariance covering, Multiterminal source coding, Lossy source coding

I. Introduction

Data compression, a.k.a., source coding, was pioneered by Shannon [1] in this seminal paper. Specifically, Shannon showed that the entropy is the minimal rate for lossless compression of a discrete memoryless source (DMS) to ensure almost exact recovery with vanishing error probability as the blocklength tends to infinity. A decade later, Strassen [2] refined Shannon's result by deriving the second-order asymptotics that approximates the finite blocklength performance of optimal codes. Specifically, second-order asymptotics characterizes the order and the coefficient of the second dominant term for the non-asymptotic coding rate as a function of the blocklength and a given error probability. Recently, the result of Strassen was further refined by Kontoyiannis and Verdú who derived further refined third-order asymptotics [3]. As demonstrated in [2], [3], refined asymptotics approximates the performance of optimal codes at finite blocklengths.

Although insightful, lossless compression is valid only for a DMS. For continuous memoryless sources, lossy source coding, a.k.a., rate-distortion was introduced by [4]. In lossy source coding, the reconstruction is no longer required to be exactly the same as the original source sequence and the imperfection is measured via a distortion function. Shannon [4] derived the first-order asymptotics for lossy source coding under the expected distortion criterion when the blocklength tends to infinity and named the coefficient of the first dominant term the rate-distortion function. The same result also holds when one uses a vanishing excess-distortion probability. Several decades later, second-order asymptotics was obtained by Ingber and Kochman [5] and Kostina and Verdú [6]. Furthermore, Kostina and Verdú also obtained the third-order asymptotics for an equiprobable DMS under Hamming distortion [6, Theorem 27].

Beyond point-to-point (P2P) data compression, motivated by practical applications in the storage and retrieval of correlated sources [7, Section V-C] and cache-aided delivery of correlated content [8], Gray and Wyner [9] proposed a source coding problem with one common compression message and two private compression messages so that a pair of correlated source sequences can be compressed at a single transmitter and recovered at two different receivers, in either a lossless or lossy manner. The first-order asymptotics, a.k.a., the rate region for the lossless case and the rate-distortion region for the lossy case, was characterized in [9]. Ever since, the problem is known as the Gray-Wyner problem, a classical multiterminal source coding problem concerning compression of correlated memoryless sources. Several decades later, for the DMS, the second-order asymptotics for the Gray-Wyner problem was derived by Watanabe [10] for the lossless case and by Zhou, Tan and Motani [11] for the lossy case. Furthermore, Liu et al. [12, Theorem 17] derived second-order converse bounds both discrete correlated sources under bounded distortion measures and bivariate Gaussian sources under the quadratic distortion measure. Recently, for Gaussian sources under the quadratic distortion measure, the explicit form of the rate-distortion region was characterized by Yu [13, Theorem 3 and Corollary 2]. Throughout the paper, we refer to the Gray-Wyner problem for compression of correlated Gaussian memoryless sources under the quadratic distortion measure as the Gaussian Gray-Wyner problem.

In contrast to the discrete case, the exact second-order asymptotics for the Gaussian Gray-Wyner problem remains open till now. The key challenge lies in how to express the common information for correlated Gaussian memoryless sources and derive the Gaussian analogue of type-covering lemma [11, Lemma 8]. The method of types [14], tailed and used extensively for the DMS, cannot be applied to continuous sources like Gaussian sources.

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A. Main Contributions

In this paper, we address the above challenge and derive exact second-order asymptotics for the Gaussian Gray-Wyner problem. In particular, we consider every boundary rate tuple of the first-order rate-distortion region (cf. Definition 3). We find that these boundary rate tuples have two cases: with nonzero common rate and with zero common rate. For the zero common rate case, both private rates equal the rate-distortion function of the P2P case but the second-order asymptotics differs from the P2P case. We would like to emphasize that the boundary rate tuple with zero common rate was ignored in all previous second-order analyses [10], [11], [12].

For a boundary rate tuple with a nonzero common rate, our main contribution is an achievability bound that matches the existing converse bound of Liu et al. [12, Theorem 17]. In particular, we derive a uniform covariance-shell covering lemma that is parallel to the type covering lemma for the DMS. To do so, we first define Gaussian type classes by grouping pairs of correlated Gaussian source sequences according to their empirical powers and correlation parameters, generalizing the power type class for a single Gaussian source in the successive refinement problem [15, Section VI-A]. Subsequently, for each Gaussian type class, we use random covering for the common message and use Euclidean ball covering for the private messages. In particular, we show that the achievable common rate equals the minimum common rate in the first-order Gaussian region with respect to the Gaussian type class when two private rates are set as desired values. Subsequently, we use the Berry–Esseen theorem [16, Theorem 1.1] to obtain the achievability second-order asymptotics. For the converse part, we combine the second-order converse bounds for the Gaussian Gray–Wyner problem of Liu et al. [12, Theorem 17] and the P2P Gaussian rate-distortion problem of Kostina and Verdú [6, Theorem 40].

For the boundary rate tuple with the zero common rate, our achievability and converse bounds are both new. Note that even if the first-order common rate equals zero, the common compression message can still convey $O(\sqrt{n})$ nats when the blocklength equals n via the second-order common rate. When the second-order common rate is positive, we use the same covariance-shell covering lemma as the nonzero common rate case to bound how the common compression message helps reduce the private compression rates. When the second-order common rate is also zero, we use two P2P coverings without a common compression message. To derive the matching converse result, we bound the non-excess-distortion probability of any code using Gaussian hypercontractivity [17], [18] and a bivariate central limit theorem [16].

B. Other Related Works

Beyond the Gray–Wyner problem, second-order asymptotics has also been studied for other multiterminal source coding problems. Specifically, Tan and Kosut [19] studied the Slepian–Wolf problem of lossless compression of correlated sources while Nomura and Han [20] generalized their results to mixed sources. Furthermore, the successive refinement problem, whose first-order rate region was characterized by Rimoldi [21], was studied by No, Ingber, and Weissman [22] under separate excess-distortion probabilities criteria and by Zhou, Tan, and Motani [15] under the joint excess-distortion probability criterion. For the Wyner-Ziv problem of lossy source coding with decoder side information, achievability result was obtained by Watanabe, Kuzuoka, and Tan [23], which was recently refined by Li and Li [24] and exactly characterized by Zhou [25] under mild conditions. For a more comprehensive summary of second-order asymptotics of lossy source coding, refer to the monograph [26].

The Gray–Wyner problem is also closely related to the study of common information. This line of research was pioneered by Wyner [27], who characterized the minimum common rate for the lossless Gray-Wyner problem when the sum rate of three encoders equals the joint entropy of the two correlated sources. Subsequently, Viswanatha, Akyol, and Rose [7] extended [27] to the lossy case and explicitly calculated the lossy common information for bivariate Gaussian sources under the quadratic distortion measure. For bivariate Gaussian sources, Sula and Gastpar [28] determined the minimum common rate for a prescribed sum of the two private rates using Gaussian auxiliary random variables.

C. Organization for the Rest of the Paper

In Section II, we formulate the Gaussian Gray-Wyner problem, define the first- and second-order asymptotics and recall existing result. In Section III, we present and discuss our main results. The achievability and converse proof of our main result is presented in Sections IV and V, respectively. Finally, Section VI concludes the paper and discusses future directions. For smooth presentation, the proofs of supporting lemmas are deferred to the appendices.

Notation

Upper-case letters denote random variables, and lower-case letters denote their realizations. Bold upper-case letters denote random vectors and matrices, while bold lower-case letters denote deterministic vectors and matrices. We use $(\mathbb{N}, \mathbb{R}, \mathbb{R}_+, \mathbb{R}_{+,0})$ to denote the sets of positive integers, real numbers, positive real numbers and non-negative real numbers, respectively. For any real number $x \in \mathbb{R}$, we define $[x]^+ := \max\{x, 0\}$. For any two integers $(a, b) \in \mathbb{N}^2$ such that $a \leq b$, we define $[a : b] := \{a, a+1, \ldots, b\}$ and use $[a]$ to denote $[1 : a]$ when $a \geq 1$. Given $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\boldsymbol{\sigma} \in \mathbb{S}_+^d$, we use $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma})$ to denote the Gaussian distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\sigma}$. We use Φ to denote the cumulative distribution function (cdf) for the standard Gaussian random variable with mean zero and variance one and use $\mathrm{Q} := 1 - \Phi$ to denote its complementary

cdf. We write $(\mathsf{E}_Q[\cdot], \mathsf{Var}_Q[\cdot], \mathsf{Cov}_Q(\cdot,\cdot))$ for expectation, variance, and covariance with respect to a distribution Q. We use the standard asymptotic notation of [11, Section II-A] such as $(O(\cdot), o(\cdot))$. All logarithms are natural. We use the scalar symbols (R_j, L_j, D_j, M_j, V) for rates, second-order coding rates, distortions, alphabet sizes, and dispersion, respectively.

Fix an integer $d \in \mathbb{N}$. We use $(\mathbb{S}^d, \mathbb{S}^d_+, \mathbb{S}^d_{++})$ to denote the real symmetric, positive-semidefinite, and positive-definite $d \times d$ matrices. Fix any two matrices $(\mathbf{a}, \mathbf{b}) \in (\mathbb{S}^d)^2$, we write $\mathbf{a} \preceq \mathbf{b}$ when $\mathbf{b} - \mathbf{a} \in \mathbb{S}^d_+$ and define $\langle \mathbf{a}, \mathbf{b} \rangle := \mathrm{tr}(\mathbf{a}^\mathsf{T}\mathbf{b})$. Furthermore, we use $|\mathbf{a}|$ to denote the determinant, $\mathrm{tr}(\mathbf{a})$ to denote the trace, $\mathbf{a}^\mathsf{T}$ to denote the transpose and $\|\mathbf{a}\|$ to denote the norm. For each $j \in [d]$, we use $\mathbf{e}_j \in \mathbb{R}^d$ to denote the d-dimensional vector with one being the j-th element and zero elsewhere. For a vector $\boldsymbol{\mu} \in \mathbb{R}^d$ and a real number $r \in \mathbb{R}_+$, we use $\mathcal{B}_d(\boldsymbol{\mu}, r) := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x} - \boldsymbol{\mu}\| \le r\}$ to denote the d-dimensional ball with center $\boldsymbol{\mu}$ and radius r; with zero radius, $\mathcal{B}_d(\boldsymbol{\mu}, 0) := \{\boldsymbol{\mu}\}$. For each $j \in [d]$, given a vector $\mathbf{v} \in \mathbb{R}^d$, we use $[\mathbf{v}]_j$ to denote its jth coordinate. Given another integer $n \in \mathbb{N}$ and a matrix $\mathbf{x}^n$ of dimension $d \times n$, we use $(\mathbf{x}^n)_j$ to denote its j-th row. We write $\mathbf{0}_d := (0, \ldots, 0)^\mathsf{T} \in \mathbb{R}^d$ and $\mathbf{0}_{d\times d} \in \mathbb{R}^{d\times d}$ for the all zero vector and all zero matrix, respectively, and use $\mathbf{1}_d$ for the $d \times d$ identity matrix. Norms are Euclidean for vectors and Frobenius for matrices.

II. Problem Formulation and Existing Results

A. System Model and Definitions

Fix positive real numbers $(\sigma_X, \sigma_Y) \in \mathbb{R}^2_+$, a correlation coefficient $\rho \in (0,1)$, two distortion levels $(D_1, D_2) \in (0, \sigma_X^2) \times (0, \sigma_Y^2)$ and three positive integers $(M_0, M_1, M_2) \in \mathbb{N}^3$. Define the joint source distribution P_{XY}:

$$P_{XY} := \mathcal{N}(\mathbf{0}_2, \boldsymbol{\sigma}_{XY}), \tag{1}$$

$$\boldsymbol{\sigma}_{XY} := \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix} \in \mathbb{S}^2_{++}. \tag{2}$$

Let the induced marginal distributions be (P_X, P_Y). Fix a positive integer $n \in \mathbb{N}$. We consider memoryless sources $(X^n, Y^n) \in (\mathbb{R}^n)^2$ that is generated i.i.d. from P_{XY}. As shown in Fig. 1, in the Gaussian Gray-Wyner problem, three encoders compress the pair of correlated source sequences (X^n, Y^n) via three encoders (f_0, f_1, f_2), which are used by two decoders (ϕ_1, ϕ_2) to generate source sequence estimates $(\hat{X}^n, \hat{Y}^n)$, respectively. In particular, encoder f_0 provides the common message S_0 to both decoders while encoder f_i, for each $i \in [2]$, provides a private message S_i to decoder ϕ_i. To evaluate the reconstruction quality, we use the quadratic distortion measure. Specifically, given any $(x, \hat{x}) \in \mathbb{R}^2$ and $(x^n, \hat{x}^n) \in \mathbb{R}^n \times \mathbb{R}^n$, the quadratic distortion measure and its block extension are defined as follows:

$$d(x, \hat{x}) := (x - \hat{x})^2, \tag{3}$$

$$d(x^n, \hat{x}^n) := \frac{1}{n} \sum_{i \in [n]} d(x_i, \hat{x}_i) = \frac{1}{n} \|x^n - \hat{x}^n\|^2. \tag{4}$$

We can now define a code that mathematically describes the encoders and decoders.

Definition 1. *An (n, M_0, M_1, M_2)-code consists of three encoders f_i, $i \in [0:2]$, and two decoders:*

$$f_i : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow [M_i], \tag{5}$$

$$\phi_1 : [M_0] \times [M_1] \longrightarrow \mathbb{R}^n, \tag{6}$$

$$\phi_2 : [M_0] \times [M_2] \longrightarrow \mathbb{R}^n. \tag{7}$$

Given a code and a pair of sequences $\mathbf{x}^n = (x^n, y^n) \in \mathbb{R}^{2\times n}$, define the reconstructed sequences by

$$\hat{x}^n(\mathbf{x}^n) := \phi_1(f_0(\mathbf{x}^n), f_1(\mathbf{x}^n)), \tag{8}$$

$$\hat{y}^n(\mathbf{x}^n) := \phi_2(f_0(\mathbf{x}^n), f_2(\mathbf{x}^n)). \tag{9}$$

To define the performance criterion, define the following excess-distortion and non-excess-distortion events:

$$\mathcal{E}_{1,n} := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : d(x^n, \hat{x}^n(\mathbf{x}^n)) > D_1\}, \tag{10}$$

$$\mathcal{E}_{2,n} := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : d(y^n, \hat{y}^n(\mathbf{x}^n)) > D_2\}, \tag{11}$$

$$\mathcal{E}_n := \mathcal{E}_{1,n} \cup \mathcal{E}_{2,n}, \tag{12}$$

$$\mathcal{A}_n := \mathbb{R}^{2\times n} \setminus \mathcal{E}_n. \tag{13}$$

Define the joint excess-distortion probability as

$$\varepsilon_n(D_1, D_2) := P^n_{XY}(\mathcal{E}_{1,n} \cup \mathcal{E}_{2,n}) = P^n_{XY}(\mathcal{E}_n) = 1 - P^n_{XY}(\mathcal{A}_n). \tag{14}$$

For ease of notation, we use ε_n when the distortions are clear. The first-order asymptotics, a.k.a., the rate-distortion region, is defined as follows.

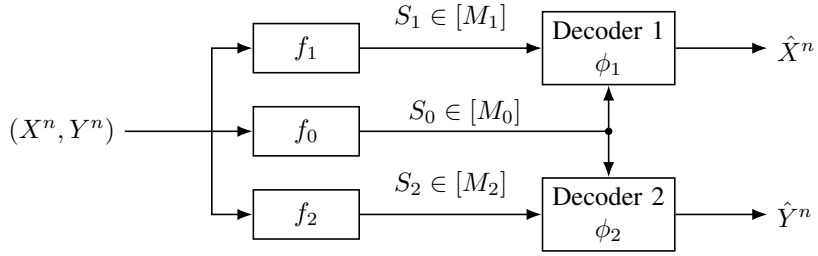


Fig. 1. System model for the Gaussian Gray–Wyner problem.

Definition 2. *A rate tuple $(R_0, R_1, R_2) \in \mathbb{R}^3_{+,0}$ is said to be first-order achievable if there exists a sequence of (n, M_0, M_1, M_2) codes such that for each $j \in [0:2]$,*

$$\limsup_{n\to\infty} \frac{\log M_{j,n}}{n} \le R_j, \tag{15}$$

$$\lim_{n\to\infty} \varepsilon_n = 0. \tag{16}$$

The convex closure of all first-order achievable rate tuples is the first-order rate region, denoted by $\mathcal{R}(D_1, D_2 \mid P_{XY})$.

In existing studies including the original paper of Gray and Wyner [9], the performance criterion is not (16) but the following expected distortion criteria:

$$\limsup_{n\to\infty} \mathsf{E}_{P^n_{XY}}[d(X^n, \hat{X}^n)] \le D_1, \tag{17}$$

$$\limsup_{n\to\infty} \mathsf{E}_{P^n_{XY}}[d(Y^n, \hat{Y}^n)] \le D_2. \tag{18}$$

Denote the corresponding first-order rate region as $\mathcal{R}_{\mathrm{E}}(D_1, D_2 \mid P_{XY})$ under expected distortion. Appendix A proves that the two distortion criteria have the same first-order rate region.

Lemma 1. *We have*

$$\mathcal{R}(D_1, D_2 \mid P_{XY}) = \mathcal{R}_{\mathrm{E}}(D_1, D_2 \mid P_{XY}). \tag{19}$$

Thus, all existing studies on first-order asymptotics are valid under the joint excess-distortion probability criterion.

Given private rates $(R_1, R_2) \in \mathbb{R}^2_{+,0}$, define the minimal common rate as

$$\mathsf{R}_0(R_1, R_2, D_1, D_2 \mid P_{XY}) := \inf\left\{r_0 \in \mathbb{R}_{+,0} : (r_0, R_1, R_2) \in \mathcal{R}(D_1, D_2 \mid P_{XY})\right\}. \tag{20}$$

In the next subsection, we recall existing results that characterize $\mathcal{R}(D_1, D_2 \mid P_{XY})$ and $\mathsf{R}_0(R_1, R_2, D_1, D_2 \mid P_{XY})$.

B. First-Order Boundary and Second-Order Criterion

For a covariance matrix $\boldsymbol{\sigma} = [\sigma_{jk}]_{j,k\in[2]} \in \mathbb{S}^2_{++}$ with $\sigma_{jj} > D_j$ and $\sigma_{12} > 0$, let $Q^{\boldsymbol{\sigma}} := \mathcal{N}(\mathbf{0}_2, \boldsymbol{\sigma})$ and denote its marginals by $(Q^{\boldsymbol{\sigma}}_X, Q^{\boldsymbol{\sigma}}_Y)$. For $(r_1, r_2) \in \mathbb{R}^2_{+,0}$, we use the covariance shorthand

$$\mathsf{R}_0(\boldsymbol{\sigma}; r_1, r_2) := \mathsf{R}_0(r_1, r_2, D_1, D_2 \mid Q^{\boldsymbol{\sigma}}). \tag{21}$$

The marginal Gaussian rate-distortion functions and the correlation are

$$R_X(Q^{\boldsymbol{\sigma}}_X, D_1) = \frac{1}{2}\log\frac{\sigma_{11}}{D_1}, \tag{22}$$

$$R_Y(Q^{\boldsymbol{\sigma}}_Y, D_2) = \frac{1}{2}\log\frac{\sigma_{22}}{D_2}, \tag{23}$$

$$\rho(\boldsymbol{\sigma}) := \frac{\sigma_{12}}{\sqrt{\sigma_{11}\sigma_{22}}}. \tag{24}$$

Define the marginal information requirements by

$$\alpha(\boldsymbol{\sigma}; r_1) := \left[R_X(Q^{\boldsymbol{\sigma}}_X, D_1) - r_1\right]^+, \tag{25}$$

$$\beta(\boldsymbol{\sigma}; r_2) := \left[R_Y(Q^{\boldsymbol{\sigma}}_Y, D_2) - r_2\right]^+. \tag{26}$$

For $t \in \mathbb{R}_{+,0}$, define

$$\nu_t := \sqrt{1 - e^{-2t}}. \tag{27}$$

For fixed $\rho \in (0,1)$ and $(\alpha,\beta) \in \mathbb{R}^2_{+,0}$, define

$$r_{\alpha,\beta} := (\rho - \nu_\alpha \nu_\beta) e^{\alpha+\beta}. \tag{28}$$

For positive marginal requirements, the four regimes are

$$\mathcal{P}_1 := \{(\alpha,\beta) \in \mathbb{R}^2_+ : \nu_\alpha \nu_\beta \le \rho \le \min\{\nu_\beta/\nu_\alpha, \nu_\alpha/\nu_\beta\}\}, \tag{29}$$

$$\mathcal{P}_2 := \{(\alpha,\beta) \in \mathbb{R}^2_+ : \rho \le \nu_\alpha \nu_\beta\}, \tag{30}$$

$$\mathcal{P}_3 := \{(\alpha,\beta) \in \mathbb{R}^2_+ : \rho > \nu_\beta/\nu_\alpha\}, \tag{31}$$

$$\mathcal{P}_4 := \{(\alpha,\beta) \in \mathbb{R}^2_+ : \rho > \nu_\alpha/\nu_\beta\}. \tag{32}$$

The neighboring expressions agree at every shared interface. For positive arguments, define

$$\Upsilon_\rho(\alpha,\beta) := \begin{cases} \alpha+\beta - \dfrac{1}{2}\log\dfrac{1-r^2_{\alpha,\beta}}{1-\rho^2}, & \text{if } (\alpha,\beta) \in \mathcal{P}_1, \\ \alpha+\beta + \dfrac{1}{2}\log(1-\rho^2), & \text{if } (\alpha,\beta) \in \mathcal{P}_2, \\ \alpha, & \text{if } (\alpha,\beta) \in \mathcal{P}_3, \\ \beta, & \text{if } (\alpha,\beta) \in \mathcal{P}_4 \end{cases}. \tag{33}$$

For $(\alpha,\beta) \in \mathbb{R}^2_{+,0}$ with $\alpha\beta = 0$, define its continuous extension on the coordinate axes by

$$\Upsilon_\rho(\alpha,\beta) := \max\{\alpha,\beta\}. \tag{34}$$

Proposition 1. *For every covariance $\boldsymbol{\sigma}$ specified above and every $(r_1,r_2) \in \mathbb{R}^2_{+,0}$,*

$$\mathsf{R}_0(\boldsymbol{\sigma}; r_1, r_2) = \Upsilon_{\rho(\boldsymbol{\sigma})}\big(\alpha(\boldsymbol{\sigma}; r_1), \beta(\boldsymbol{\sigma}; r_2)\big). \tag{35}$$

In particular, the operational lower boundary is

$$\mathsf{R}_0(R_1,R_2,D_1,D_2 \mid P_{XY}) = \Upsilon_\rho\big([R_X(P_X,D_1)-R_1]^+, [R_Y(P_Y,D_2)-R_2]^+\big). \tag{36}$$

Lemma 1 transfers Yu's expected-distortion boundary to (36). The Gaussian residual representation used by the coding scheme is recalled in (231) in Appendix A. A nonnegative rate tuple belongs to $\mathcal{R}(D_1,D_2 \mid P_{XY})$ precisely when its common rate is at least (36).

Equivalently, the three-dimensional rate region is

$$\mathcal{R}(D_1,D_2 \mid P_{XY}) = \big\{(R_0,R_1,R_2) \in \mathbb{R}^3_{+,0} : R_0 \ge \Upsilon_\rho\big([R_X(P_X,D_1)-R_1]^+, [R_Y(P_Y,D_2)-R_2]^+\big)\big\}. \tag{37}$$

For each fixed private-rate pair (R_1,R_2), equality in (37) determines its minimum common rate. All nonnegative rate tuples on or above this graph are achievable. Fig. 2 also displays private rates larger than their marginal rate-distortion functions. For example, if $R_1 \ge R_X(P_X,D_1)$, the first positive-part argument in (37) is zero, so only the second source contributes to the minimum common rate. If both private rates exceed their marginal rate-distortion functions, the minimum common rate is zero. These points belong to the lower common-rate graph, but tuples with excess private rate are excluded by Definition 3.

Definition 3. *A tuple $\mathbf{r}^* = (R_0^*, R_1^*, R_2^*) \in \mathcal{R}(D_1,D_2 \mid P_{XY}) \subset \mathbb{R}^3_{+,0}$ is a boundary rate tuple if it is Pareto-minimal, meaning that reducing any positive coordinate with the other two fixed leaves the rate region. For $j \in [0:2]$, the rate R_j^* is coordinate $j+1$ of $\mathbf{r}^*$. Thus, for every $j \in [0:2]$ and $0 < t \le R_j^*$,*

$$\mathbf{r}^* - t\mathbf{e}_{j+1} \notin \mathcal{R}(D_1,D_2 \mid P_{XY}). \tag{38}$$

Recall the fixed source covariance $\boldsymbol{\sigma}_{XY}$ in (2), with diagonal entries $[\boldsymbol{\sigma}_{XY}]_{11} = \sigma_X^2$ and $[\boldsymbol{\sigma}_{XY}]_{22} = \sigma_Y^2$. For a tuple whose private rates satisfy, for each $j \in [2]$,

$$0 \le R_j^* \le \frac{1}{2}\log\frac{[\boldsymbol{\sigma}_{XY}]_{jj}}{D_j}, \tag{39}$$

define its selected marginal requirements by

$$(\alpha^*,\beta^*) := \big(R_X(P_X,D_1) - R_1^*, R_Y(P_Y,D_2) - R_2^*\big). \tag{40}$$

The unstarred pair (α,β) ranges over the feasible rectangle $\mathcal{I}_{\boldsymbol{\sigma}_{XY}} := [0, R_X(P_X,D_1)] \times [0, R_Y(P_Y,D_2)]$. By Proposition 1, its lower-boundary tuple is

$$\mathbf{r}(\alpha,\beta) := \big(\Upsilon_\rho(\alpha,\beta), R_X(P_X,D_1) - \alpha, R_Y(P_Y,D_2) - \beta\big). \tag{41}$$

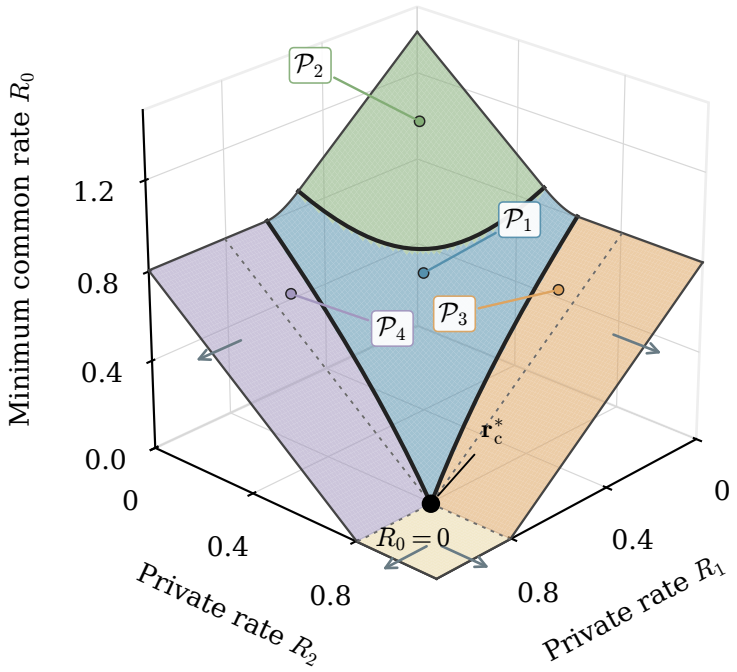


Fig. 2. Minimum common rate in (36), for $\sigma_X^2 = \sigma_Y^2 = 1$, $D_1 = D_2 = 0.2$, and $\rho = 0.6$. The colored branches and their continuous extensions correspond to $(\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4)$ in (29)–(32), with black curves marking the interfaces. The pale-gold sheet has $R_0 = 0$, and the black point is $\mathbf{r}_c^* = (0, \frac{1}{2}\log 5, \frac{1}{2}\log 5)$.

Thus, (α^*, β^*) denotes one fixed pair of these parameters, associated with the operating tuple $\mathbf{r}^* = \mathbf{r}(\alpha^*, \beta^*)$. The four sets $(\mathcal{P}_1, \ldots, \mathcal{P}_4)$ in (29)–(32) describe the branches of this same parametrization, rather than different types of marginal requirements. Define the zero-common-rate tuple by

$$\mathbf{r}_c^* := \big(0, R_X(P_X, D_1), R_Y(P_Y, D_2)\big). \tag{42}$$

The following lemma identifies which tuples on the lower-boundary graph are boundary rate tuples.

Lemma 2. *Every boundary rate tuple satisfies* (39)*. Within this private-rate rectangle, a rate tuple is a boundary rate tuple if and only if one of the following cases holds:*

(i) The common rate is zero and $\mathbf{r}^* = \mathbf{r}_c^*$.

(ii) The common rate is positive, $\mathbf{r}^* = \mathbf{r}(\alpha^*, \beta^*)$*, and* $(\alpha^*, \beta^*) \in \mathbb{R}_+^2$*. The pair* (α^*, β^*) *belongs to one of the following four regimes:*

(a) If $(\alpha^*, \beta^*) \in \mathcal{P}_1$*, then*

$$R_0^* + R_1^* + R_2^* = R_X(P_X, D_1) + R_Y(P_Y, D_2) - \frac{1}{2}\log\frac{1 - r_{\alpha^*,\beta^*}^2}{1 - \rho^2}. \tag{43}$$

(b) If $(\alpha^*, \beta^*) \in \mathcal{P}_2$*, then*

$$R_0^* + R_1^* + R_2^* = R_X(P_X, D_1) + R_Y(P_Y, D_2) + \frac{1}{2}\log(1 - \rho^2). \tag{44}$$

(c) If $(\alpha^*, \beta^*) \in \mathcal{P}_3$*, then* $R_2^* = 0$ *and* $R_0^* + R_1^* = R_X(P_X, D_1)$.

(d) If $(\alpha^*, \beta^*) \in \mathcal{P}_4$*, then* $R_1^* = 0$ *and* $R_0^* + R_2^* = R_Y(P_Y, D_2)$.

The proof of Lemma 2 is provided in Appendix C. In Fig. 2, the black point is the zero-common-rate tuple $\mathbf{r}_c^*$, and every other boundary rate tuple belongs to one of the four regimes in the lemma. The plotted minimum common rate is given by (36), obtained by taking equality in (37).

Definition 4. *Fix a boundary rate tuple $\mathbf{r}^*$ from Definition 3 and $\varepsilon \in (0,1)$. A vector $(L_0, L_1, L_2) \in \mathbb{R}^3$ is second-order $(R_0^*, R_1^*, R_2^*, D_1, D_2, \varepsilon)$-achievable if a code sequence satisfies, for each $j \in [0:2]$,*

$$\limsup_{n\to\infty} \frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \le L_j, \tag{45}$$

$$\limsup_{n\to\infty} \varepsilon_n \le \varepsilon. \tag{46}$$

The closure of these vectors is denoted by $\mathcal{L}(R_0^*, R_1^*, R_2^*, D_1, D_2, \varepsilon)$, abbreviated as $\mathcal{L}(\mathbf{r}^*, D_1, D_2, \varepsilon)$.

Define the zero-rate index set by

$$\mathcal{Z}(\mathbf{r}^*) := \{j \in [0:2] : R_j^* = 0\}. \tag{47}$$

Nonempty message alphabets require $L_j \ge 0$ for $j \in \mathcal{Z}(\mathbf{r}^*)$. When $R_0^* = 0$, a common description may still contain order-$\sqrt{n}$ nats. The definition therefore permits $M_{0,n} > 1$ at the zero-common-rate tuple.

III. Second-Order Asymptotics

A. Preliminaries

We first collect the boundary derivatives and fluctuation statistics used in Theorem 1. The theorem has two cases. At positive common rate, the supporting slopes determine a scalar dispersion through the distortion-tilted information density. At zero common rate, a nonlinear allocation function acts on the two marginal-power fluctuations, whose joint Gaussian limit determines the second-order region.

Proposition 2. *For every boundary rate tuple $\mathbf{r}^*$ with $R_0^* > 0$, the common-rate boundary $\mathsf{R}_0(\boldsymbol{\sigma}; r_1, r_2)$ is continuously differentiable near $(\boldsymbol{\sigma}_{XY}, R_1^*, R_2^*)$. At zero private rates, differentiability refers to the extension established in Appendix B.*

Proposition 2 is proved in Appendix B.

Fix a boundary rate tuple $\mathbf{r}^*$ with $R_0^* > 0$ and recall its marginal requirements (α^*, β^*) from (40). Throughout this subsection, (α^*, β^*) denotes these same fixed marginal requirements. Define the supporting slopes by

$$\lambda_1^* := -\frac{\partial}{\partial r_1}\mathsf{R}_0(r_1, R_2^*, D_1, D_2 \mid P_{XY})\Big|_{r_1=R_1^*} = \partial_\alpha \Upsilon_\rho(\alpha^*, \beta^*), \tag{48}$$

$$\lambda_2^* := -\frac{\partial}{\partial r_2}\mathsf{R}_0(R_1^*, r_2, D_1, D_2 \mid P_{XY})\Big|_{r_2=R_2^*} = \partial_\beta \Upsilon_\rho(\alpha^*, \beta^*). \tag{49}$$

Write $\boldsymbol{\lambda}^* := (\lambda_1^*, \lambda_2^*) \in [0,1]^2$. As in [11, Eq. (22)], $(1, \lambda_1^*, \lambda_2^*)$ is the supporting normal in rate coordinates. Recall the source covariance $\boldsymbol{\sigma}_{XY}$ and define its boundary gradient by

$$\mathbf{g} := \nabla_{\boldsymbol{\sigma}} \mathsf{R}_0(\boldsymbol{\sigma}; R_1^*, R_2^*)|_{\boldsymbol{\sigma}=\boldsymbol{\sigma}_{XY}} \in \mathbb{S}^2. \tag{50}$$

For $(x,y) \in \mathbb{R}^2$, write $\mathbf{x} := (x,y)^\mathsf{T}$ and define the Gaussian Gray–Wyner distortion-tilted information density by

$$\jmath_{XY}(x,y \mid R_1^*, R_2^*, D_1, D_2, P_{XY}) := R_0^* + \mathbf{x}^\mathsf{T}\mathbf{g}\mathbf{x} - \operatorname{tr}(\mathbf{g}\boldsymbol{\sigma}_{XY}). \tag{51}$$

We abbreviate this density by $\jmath_{XY}(x,y)$. In the discrete Gray–Wyner problem, its centered value represents the derivative of the common-rate requirement with respect to the source probabilities [11, Lemma 3]. Here it represents the corresponding derivative with respect to the source covariance. Appendix D proves that (51) also equals the Gaussian evaluation of the log-Laplace definition in [11, Eq. (26)], with the finite sum over the common auxiliary replaced by a Gaussian expectation. Define the dispersion by

$$V(R_1^*, R_2^*, D_1, D_2 \mid P_{XY}) := \mathsf{Var}_{P_{XY}}[\jmath_{XY}(X,Y)]. \tag{52}$$

We abbreviate the dispersion by V. For $(\alpha^*, \beta^*) \in \mathcal{P}_1$, recall $\nu_{\alpha^*} = \sqrt{1 - e^{-2\alpha^*}}$ and $\nu_{\beta^*} = \sqrt{1 - e^{-2\beta^*}}$ from (27), and the residual correlation from (28). Define the ratio u by

$$r := r_{\alpha^*,\beta^*} = (\rho - \nu_{\alpha^*}\nu_{\beta^*}) \exp\big(\alpha^* + \beta^*\big), \tag{53}$$

$$u := \frac{\nu_{\beta^*}}{\nu_{\alpha^*}} \exp\big(\beta^* - \alpha^*\big). \tag{54}$$

Recall the Gaussian distribution $Q^{\boldsymbol{\sigma}} = \mathcal{N}(\mathbf{0}_2, \boldsymbol{\sigma})$ from Section II-B. The following lemma states the properties used in the second-order analysis.

Lemma 3. *For the fixed boundary tuple $\mathbf{r}^*$ with $R_0^* > 0$, the following claims hold.*

(i) The expected value of the distortion-tilted information density satisfies

$$\mathsf{E}_{P_{XY}}[\jmath_{XY}(X,Y)] = R_0^*. \tag{55}$$

(ii) The dispersion satisfies

$$V = 2\,\mathrm{tr}[(\mathbf{g}\boldsymbol{\sigma}_{XY})^2] = \frac{1+(\lambda_1^*+\lambda_2^*-1)^2}{2}. \tag{56}$$

Its values in the four regimes are

$$V = \begin{cases} 0.5\left[1+\left(\dfrac{(u-r)(1-ru)}{u(1-r^2)}\right)^2\right], & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_1, \\ 1, & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_2, \\ 0.5, & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_3, \\ 0.5, & \text{if } (\alpha^*,\beta^*)\in\mathcal{P}_4 \end{cases}. \tag{57}$$

The values agree at the shared interfaces. At the sum-rate interface, the residual correlation $r = r_{\alpha^*,\beta^*}$ equals zero, so $V = 1$. At a one-source interface, $r = u$ or $r = u^{-1}$, so $V = 0.5$.

Appendix D proves Lemma 3. Claim (i) gives the mean, while claim (ii) gives the variance of the Gaussian fluctuation and its value in each first-order regime.

At the zero-common-rate tuple $\mathbf{r}_{\mathrm{c}}^*$ (cf. (42)), a common description supplies the two residual private-rate requirements jointly. For $\rho \in (0,1)$ and $(\delta_1,\delta_2) \in \mathbb{R}_{+,0}^2$, define the allocation function by

$$\psi_\rho(\delta_1,\delta_2) := \begin{cases} \dfrac{\delta_1+\delta_2-2\rho\sqrt{\delta_1\delta_2}}{1-\rho^2}, & \rho^2\delta_1 \le \delta_2 \le \delta_1/\rho^2, \\ \delta_1, & \delta_2 < \rho^2\delta_1, \\ \delta_2, & \delta_1 < \rho^2\delta_2 \end{cases}. \tag{58}$$

In particular, $\psi_\rho(0,0) = 0$. Appendix G derives the equivalent optimization used in the converse.

Let $\mathbf{Z} \in \mathbb{R}^2$ have distribution

$$\mathbf{Z} = (Z_1,Z_2)^{\mathsf{T}} \sim \mathcal{N}\left(\mathbf{0}_2, \frac{1}{2}\begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix}\right), \tag{59}$$

and denote its distribution by $P_{\mathbf{Z}}$. Its coordinates are the limiting marginal-power fluctuations; the allocation function in (58) acts on their two positive private-rate requirements.

B. Main Results and Discussions

Recall the Gaussian source $P_{XY} = \mathcal{N}(\mathbf{0}_2, \boldsymbol{\sigma}_{XY})$ and distortions of Section II-A, the second-order region $\mathcal{L}(\mathbf{r}^*, D_1, D_2, \varepsilon)$ of Definition 4, and the zero-rate index set $\mathcal{Z}(\mathbf{r}^*)$ in (47). At positive common rate, the supporting slopes and dispersion are specified by (48)–(49) and (56). At the zero-common-rate tuple (42), the allocation function ψ_ρ and Gaussian distribution $P_{\mathbf{Z}}$ are specified by (58) and (59). The source distribution, distortions, and operating boundary tuple are held fixed as $n \to \infty$.

Theorem 1. *Given a boundary rate tuple $\mathbf{r}^* = (R_0^*, R_1^*, R_2^*)$ and $\varepsilon \in (0,1)$, we have the following.*

(i) If $R_0^* > 0$,

$$\mathcal{L}(\mathbf{r}^*, D_1, D_2, \varepsilon) = \Big\{(L_0,L_1,L_2) \in \mathbb{R}^3 : L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 \ge \sqrt{V}\,\mathrm{Q}^{-1}(\varepsilon), \quad L_j \ge 0 \ \forall j \in \mathcal{Z}(\mathbf{r}^*)\Big\}. \tag{60}$$

(ii) If $\mathbf{r}^* = \mathbf{r}_{\mathrm{c}}^*$,

$$\mathcal{L}(\mathbf{r}_{\mathrm{c}}^*, D_1, D_2, \varepsilon) = \Big\{(L_0,L_1,L_2) \in \mathbb{R}^3 : L_0 \ge 0, \quad P_{\mathbf{Z}}\big\{\psi_\rho([Z_1-L_1]^+, [Z_2-L_2]^+) \le L_0\big\} \ge 1-\varepsilon\Big\}. \tag{61}$$

Proof sketch: The proof of Theorem 1 is provided in Sections IV and V. For achievability, we group source blocks by their empirical covariance and construct a common description with two private refinements for each group. Our uniform covering lemma ensures simultaneous reconstruction whenever the local common-rate requirement fits the available budget. At positive common rate, a differentiable expansion of this requirement yields a scalar Gaussian limit. At zero common rate, a directional expansion yields a bivariate Gaussian probability; when $L_0 = 0$, separate private coverings attain the same limit with $M_{0,n} = 1$. For the converse, we combine the existing positive-weight Gaussian Gray–Wyner bound with the scalar P2P bound at positive common rate. At zero common rate, our new covariance-cell bound controls simultaneous successful decoding before taking the joint Gaussian limit. These bounds yield the same two Gaussian probabilities as achievability.

We make the following two remarks. Firstly, (60) has the same weighted-half-space form as the discrete lossy Gray–Wyner characterization [11, Theorem 4]. In both cases, the negative private-rate derivatives determine the supporting slopes, and

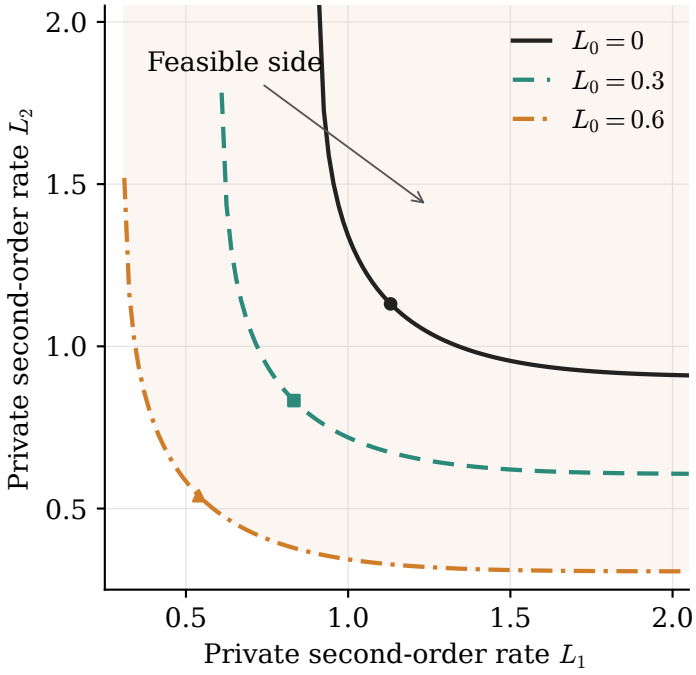


Fig. 3. Private second-order regions from (61), for $\sigma_X^2 = \sigma_Y^2 = 1$, $D_1 = D_2 = 0.2$, $\rho = 0.6$, and $\varepsilon = 0.1$. Solid, dashed, and dash-dotted curves correspond to $L_0 = 0, 0.3, 0.6$, respectively. Each feasible region lies above and to the right of its curve. The $L_0 = 0$ curve is the exact jointly evaluated two-P2P benchmark; markers indicate $L_1 = L_2$. Probabilities are evaluated by one-dimensional conditional Gaussian integration and scalar root inversion.

the variance of the centered distortion-tilted information density determines the dispersion. The Gaussian setting replaces the joint type by the empirical covariance and uses the quadratic covariance-gradient representation in (51). Our covariance-shell covering lemma supplies the counterpart of the discrete type-covering lemma, while the positive-weight converse is inherited from [12, Theorem 17]. The Gaussian result also includes finite interfaces and one-source cases with a zero supporting slope, where the scalar P2P converse [6, Theorem 40] applies. Corollary 2 makes the four Gaussian expressions explicit. Lemma 2 distinguishes a redundant private description inside a strict one-source face from a necessary private description at an adjoining interface. Appendix P provides the spectral identities and a detailed discrete/Gaussian comparison.

Secondly, a positive second-order common coding rate L_0 in (61) reduces the private second-order coding rates even though $R_0^* = 0$. Increasing L_0 enlarges the success event, so the feasible private-rate region expands and its boundary moves towards smaller (L_1, L_2). When $L_0 = 0$, two P2P coverings with $M_{0,n} = 1$ attain the entire slice under the same joint excess-distortion criterion; their source-power fluctuations retain correlation ρ^2 (cf. Section IV-E2). Fig. 3 illustrates how increasing L_0 shifts the private-rate boundary down and left relative to this exact private-only benchmark.

For the positive-common-rate case, recall the selected marginal requirements (α^*, β^*) in (40), the four regimes in (29)–(32), and, on $\mathcal{P}_1$, the residual correlation $r = r_{\alpha^*,\beta^*}$ and ratio u in (54). The following corollary writes the region explicitly for each regime.

Corollary 2. *Under the hypotheses of Theorem 1, suppose that $R_0^* > 0$. The region consists exactly of the offsets satisfying $L_j \geq 0$ for $j \in \mathcal{Z}(\mathbf{r}^*)$ and the corresponding condition below:*

(i) If $(\alpha^*, \beta^*) \in \mathcal{P}_1$,

$$L_0 + \frac{1 - ru}{1 - r^2} L_1 + \frac{1 - r/u}{1 - r^2} L_2 \geq \sqrt{V}\, \mathrm{Q}^{-1}(\varepsilon), \tag{62}$$

with V specified by the first branch of (57).

(ii) If $(\alpha^*, \beta^*) \in \mathcal{P}_2$,

$$L_0 + L_1 + L_2 \geq \mathrm{Q}^{-1}(\varepsilon). \tag{63}$$

(iii) If $(\alpha^*, \beta^*) \in \mathcal{P}_3$,

$$L_0 + L_1 \geq \frac{\mathrm{Q}^{-1}(\varepsilon)}{\sqrt{2}}. \tag{64}$$

(iv) If $(\alpha^*, \beta^*) \in \mathcal{P}_4$,

$$L_0 + L_2 \geq \frac{\mathrm{Q}^{-1}(\varepsilon)}{\sqrt{2}}. \tag{65}$$

The neighboring conditions agree at every finite interface. In particular, the one-source expressions also hold at their adjoining $\mathcal{P}_1$ *interfaces.*

The proof of Corollary 2 is provided in Appendix O.

IV. Achievability Proof

The proof follows the coding scheme and error analysis of the discrete Gray–Wyner argument [11, Section IV-A]. We replace type covering by simultaneous covering of the two residual blocks in each empirical covariance cell. We first introduce the covariance cells and residual covariances, then specify the coding scheme and establish its covering and rate bounds. Finally, we analyze the nonzero and zero common rate cases separately.

A. Preliminaries

For a deterministic source block $\mathbf{x}^n \in \mathbb{R}^{2\times n}$ with rows (x^n, y^n), define its empirical second-moment matrix by

$$\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) := \frac{1}{n}\mathbf{x}^n(\mathbf{x}^n)^{\mathsf{T}}. \tag{66}$$

Its diagonal entries are the empirical powers $\|x^n\|^2/n$ and $\|y^n\|^2/n$, while the off-diagonal entry is their empirical cross moment. For $\mathbf{X}_i := (X_i, Y_i)^{\mathsf{T}}$ and $\mathbf{X}^n := [\mathbf{X}_1 \ \cdots \ \mathbf{X}_n]$, the random matrix is $\hat{\boldsymbol{\Sigma}}_n := \hat{\boldsymbol{\sigma}}(\mathbf{X}^n)$. Since the source mean is known to be zero, these uncentered second moments describe the covariance cells used by the code. Fix nonnegative design rates (r_0, r_1, r_2) and recall the correlation $\rho(\boldsymbol{\sigma}) = \sigma_{12}/\sqrt{\sigma_{11}\sigma_{22}}$ from (24). Write $\lambda_{\min}(\boldsymbol{\sigma}_{XY})$ for the smallest eigenvalue of the source covariance in (2). Define the positive constant

$$c_{\mathrm{cor}} := \sigma_X^{-2} + \sigma_Y^{-2} + \frac{2}{\sigma_X\sigma_Y}, \tag{67}$$

and choose the covariance-neighborhood radius explicitly as

$$\eta_{\mathrm{cov}} := \frac{1}{8}\min\left\{\lambda_{\min}(\boldsymbol{\sigma}_{XY}), \sigma_X^2 - D_1, \sigma_Y^2 - D_2, \frac{\rho}{c_{\mathrm{cor}}}, \frac{1-\rho}{c_{\mathrm{cor}}}\right\} > 0. \tag{68}$$

Define the two closed covariance neighborhoods by

$$\mathcal{K}_0 := \{\boldsymbol{\sigma} \in \mathbb{S}^2 : \|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\| \leq \eta_{\mathrm{cov}}\}, \tag{69}$$

$$\mathcal{K} := \{\boldsymbol{\sigma} \in \mathbb{S}^2 : \|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\| \leq 2\eta_{\mathrm{cov}}\}. \tag{70}$$

The chosen radius makes $\mathcal{K}$ a compact subset of $\{\boldsymbol{\sigma} \in \mathbb{S}^2_{++} : \sigma_{jj} > D_j \ (j \in [2]),\ 0 < \rho(\boldsymbol{\sigma}) < 1\}$, with $\mathcal{K}_0 \subset \mathrm{int}(\mathcal{K})$. Thus every quantized center used below lies in the domain of the Gaussian boundary formula. Appendix K verifies the uniform covariance margins when establishing the optimizer bounds.

For $n \in \mathbb{N}$ and $t \in \mathbb{R}$, let $\mathrm{rnd}_n(t) := \lfloor nt + 1/2 \rfloor/n$; a midpoint is rounded upward. For $\boldsymbol{\sigma} \in \mathbb{S}^2$, define the covariance quantizer by

$$q_n(\boldsymbol{\sigma}) := \begin{bmatrix} \mathrm{rnd}_n(\sigma_{11}) & \mathrm{rnd}_n(\sigma_{12}) \\ \mathrm{rnd}_n(\sigma_{12}) & \mathrm{rnd}_n(\sigma_{22}) \end{bmatrix}. \tag{71}$$

Let $\{\boldsymbol{\sigma}_q : q \in [N_n]\} := q_n(\mathcal{K}_0)$ enumerate its distinct quantized centers in lexicographic order of the three independent entries. Boundedness of $\mathcal{K}_0$ implies $N_n \leq Cn^3$. For $\boldsymbol{\sigma} \in \mathcal{K}_0$, the Frobenius norm satisfies

$$\|q_n(\boldsymbol{\sigma}) - \boldsymbol{\sigma}\|^2 = |\,\mathrm{rnd}_n(\sigma_{11}) - \sigma_{11}|^2 + 2|\,\mathrm{rnd}_n(\sigma_{12}) - \sigma_{12}|^2 + |\,\mathrm{rnd}_n(\sigma_{22}) - \sigma_{22}|^2, \tag{72}$$

$$\leq \frac{1+2+1}{4n^2} = \frac{1}{n^2}, \tag{73}$$

where each scalar rounding error is at most $1/(2n)$. Thus $\|q_n(\boldsymbol{\sigma}) - \boldsymbol{\sigma}\| \leq 1/n$. The triangle inequality implies $q_n(\mathcal{K}_0) \subset \mathcal{K}$ for $1/n \leq \eta_{\mathrm{cov}}$.

For a fixed shell-width constant $C_0 \geq 1$ and $\boldsymbol{\sigma} \in \mathcal{K}$, recall the source-block representation $\mathbf{x}^n \in \mathbb{R}^{2\times n}$ and define

$$\mathcal{T}_n(\boldsymbol{\sigma}) := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : \|\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) - \boldsymbol{\sigma}\| \leq C_0/n\}. \tag{74}$$

For each quantized center $q \in [N_n]$, its cell is

$$\mathcal{T}_{n,q} := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : \hat{\boldsymbol{\sigma}}(\mathbf{x}^n) \in \mathcal{K}_0,\ q_n(\hat{\boldsymbol{\sigma}}(\mathbf{x}^n)) = \boldsymbol{\sigma}_q\}. \tag{75}$$

Equation (73) implies $\mathcal{T}_{n,q} \subseteq \mathcal{T}_n(\boldsymbol{\sigma}_q)$. Recall the common-rate function $\mathsf{R}_0(\boldsymbol{\sigma}; r_1, r_2)$ in (35). Retain the cells indexed by

$$\mathcal{J}_n := \{q \in [N_n] : \mathsf{R}_0(\boldsymbol{\sigma}_q; r_1, r_2) \le r_0\}. \tag{76}$$

The cells indexed by $\mathcal{J}_n$ will be covered by the coding scheme below; all other source blocks use a fallback reconstruction.

For a center $\boldsymbol{\sigma}_q$, a residual covariance $\mathbf{k} = [k_{jk}]_{j,k\in[2]} \in \mathbb{S}_+^2$ belongs to the feasible set

$$\mathcal{F}_q := \{\mathbf{k} \in \mathbb{S}_+^2 : \mathbf{k} \preceq \boldsymbol{\sigma}_q,\ k_{jj} \le D_j e^{2r_j} \text{ for } j \in [2]\}. \tag{77}$$

Choose the residual optimizer

$$\mathbf{k}_q := \operatorname*{arg\,max}_{\mathbf{k}\in\mathcal{F}_q} |\mathbf{k}|. \tag{78}$$

The Gaussian covariance representation (231) uses the decomposition $\mathbf{X} = \mathbf{W} + \mathbf{N}$, where $(\mathbf{W}, \mathbf{N}) \in (\mathbb{R}^2)^2$ are independent centered Gaussian vectors with covariances $(\boldsymbol{\sigma}_q - \mathbf{k}_q, \mathbf{k}_q)$, respectively. Its source marginal is $Q^{\boldsymbol{\sigma}_q}$, and $\mathbf{k}_q$ is the residual covariance. Maximizing its determinant minimizes the common mutual information, while the diagonal constraints ensure that each residual can be described within its private design rate. Lemma 4 below proves positive definiteness and uniform eigenvalue bounds for this optimizer. Strict concavity of $\log|\mathbf{k}|$ then ensures uniqueness. Its explicit form is specified in (316)–(317).

Choose $0 < m < M$ with $m\mathbf{1}_2 \preceq \boldsymbol{\sigma} \preceq M\mathbf{1}_2$ on $\mathcal{K}$, take $0 < \delta < \min\{m, D_1, D_2\}$, and fix $0 < \kappa \le \min\{\delta^2/M, 1/M\}$. The optimizer bound in Lemma 4 implies, for every retained center,

$$\kappa\mathbf{1}_2 \preceq \mathbf{k}_q \preceq \boldsymbol{\sigma}_q \preceq \kappa^{-1}\mathbf{1}_2. \tag{79}$$

For $n \ge 3$, set

$$s_n := \frac{\sqrt{n}}{\log n}. \tag{80}$$

Define the fixed residual-margin constant by

$$\chi_\kappa := 2 + 2\sqrt{\kappa^{-1} + 1}. \tag{81}$$

B. Coding Scheme

1) Codebook Generation: For every retained cell $q \in \mathcal{J}_n$, we generate one common codebook and two private codebooks. The encoder and both decoders know these codebooks and the quantized centers before transmission. Let $\mathbf{b}_q := (\boldsymbol{\sigma}_q - \mathbf{k}_q)^{1/2} \in \mathbb{S}_+^2$ be the principal square root and choose the common codebook size

$$M_{0,q} := \left\lceil \exp\left\{ \frac{n}{2} \log \frac{|\boldsymbol{\sigma}_q|}{|\mathbf{k}_q|} + \frac{s_n}{8} \right\} \right\rceil. \tag{82}$$

For every $m_0 \in [M_{0,q}]$, independently generate a block $\mathbf{U}_q^n(m_0) \in \mathbb{R}^{2\times n}$ with independent $\mathcal{N}(\mathbf{0}_2, \mathbf{1}_2)$ columns and set

$$\mathbf{W}_q^n(m_0) := \mathbf{b}_q \mathbf{U}_q^n(m_0). \tag{83}$$

Thus each column of a common codeword has distribution $\mathcal{N}(\mathbf{0}_2, \boldsymbol{\sigma}_q - \mathbf{k}_q)$. The nonsingular normalized draw $\mathbf{U}_q^n$ also specifies the construction when this covariance is singular. Write $(\mathbf{u}_q^n(m_0), \mathbf{w}_q^n(m_0))$ for the realized blocks. Lemma 4 guarantees a realization with the required simultaneous residual covering, and we fix such a realization separately for every retained center.

For $q \in \mathcal{J}_n$ and $j \in [2]$, the residual covering radius is

$$\xi_{j,q,n} := \sqrt{n[\mathbf{k}_q]_{jj} + \chi_\kappa s_n}. \tag{84}$$

Recall that $\mathcal{B}_n(\mathbf{0}_n, r)$ denotes the Euclidean ball of radius r defined in Notation. Choose ordered private offset codewords $v_{j,q}^n(m_j) \in \mathbb{R}^n$, $m_j \in [M_{j,q}]$, whose balls cover the residual ball:

$$\mathcal{B}_n(\mathbf{0}_n, \xi_{j,q,n}) \subseteq \bigcup_{m_j\in[M_{j,q}]} \mathcal{B}_n(v_{j,q}^n(m_j), \sqrt{nD_j}). \tag{85}$$

Their sizes can be chosen to satisfy

$$M_{j,q} \le C_{\mathrm{cov}} n^4 \max\left\{1, \left(\frac{\xi_{j,q,n}^2}{nD_j}\right)^{n/2}\right\}, \tag{86}$$

where the uniform Euclidean covering bound of Lemma 10 supplies a constant $C_{\mathrm{cov}} \ge 1$ depending only on (κ, D_1, D_2). Its proof in Appendix H uses the method of [15, Lemma 16], including a polynomial prefactor uniform as the radius ratio approaches one. A single zero offset suffices when $\xi_{j,q,n}^2 \le nD_j$. These offsets are shared by all common codewords of center q.

2) Encoding and Decoding: We now define the encoders and decoders using the fixed codebooks. For a center $q \in \mathcal{J}_n$, source block $\mathbf{x}^n$, common index $m_0 \in [M_{0,q}]$, and $j \in [2]$, let the eligible-index sets be

$$\mathcal{A}_{0,n,q}(\mathbf{x}^n) := \{m_0 \in [M_{0,q}] : \left\|(\mathbf{x}^n)_j - \mathbf{w}_q^n(m_0)_j\right\|^2 \le \xi_{j,q,n}^2 \text{ for every } j \in [2]\}, \tag{87}$$

$$\mathcal{A}_{j,n,q}(\mathbf{x}^n, m_0) := \{m_j \in [M_{j,q}] : \left\|(\mathbf{x}^n)_j - \mathbf{w}_q^n(m_0)_j - v_{j,q}^n(m_j)\right\|^2 \le nD_j\}. \tag{88}$$

The common set $\mathcal{A}_{0,n,q}(\mathbf{x}^n)$ contains the indices of common codewords that leave both residual blocks inside their private covering balls. For a selected common index m_0, the private set $\mathcal{A}_{j,n,q}(\mathbf{x}^n, m_0)$ contains the indices of offsets that reconstruct row j within its distortion constraint D_j. Write $\mathcal{H}_{n,q} := \{\mathbf{x}^n : \mathcal{A}_{0,n,q}(\mathbf{x}^n) \neq \emptyset\}$. On this set, let $\mu_{0,q}(\mathbf{x}^n)$ be the smallest eligible common index and $\mu_{j,q}(\mathbf{x}^n)$ the smallest index in $\mathcal{A}_{j,n,q}(\mathbf{x}^n, \mu_{0,q}(\mathbf{x}^n))$. The private coverings (85) make these latter sets nonempty. Reserve one fallback common symbol, denoted by 0, and define the alphabet before relabeling by

$$\mathcal{M}_{0,n} := \{0\} \cup \bigcup_{q \in \mathcal{J}_n} (\{q\} \times [M_{0,q}]). \tag{89}$$

Let $\iota_n : \mathcal{M}_{0,n} \to [M_0]$ be a fixed bijection, where

$$M_0 := 1 + \sum_{q \in \mathcal{J}_n} M_{0,q}. \tag{90}$$

For each $j \in [2]$, use a private alphabet large enough for every retained cell:

$$M_j := \max\left(\{1\} \cup \{M_{j,q} : q \in \mathcal{J}_n\}\right). \tag{91}$$

The local encoder $f_q := f^{\boldsymbol{\sigma}_q, \mathbf{k}_q}$ is

$$f_q(\mathbf{x}^n) := \begin{cases} (\mu_{0,q}(\mathbf{x}^n), \mu_{1,q}(\mathbf{x}^n), \mu_{2,q}(\mathbf{x}^n)), & \text{if } \mathbf{x}^n \in \mathcal{H}_{n,q}, \\ (1,1,1), & \text{otherwise} \end{cases}. \tag{92}$$

Write $f_{i,q}$ for its component $i \in [0:2]$. The global encoder $f = (f_0, f_1, f_2)$ is

$$f(\mathbf{x}^n) := \begin{cases} (\iota_n(q, f_{0,q}(\mathbf{x}^n)), f_{1,q}(\mathbf{x}^n), f_{2,q}(\mathbf{x}^n)), & \text{if } \mathbf{x}^n \in \mathcal{T}_{n,q} \cap \mathcal{H}_{n,q},\ q \in \mathcal{J}_n, \\ (\iota_n(0), 1, 1), & \text{otherwise} \end{cases}. \tag{93}$$

For a fixed cell $q \in \mathcal{J}_n$, the local decoder $\phi_{j,q} := \phi_j^{\boldsymbol{\sigma}_q, \mathbf{k}_q}$ uses the codebooks for that cell. For $j \in [2]$ and valid indices, it is

$$\phi_{j,q}(m_0, m_j) := \mathbf{w}_q^n(m_0)_j + v_{j,q}^n(m_j). \tag{94}$$

The global decoder for the full code recovers q from the common message $\iota_n(q, m_0)$ and applies the corresponding local decoder:

$$\phi_j(\iota_n(q, m_0), m_j) := \phi_{j,q}(m_0, m_j). \tag{95}$$

It outputs $\mathbf{0}_n$ on the fallback common symbol and on an unused padded private index.

For an input in a retained cell, the encoder selects an eligible common codeword and a distortion-feasible private offset for each decoder. The common message carries the cell and codeword indices; decoder j adds the offset indexed by its private message to row j of that common codeword. Otherwise, both decoders output zero.

The finite norm tests and smallest-index rules make the encoder Borel. Use singleton alphabets and zero reconstructions before the quantizer and covering conditions hold, and when $\mathcal{J}_n = \emptyset$.

C. Covering Bounds and Rate Budgets

For the covering analysis of one pair $(\boldsymbol{\sigma}, \mathbf{k})$, suppress the cell index and write $\mathbf{b} := (\boldsymbol{\sigma} - \mathbf{k})^{1/2}$ and $\xi_{j,n} := \sqrt{nk_{jj} + \chi_\kappa s_n}$. Denote the codewords and alphabet sizes by $(\mathbf{w}^n(m_0), v_j^n(m_j), M_i)$. The same one-cell construction applies to any covariance pair satisfying the uniform bounds below. Its local maps are denoted by $(f^{\boldsymbol{\sigma},\mathbf{k}}, \phi_j^{\boldsymbol{\sigma},\mathbf{k}})$. The Gaussian posterior and information-density tests used in the proof establish the existence of these fixed codebooks.

Recall the fixed distortions, the compact covariance neighborhood $\mathcal{K}$ in (70), and the constants (m, M, δ, κ) chosen in Section IV-A. For a covariance $\boldsymbol{\sigma} \in \mathcal{K}$ and nonnegative private rates, its residual optimizer maximizes $|\mathbf{k}|$ subject to $\mathbf{0}_{2\times 2} \preceq \mathbf{k} \preceq \boldsymbol{\sigma}$ and $k_{jj} \le D_j e^{2r_j}$, as in (78). For the one-shell assertion, consider all pairs in the compact family

$$\mathcal{F}_\kappa := \{(\boldsymbol{\sigma}, \mathbf{k}) \in (\mathbb{S}_{++}^2)^2 : \kappa \mathbf{1}_2 \preceq \mathbf{k} \preceq \boldsymbol{\sigma} \preceq \kappa^{-1}\mathbf{1}_2\}. \tag{96}$$

Recall $\mathcal{T}_n(\boldsymbol{\sigma})$ in (74), s_n in (80), and the local encoder and decoders in (92) and (94).

Lemma 4. *For every $\boldsymbol{\sigma} \in \mathcal{K}$ and nonnegative private-rate pair, the determinant maximization in* (78)*, with the corresponding covariance and private caps, has a positive definite optimizer $\mathbf{k}^*$ satisfying*

$$\kappa \mathbf{1}_2 \preceq \mathbf{k}^* \preceq \boldsymbol{\sigma} \preceq \kappa^{-1}\mathbf{1}_2. \tag{97}$$

There are a finite positive constant C and an integer $n_0 \in \mathbb{N}$, uniform over $\mathcal{F}_\kappa$, such that every $n \geq n_0$ and every $(\boldsymbol{\sigma}, \mathbf{k}) \in \mathcal{F}_\kappa$ admit fixed codebooks with the following property: for every $\mathbf{x}^n \in \mathcal{T}_n(\boldsymbol{\sigma})$ and $j \in [2]$,

$$d\big((\mathbf{x}^n)_j, \phi_j^{\boldsymbol{\sigma},\mathbf{k}}(f_0^{\boldsymbol{\sigma},\mathbf{k}}(\mathbf{x}^n), f_j^{\boldsymbol{\sigma},\mathbf{k}}(\mathbf{x}^n))\big) \leq D_j. \tag{98}$$

Their alphabet sizes satisfy

$$\log M_0 \leq \frac{n}{2} \log \frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|} + C s_n. \tag{99}$$

For each $j \in [2]$,

$$\log M_j \leq \frac{n}{2} \left[\log \frac{k_{jj}}{D_j}\right]^+ + C s_n + C \log n. \tag{100}$$

In particular, codebooks can be fixed separately for any finite collection of covariance centers, with the same constants and both distortion constraints holding for every block in each corresponding cell.

The proof of Lemma 4 is provided in Appendix K.

The lemma covers every source block in every retained covariance cell, because the same constants apply to all pairs $(\boldsymbol{\sigma}_q, \mathbf{k}_q)$ and $\mathcal{T}_{n,q} \subseteq \mathcal{T}_n(\boldsymbol{\sigma}_q)$. The quantized covariance neighborhood itself has probability tending to one by the law of large numbers. Whether a center is retained is determined by its common-rate requirement; the resulting probability of an unretained cell is exactly the boundary event below.

Recall the empirical random covariance $\hat{\boldsymbol{\Sigma}}_n = \hat{\boldsymbol{\sigma}}(\mathbf{X}^n)$ in (66) and the quantizer q_n in (71), which rounds each independent covariance entry to the nearest multiple of $1/n$, with midpoint ties rounded upward. For the error analysis, extend q_n to a $\mathcal{K}$-valued map on the positive-semidefinite matrices:

$$\bar{q}_n(\boldsymbol{\sigma}) := \begin{cases} q_n(\boldsymbol{\sigma}), & \text{if } \boldsymbol{\sigma} \in \mathcal{K}_0, \\ \boldsymbol{\sigma}_{XY}, & \text{otherwise} \end{cases}. \tag{101}$$

Set

$$\tilde{\boldsymbol{\Sigma}}_n := \bar{q}_n(\hat{\boldsymbol{\Sigma}}_n). \tag{102}$$

Before $q_n(\mathcal{K}_0) \subset \mathcal{K}$, use $\bar{q}_n \equiv \boldsymbol{\sigma}_{XY}$. This map agrees with the transmitted representative on a retained cell and keeps the covariance boundary defined on every source input.

For the following operational bound, recall the nonnegative design rates (r_0, r_1, r_2) fixed at the beginning of Section IV-A, the joint excess-distortion probability ε_n in (14), and s_n in (80). Let $h_n := C(s_n + \log n)$, where C is a sufficiently large constant depending only on the fixed source, distortions, and neighborhoods.

Lemma 5. *For every nonnegative target-rate triple and all sufficiently large n, the coding scheme admits a code with*

$$\log M_j \leq n r_j + h_n \qquad \textit{for every } j \in [0:2], \tag{103}$$

and

$$\varepsilon_n \leq P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\} + P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0,\ \mathsf{R}_0(\tilde{\boldsymbol{\Sigma}}_n; r_1, r_2) > r_0\}. \tag{104}$$

The same constant C applies to every target-rate triple.

Proof. For every retained cell $q \in \mathcal{J}_n$, the covariance choice (78), its common-rate value in (231), and the definition of $\mathcal{J}_n$ imply

$$\frac{1}{2} \log \frac{|\boldsymbol{\sigma}_q|}{|\mathbf{k}_q|} = \mathsf{R}_0(\boldsymbol{\sigma}_q; r_1, r_2) \leq r_0. \tag{105}$$

For each $j \in [2]$, the diagonal constraint also implies

$$\frac{1}{2} \left[\log \frac{[\mathbf{k}_q]_{jj}}{D_j}\right]^+ \leq r_j. \tag{106}$$

Lemma 4 supplies one κ and codebooks satisfying the same size bounds in every retained cell. There are at most Cn^3 covariance representatives, because $\mathcal{K}_0$ is bounded and the grid has three independent entries. For a nonempty retained set,

$$M_0 = 1 + \sum_{q \in \mathcal{J}_n} M_{0,q}, \tag{107}$$

$$\leq 1 + N_n e^{nr_0 + Cs_n}, \tag{108}$$

$$\leq \exp\bigl(nr_0 + Cs_n + C\log n\bigr), \tag{109}$$

where the first line follows from (90), the second follows from (105) and (99), and the third follows since $N_n \leq Cn^3$. For each $j \in [2]$,

$$M_j = \max_{q \in \mathcal{J}_n} M_{j,q}, \tag{110}$$

$$\leq \exp\bigl(nr_j + Cs_n + C\log n\bigr), \tag{111}$$

where the first line uses the private padding (91), and the second uses (106) and (100). Thus only the common size acquires a factor for the cell index. If $\mathcal{J}_n = \emptyset$, the singleton fallback code satisfies the same bounds.

Every input in a retained covariance cell belongs to its shell by (73) and (74). The distortion guarantee (98) therefore applies to both reconstructions of the global code. An error can occur only outside $\mathcal{K}_0$ or in an unretained cell, and hence

$$\varepsilon_n \leq P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\} + \sum_{q \in [N_n] \setminus \mathcal{J}_n} P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0, q_n(\hat{\boldsymbol{\Sigma}}_n) = \boldsymbol{\sigma}_q\}, \tag{112}$$

$$= P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\} + P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0, \mathsf{R}_0(\tilde{\boldsymbol{\Sigma}}_n; r_1, r_2) > r_0\}, \tag{113}$$

where Eq. (112) follows from the simultaneous covering of every retained cell, and Eq. (113) follows from the cell partition and (76). Enlarging the uniform constant proves (103) and (104). □

The two error terms in (104) correspond to an empirical covariance outside the chosen neighborhood and to an unretained covariance center. Fix an offset satisfying the one-sided constraints in Theorem 1. For the cell-indexed construction, let $b_n := (\log n)^{-1/4}$ and choose

$$r_{0,n} := R_0^* + (L_0 + b_n)/\sqrt{n}. \tag{114}$$

For each $j \in [2]$, let

$$r_{j,n} := \max\{0, R_j^* + L_j/\sqrt{n}\}. \tag{115}$$

These targets are eventually nonnegative. The maximum is inactive at positive private rates and equals $L_j/\sqrt{n}$ when $R_j^* = 0$. For all sufficiently large n and each $j \in [0:2]$, Lemma 5 implies

$$\frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \leq L_j + b_n + \frac{h_n}{\sqrt{n}}. \tag{116}$$

Since $b_n \to 0$ and $h_n = o(\sqrt{n})$, it follows that, for each $j \in [0:2]$,

$$\limsup_{n\to\infty} \frac{\log M_{j,n} - nR_j^*}{\sqrt{n}} \leq L_j. \tag{117}$$

The corresponding finite-blocklength error bound is

$$\varepsilon_n \leq P_{XY}^n\left\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0, \sqrt{n}[\mathsf{R}_0(\tilde{\boldsymbol{\Sigma}}_n; r_{1,n}, r_{2,n}) - R_0^*] > L_0 + b_n\right\} + P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{118}$$

where Eq. (118) follows from substituting (114) and (115) into (104). Recall that $\mathcal{K}_0$ is a neighborhood of $\boldsymbol{\sigma}_{XY}$ as specified in (69). Thus the law of large numbers implies

$$P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\} \longrightarrow 0. \tag{119}$$

It remains to evaluate the first probability in (118). At the zero common rate tuple with $L_0 = 0$, we use the private-only construction specified in Section IV-E.

D. Nonzero Common Rate

For $R_0^* > 0$, recall the distortion-tilted information density in (51). Its Gaussian expectation is $R_0^* + \langle \mathbf{g}, \boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY} \rangle$. Proposition 2 therefore gives

$$\mathsf{R}_0(\boldsymbol{\sigma}; R_1^*, R_2^*) = \mathsf{E}_{Q^{\boldsymbol{\sigma}}}[\jmath_{XY}(X,Y)] + o(\|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\|). \tag{120}$$

Allowing the private rates to vary as well, the same differentiability property implies

$$\mathsf{R}_0(\boldsymbol{\sigma}; r_1, r_2) = R_0^* + \langle \mathbf{g}, \boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY} \rangle - \sum_{j\in[2]} \lambda_j^*(r_j - R_j^*) + o\big(\|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\| + \sum_{j\in[2]} |r_j - R_j^*|\big). \tag{121}$$

The covariance fluctuation in (121) can be written directly as a sum of centered information quantities. Recall $\jmath_{XY}$ from (51) and define the scalar statistic by

$$\mathsf{A}_n := \sqrt{n} \left\langle \mathbf{g}, \hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\sigma}_{XY} \right\rangle, \tag{122}$$

$$= \frac{1}{\sqrt{n}} \sum_{i\in[n]} \big[\mathbf{X}_i^{\mathsf{T}} \mathbf{g} \mathbf{X}_i - \operatorname{tr}(\mathbf{g}\boldsymbol{\sigma}_{XY})\big], \tag{123}$$

$$= \frac{1}{\sqrt{n}} \sum_{i\in[n]} \big[\jmath_{XY}(X_i, Y_i) - R_0^*\big], \tag{124}$$

where Eq. (123) follows from (66), and Eq. (124) follows from (51). The summands are independent and centered under P_{XY}^n, with variance $V > 0$. Define their third absolute moment by $\beta_3 := \mathsf{E}_{P_{XY}}[|\jmath_{XY}(X,Y) - R_0^*|^3]$. Since (51) is a quadratic function of a Gaussian vector,

$$\beta_3 \le 4 \|\mathbf{g}\|^3 \, \mathsf{E}_{P_{XY}}[\|\mathbf{X}\|^6] + 4|\operatorname{tr}(\mathbf{g}\boldsymbol{\sigma}_{XY})|^3 < \infty. \tag{125}$$

The last moment is finite because Gaussian random variables have finite sixth moments. The scalar Berry–Esseen bound, which follows by taking dimension one in [16, Theorem 1.1], therefore yields a finite positive universal constant C_{BE} such that

$$\sup_{t\in\mathbb{R}} \left| P_{XY}^n\{\mathsf{A}_n \le t\} - \Phi(t/\sqrt{V}) \right| \le \frac{C_{\mathrm{BE}} \beta_3}{V^{3/2}\sqrt{n}}. \tag{126}$$

Define the global remainder by

$$E_n := \sqrt{n}[\mathsf{R}_0(\tilde{\boldsymbol{\Sigma}}_n; r_{1,n}, r_{2,n}) - R_0^*] - \mathsf{A}_n + \lambda_1^* L_1 + \lambda_2^* L_2. \tag{127}$$

It is finite because the safe representative lies in $\mathcal{K}$. Fix $B > 0$. Whenever $\sqrt{n} \left\| \hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\sigma}_{XY} \right\| \le B$, the empirical covariance belongs to $\mathcal{K}_0$ for all sufficiently large n. Continuous differentiability in (121) and the $1/n$ quantization error then give a deterministic bound $|E_n| \le e_n(B)$ on this event, with $e_n(B) \to 0$. Furthermore, $\sup_n \mathsf{E}_{P_{XY}^n}[n \left\| \hat{\boldsymbol{\Sigma}}_n - \boldsymbol{\sigma}_{XY} \right\|^2] < \infty$. Markov's inequality therefore gives, for every fixed $\zeta > 0$ and sufficiently large n with $e_n(B) \le \zeta$, $P_{XY}^n\{|E_n| > \zeta\} \le C/B^2$. Taking $n \to \infty$ and then $B \to \infty$ proves that, for every $\zeta > 0$,

$$P_{XY}^n\{|E_n| > \zeta\} \longrightarrow 0. \tag{128}$$

No magnitude estimate for E_n outside $\mathcal{K}_0$ is required. Set $\ell := L_0 + \lambda_1^* L_1 + \lambda_2^* L_2$. For fixed $\zeta > 0$ and sufficiently large n,

$$\varepsilon_n \le P_{XY}^n\{\mathsf{A}_n + E_n > \ell + b_n\} + P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{129}$$

$$\le P_{XY}^n\{\mathsf{A}_n > \ell - \zeta\} + P_{XY}^n\{|E_n| > \zeta\} + P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{130}$$

$$\le \mathrm{Q}\left(\frac{\ell - \zeta}{\sqrt{V}}\right) + \frac{C_{\mathrm{BE}}\beta_3}{V^{3/2}\sqrt{n}} + P_{XY}^n\{|E_n| > \zeta\} + P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{131}$$

where Eq. (129) follows from (118) and the exact remainder identity (127), Eq. (130) follows since $b_n \ge 0$ and the union bound applies, and Eq. (131) follows from (126). The Berry–Esseen term is $O(n^{-1/2})$, while the two probability remainders vanish by (128) and (119). Taking $n \to \infty$ at fixed ζ leads to

$$\limsup_{n\to\infty} \varepsilon_n \le \mathrm{Q}\left(\frac{\ell - \zeta}{\sqrt{V}}\right), \tag{132}$$

where Eq. (132) follows from (131) and the vanishing remainder terms. Continuity of Q permits $\zeta \to 0$ after the blocklength limit and yields

$$\limsup_{n\to\infty} \varepsilon_n \le \mathrm{Q}\left(\frac{L_0 + \lambda_1^* L_1 + \lambda_2^* L_2}{\sqrt{V}}\right). \tag{133}$$

For every offset in (60), this bound is at most ε. Together with (117), it proves the direct part of case (i). The expansion uses only continuous differentiability, including at interfaces where bounded second derivatives are unavailable.

E. Zero Common Rate

We now prove case (ii), where $\mathbf{r}^* = \mathbf{r}_{\mathrm{c}}^*$. For $L_0 > 0$, we use exactly the covariance-cell codebooks, encoder, and decoders of Section IV-B, with the targets in (114) and (115). The change is in the error analysis: the common-rate boundary at this tuple is described by the two marginal information requirements, and the scalar differentiable expansion in Section IV-D no longer applies. The following lemma approximates that boundary uniformly while both marginal requirements and the empirical correlation fluctuate. Its positive-level continuity then converts the joint Gaussian limit into the required error probability. When $L_0 = 0$, the limiting statistic can have an atom at zero. We instead use the private-only code specified below, with one common message and no covariance-cell index.

1) Positive Second-order Common Rate: Recall ψ_ρ from (58) and the Gaussian vector $\mathbf{Z}$ from (59). Fix $\rho \in (0,1)$, $(M,H) \in \mathbb{R}^2_{+,0}$, and $(L_1, L_2) \in \mathbb{R}^2$. For $\mathbf{z} = (z_1, z_2)^{\mathsf{T}} \in \mathbb{R}^2$, define $T : \mathbb{R}^2 \to \mathbb{R}_{+,0}$ by

$$T(\mathbf{z}) := \psi_\rho([z_1 - L_1]^+, [z_2 - L_2]^+). \tag{134}$$

Lemma 6. *There exist finite positive real numbers $(C_{M,H}, t_{M,H})$ such that, for every $0 < t < t_{M,H}$,*

$$\sup_{\substack{0 \le \delta_1, \delta_2 \le M \\ |h| \le H}} |\Upsilon_{\rho+th}(t\delta_1, t\delta_2) - t\psi_\rho(\delta_1, \delta_2)| \le C_{M,H} t^2. \tag{135}$$

The statistic $T(\mathbf{Z})$ has no atom at a positive level. Its mass at zero is

$$P_{\mathbf{Z}}\{T(\mathbf{Z}) = 0\} = P_{\mathbf{Z}}\{Z_1 \le L_1,\ Z_2 \le L_2\}. \tag{136}$$

The proof is provided in Appendix G. In (135), uniformity across branch interfaces and bounded correlation perturbations is essential because the empirical covariance can cross these interfaces at each blocklength. For each $j \in [2]$, define the powers, correlation, and marginal fluctuations by

$$\hat{\Sigma}_{jj,n} := [\hat{\mathbf{\Sigma}}_n]_{jj}, \tag{137}$$

$$\rho(\hat{\mathbf{\Sigma}}_n) = \frac{[\hat{\mathbf{\Sigma}}_n]_{12}}{\sqrt{\hat{\Sigma}_{11,n}\hat{\Sigma}_{22,n}}}, \tag{138}$$

$$Z_{j,n} := \frac{\sqrt{n}}{2} \log \frac{\hat{\Sigma}_{jj,n}}{[\boldsymbol{\sigma}_{XY}]_{jj}}. \tag{139}$$

On the null set of vanishing power, assign these statistics and the marginal requirements below the value zero. For each $j \in [2]$, define the exact remainder $E'_{j,n}$ by

$$E'_{j,n} := Z_{j,n} - \frac{1}{2[\boldsymbol{\sigma}_{XY}]_{jj}\sqrt{n}} \sum_{i \in [n]} ([\mathbf{X}_i]_j^2 - [\boldsymbol{\sigma}_{XY}]_{jj}). \tag{140}$$

Define the centered squared-source vector $\mathbf{H}_i := \frac{1}{2}(X_i^2/\sigma_X^2 - 1, Y_i^2/\sigma_Y^2 - 1)^{\mathsf{T}}$ and its normalized sum $\mathbf{H}_n := n^{-1/2} \sum_{i \in [n]} \mathbf{H}_i$. Let $\boldsymbol{\sigma}_{\mathbf{Z}}$ denote the covariance matrix in (59). Gaussian fourth moments give $\mathsf{Cov}_{P_{XY}}(\mathbf{H}_i, \mathbf{H}_i) = \boldsymbol{\sigma}_{\mathbf{Z}}$, which is positive definite because $0 < \rho < 1$. Its whitened third absolute moment $\beta_{3,2} := \mathsf{E}_{P_{XY}}[\left\|\boldsymbol{\sigma}_{\mathbf{Z}}^{-1/2}\mathbf{H}_1\right\|^3]$ is finite, since the coordinates of $\mathbf{H}_1$ are quadratic Gaussian polynomials and hence their third absolute moments are bounded by sixth Gaussian moments. Bentkus' Berry–Esseen theorem [16, Theorem 1.1] gives a finite positive constant C_2 such that, for every convex Borel set $\mathcal{A} \subseteq \mathbb{R}^2$,

$$|P_{XY}^n\{\mathbf{H}_n \in \mathcal{A}\} - P_{\mathbf{Z}}(\mathcal{A})| \le \frac{C_2 \beta_{3,2}}{\sqrt{n}}. \tag{141}$$

For each $j \in [2]$, write $U_{j,n} := \hat{\Sigma}_{jj,n}/[\boldsymbol{\sigma}_{XY}]_{jj} - 1$. Then $\mathsf{E}_{P_{XY}^n}[U_{j,n}^2] = 2/n$. The bound $|\log(1+u) - u| \le u^2$ for $|u| \le 1/2$, together with (140), Chebyshev's inequality, and Markov's inequality, implies that, for every $\zeta > 0$,

$$P_{XY}^n\{|E'_{j,n}| > \zeta\} \le P_{XY}^n\{|U_{j,n}| > 1/2\} + \frac{\sqrt{n}\mathsf{E}_{P_{XY}^n}[U_{j,n}^2]}{2\zeta}, \tag{142}$$

$$\le \frac{8}{n} + \frac{1}{\zeta\sqrt{n}} \longrightarrow 0. \tag{143}$$

Define $\mathbf{Z}_n := (Z_{1,n}, Z_{2,n})^{\mathsf{T}}$. The identity $\mathbf{Z}_n = \mathbf{H}_n + (E'_{1,n}, E'_{2,n})^{\mathsf{T}}$ therefore yields

$$\mathbf{Z}_n \Rightarrow \mathbf{Z}, \tag{144}$$

where Eq. (144) follows from (141), (143), and Slutsky's theorem. Differentiability of correlation also makes $\sqrt{n}(\rho(\hat{\mathbf{\Sigma}}_n) - \rho)$ tight.

For the design rates (115), set $\Delta_{j,n} := [\frac{1}{2}\log(\hat{\Sigma}_{jj,n}/D_j) - r_{j,n}]^+$. Both private first-order rates are positive, and therefore, for sufficiently large n,

$$\sqrt{n}\,\Delta_{j,n} = \left[\frac{\sqrt{n}}{2}\log\frac{\hat{\Sigma}_{jj,n}}{[\boldsymbol{\sigma}_{XY}]_{jj}} - L_j\right]^+, \tag{145}$$

$$= [Z_{j,n} - L_j]^+, \tag{146}$$

where Eq. (145) follows since $R_j^* = \frac{1}{2}\log([\boldsymbol{\sigma}_{XY}]_{jj}/D_j)$ and the maximum in (115) is eventually inactive, and Eq. (146) follows from (139). For $B > 0$, define

$$\mathcal{V}_n(B) := \left\{\max_{j\in[2]}|Z_{j,n}| \le B, \sqrt{n}|\rho(\hat{\boldsymbol{\Sigma}}_n) - \rho| \le B\right\}. \tag{147}$$

Its complement has arbitrarily small limiting probability as $B \to \infty$. Recall that $\mathcal{K}_0$ is the neighborhood of $\boldsymbol{\sigma}_{XY}$ chosen in (69). At fixed B, $\mathcal{V}_n(B)$ eventually lies in $\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0\}$, since the power ratios are $\exp\bigl(2Z_{j,n}/\sqrt{n}\bigr)$. Hence safe and raw representatives agree there. Uniformly on this event,

$$\sqrt{n}\,\mathsf{R}_0(\tilde{\boldsymbol{\Sigma}}_n; r_{1,n}, r_{2,n}) = \sqrt{n}\,\Upsilon_{\rho(\tilde{\boldsymbol{\Sigma}}_n)}(\alpha(\tilde{\boldsymbol{\Sigma}}_n; r_{1,n}), \beta(\tilde{\boldsymbol{\Sigma}}_n; r_{2,n})), \tag{148}$$

$$= \psi_\rho(\sqrt{n}\,\alpha(\tilde{\boldsymbol{\Sigma}}_n; r_{1,n}), \sqrt{n}\,\beta(\tilde{\boldsymbol{\Sigma}}_n; r_{2,n})) + O_B(n^{-1/2}), \tag{149}$$

$$= \psi_\rho([Z_{1,n} - L_1]^+, [Z_{2,n} - L_2]^+) + O_B(n^{-1/2}), \tag{150}$$

where Eq. (148) follows from Yu's representation (35), Eq. (149) follows from Lemma 6 with $t = n^{-1/2}$, and Eq. (150) follows from (146), (73), and (281). Quantization changes the scaled marginal requirements by $O(n^{-1/2})$, while the bounded scaled correlation perturbation contributes only to the remainder in Lemma 6.

For $L_0 > 0$, recall T from (134). For each fixed B and all sufficiently large n, the error bound (118) implies

$$\varepsilon_n \le P_{XY}^n\{T(\mathbf{Z}_n) > L_0 + b_n - C_B n^{-1/2}\} + P_{XY}^n(\mathcal{V}_n(B)^{\mathrm{c}}) + P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\}, \tag{151}$$

where Eq. (151) follows from (150) on $\mathcal{V}_n(B)$ and the union bound. Taking $n \to \infty$ at fixed B and then $B \to \infty$ leads to

$$\limsup_{n\to\infty}\varepsilon_n \le \lim_{B\to\infty}\left[P_{\mathbf{Z}}\{T(\mathbf{Z}) > L_0\} + \limsup_{n\to\infty} P_{XY}^n(\mathcal{V}_n(B)^{\mathrm{c}})\right], \tag{152}$$

$$= P_{\mathbf{Z}}\{\psi_\rho([Z_1 - L_1]^+, [Z_2 - L_2]^+) > L_0\}, \tag{153}$$

where Eq. (152) follows from (151), (144), (119), continuity of T, and the absence of positive atoms in Lemma 6. Equation (153) follows from tightness and the definition of T in (134). The right-hand side is at most ε for every offset in (61) with $L_0 > 0$. Together with (117), this proves the positive-L_0 part of case (ii).

2) Zero Second-order Common Rate: For $L_0 = 0$, use the following private-only code. Fix $(L_1, L_2) \in \mathbb{R}^2$ and set $M_{0,n} = 1$. For $n \ge 3$ and each $j \in [2]$, choose ordered private reproduction codewords in $\mathbb{R}^n$, denoted by $(v_{j,n}^{(0)}(1), \ldots, v_{j,n}^{(0)}(M_{j,n}^{(0)}))$, to cover the fixed balls of radii

$$\xi_{j,n}^{(0)} := \sqrt{n[\boldsymbol{\sigma}_{XY}]_{jj}}\, e^{L_j/\sqrt{n}}. \tag{154}$$

Specifically, apply the Euclidean covering bound in Lemma 10 with covering radius $\sqrt{nD_j}$. The ratio is bounded uniformly in $n \ge 3$ for these fixed parameters. For every input block and each $j \in [2]$, define

$$\mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n) := \{m \in [M_{j,n}^{(0)}] : \left\|(\mathbf{x}^n)_j - v_{j,n}^{(0)}(m)\right\|^2 \le nD_j\}, \tag{155}$$

$$f_0^{(0)}(\mathbf{x}^n) := 1, \tag{156}$$

$$f_j^{(0)}(\mathbf{x}^n) := \begin{cases}\min \mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n), & \text{if } \mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n) \ne \emptyset,\\ 1, & \text{if } \mathcal{A}_{j,n}^{(0)}(\mathbf{x}^n) = \emptyset\end{cases}, \tag{157}$$

$$\phi_j^{(0)}(1, m) := v_{j,n}^{(0)}(m). \tag{158}$$

For $n \in [2]$, use singleton alphabets and zero reconstructions. These maps are Borel and defined on every input by the smallest-index and fallback rules. Each private encoder uses only its own row, so the joint encoder can implement both without a covariance-cell index.

The fixed offsets make $\xi_{j,n}^{(0)}/\sqrt{nD_j}$ bounded and greater than one for large n, so Lemma 10 implies

$$\log M_{j,n}^{(0)} \le \frac{n}{2}\log\frac{(\xi_{j,n}^{(0)})^2}{nD_j} + 4\log n + C, \tag{159}$$

$$= nR_j^* + \sqrt{n}L_j + 4\log n + C, \tag{160}$$

where Eq. (159) follows from Lemma 10, and Eq. (160) follows from (154). Thus the common offset is zero and each private offset is at most L_j. The covered-ball event is $\mathcal{A}_n^{(0)} := \{\mathbf{x}^n \in \mathbb{R}^{2\times n} : \|(\mathbf{x}^n)_j\|^2 \le n[\boldsymbol{\sigma}_{XY}]_{jj} e^{2L_j/\sqrt{n}},\ j \in [2]\}$, and

$$1 - \varepsilon_n^{(0)} \ge P_{XY}^n(\mathcal{A}_n^{(0)}), \tag{161}$$

$$= P_{XY}^n\{Z_{1,n} \le L_1, Z_{2,n} \le L_2\}, \tag{162}$$

where Eq. (161) follows from the two private coverings, and Eq. (162) follows from the logarithmic power identity (139). For every fixed $\zeta > 0$, the event $\{[\mathbf{H}_n]_j \le L_j - \zeta,\ |E'_{j,n}| \le \zeta,\ j \in [2]\}$ is contained in $\{Z_{j,n} \le L_j,\ j \in [2]\}$. Applying (141) to this rectangle and (143) to the remainder probabilities gives

$$1 - \varepsilon_n^{(0)} \ge P_{\mathbf{Z}}\{Z_1 \le L_1 - \zeta, Z_2 \le L_2 - \zeta\} - \frac{C_2\beta_{3,2}}{\sqrt{n}} - \frac{16}{n} - \frac{2}{\zeta\sqrt{n}}. \tag{163}$$

At fixed ζ, all three remainders vanish. Taking $n \to \infty$ and then $\zeta \to 0$, using continuity of the Gaussian rectangle probability, leads to

$$\liminf_{n\to\infty}(1 - \varepsilon_n^{(0)}) \ge P_{\mathbf{Z}}\{Z_1 \le L_1, Z_2 \le L_2\}, \tag{164}$$

$$= P_{\mathbf{Z}}\{T(\mathbf{Z}) = 0\}, \tag{165}$$

where Eq. (165) follows from (136). For offsets in (61) with $L_0 = 0$, the limiting success probability is at least $1 - \varepsilon$. Together with (160) and $M_{0,n} = 1$, this proves the remaining part of case (ii) and the direct inclusion in (348). The achievability proof of Theorem 1 is complete.

V. Converse Proof

We prove the converse part of Theorem 1. At nonzero common rate, the converse follows from existing results: we specialize the positive-weight Gaussian converse of Liu et al. [12, Theorem 17] and apply the scalar Gaussian converse [6, Theorem 40] when a supporting weight is zero. At zero common rate, we establish a new joint covariance-cell bound and apply it before taking the bivariate Gaussian limit. This order retains the simultaneous decoding constraint at the corner. Fix an arbitrary code sequence satisfying (45). If $R_j^* = 0$, the inequality $M_{j,n} \ge 1$ forces $L_j \ge 0$. For every fixed $\gamma > 0$ and each $j \in [0:2]$, the remaining rate constraints imply the eventual envelope

$$\log M_{j,n} \le nR_j^* + \sqrt{n}(L_j + \gamma), \tag{166}$$

where Eq. (166) follows from the limsup constraints in (45). We use this one-sided statement throughout and send γ to zero only after the blocklength limit.

A. Nonzero Common Rate

Fix a boundary rate tuple with $R_0^* > 0$. We first suppose that both supporting weights are positive, so $\boldsymbol{\lambda}^* = (\lambda_1^*, \lambda_2^*) \in (0,1]^2$. The converse of Liu et al. [12, Theorem 17] is expressed through an entropy supporting value and a functional subgradient. Lemma 7 below identifies them with the weighted boundary rate and the fluctuation whose variance is V. For $\boldsymbol{\sigma} \in \mathbb{S}_{++}^2$ and $\mathbf{r} = (r_0, r_1, r_2)$, the first-order region $\mathcal{R}(D_1, D_2 \mid Q^{\boldsymbol{\sigma}})$ determines the minimum weighted rate. The residual representation (231) and the Gaussian private rates (221)–(222) imply

$$\inf_{\mathbf{r}\in\mathcal{R}(D_1,D_2|Q^{\boldsymbol{\sigma}})}(r_0 + \lambda_1^* r_1 + \lambda_2^* r_2) = \inf_{r_1,r_2\ge 0}[\mathsf{R}_0(\boldsymbol{\sigma}; r_1, r_2) + \lambda_1^* r_1 + \lambda_2^* r_2], \tag{167}$$

$$= \inf_{\substack{\mathbf{k}\in\mathbb{S}_{++}^2 \\ \mathbf{k}\preceq\boldsymbol{\sigma}}} \Big\{\frac{1}{2}\log\frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|} + \sum_{j\in[2]}\frac{\lambda_j^*}{2}[\log(k_{jj}/D_j)]^+\Big\}, \tag{168}$$

where the first equality minimizes the common rate at fixed private rates, and the second follows by minimizing each private budget for a feasible residual covariance. At $\boldsymbol{\sigma} = \boldsymbol{\sigma}_{XY}$, the supporting normal $(1, \lambda_1^*, \lambda_2^*)$ makes this value $R_0^* + \lambda_1^* R_1^* + \lambda_2^* R_2^*$. The published converse uses the entropy relaxation obtained by removing the positive parts. In the normalization of the present paper, define this entropy support by

$$\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma}) := \inf_{\substack{\mathbf{k}\in\mathbb{S}_{++}^2 \\ \mathbf{k}\preceq\boldsymbol{\sigma}}} \left\{\frac{1}{2}\log\frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|} + \sum_{j\in[2]}\frac{\lambda_j^*}{2}\log\frac{k_{jj}}{D_j}\right\}. \tag{169}$$

The entropy support is no greater than the operational support in (168). Lemma 7 shows that they coincide at the selected covariance, including when a private first-order rate is zero. Their equality on a neighborhood is not required. Keep the true covariance $\boldsymbol{\sigma}_{XY}$ fixed and define

$$\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}(\boldsymbol{\sigma}) := -\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma}) + \frac{1}{2}\log|\boldsymbol{\sigma}| - \frac{1}{2}\operatorname{tr}(\boldsymbol{\sigma}_{XY}^{-1}\boldsymbol{\sigma}). \tag{170}$$

A matrix $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*} \in \mathbb{S}^2$ is a *supergradient* of this function at $\boldsymbol{\sigma}_{XY}$ if, for every $\boldsymbol{\sigma} \in \mathbb{S}^2_{++}$,

$$\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}(\boldsymbol{\sigma}) \le \varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}(\boldsymbol{\sigma}_{XY}) + \langle \mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}, \boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY} \rangle . \tag{171}$$

Thus its affine plane bounds the function from above. For such a matrix and $\mathbf{x} \in \mathbb{R}^2$, define the centered quadratic function

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x}) := \mathbf{x}^{\mathsf{T}} \mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*} \mathbf{x} - \mathrm{tr}(\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*} \boldsymbol{\sigma}_{XY}). \tag{172}$$

The concavity of (170) and the correspondence between matrix and quadratic functional supergradients are supplied by Liu et al. [12, Proposition 5, Eqs. (64)–(74)]. The following lemma specializes that correspondence to the selected Gray–Wyner boundary tuple. Its proof combines Yu's unrestricted Gaussian mutual-information envelope [13, Theorem 3] with a supporting-plane argument to identify the entropy value and its covariance derivative. Equation (171) fixes the sign convention, and centering in (172) removes the additive constant.

For the fixed positive-common-rate tuple, recall the supporting weights from (48)–(49) and write $\mathbf{x} = (x, y)^{\mathsf{T}} \in \mathbb{R}^2$.

Lemma 7. *For the fixed boundary tuple with positive supporting weights, the entropy support and its covariance gradient satisfy*

$$\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma}_{XY}) = R_0^* + \lambda_1^* R_1^* + \lambda_2^* R_2^*, \tag{173}$$

$$\nabla_{\boldsymbol{\sigma}} \underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma})|_{\boldsymbol{\sigma}=\boldsymbol{\sigma}_{XY}} = \mathbf{g}. \tag{174}$$

The unique supergradient in (171) *is* $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*} = -\mathbf{g}$. *Consequently, the centered fluctuation satisfies*

$$g_{\boldsymbol{\lambda}^*}(\mathbf{x}) = -[\jmath_{XY}(x, y) - R_0^*], \tag{175}$$

$$\mathsf{Var}_{P_{XY}}[g_{\boldsymbol{\lambda}^*}(\mathbf{X})] = V. \tag{176}$$

Appendix E proves this identification, including when a private first-order rate is zero. Equations (173) and (176) are precisely the substitutions needed to express the published converse in terms of the weighted coding offsets and V. The proof uses equality at the selected covariance and a touching inequality nearby. It does not require equality of the clipped operational support and the entropy support on a whole neighborhood. We can therefore state the applicable specialization of Liu et al. [12, Theorem 17] as follows.

For the same boundary tuple, consider a sequence of codes satisfying the second-order rate envelopes in (166) eventually for every $\gamma > 0$.

Lemma 8. *The joint excess-distortion probability of every code sequence described above satisfies*

$$\liminf_{n\to\infty} \varepsilon_n \ge \mathrm{Q}\left(\frac{L_0 + \lambda_1^* L_1 + \lambda_2^* L_2}{\sqrt{V}} \right). \tag{177}$$

Lemma 8 is the imported positive-weight converse after the identifications in Lemma 7. Appendix F verifies its normalization and hypotheses, including the case in which the normalized weighted displacement tends to minus infinity. For a code sequence also satisfying (46), it yields

$$L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 \ge \sqrt{V}\, \mathrm{Q}^{-1}(\varepsilon), \tag{178}$$

where Eq. (178) follows from (177) and (46) since Q is decreasing. We now apply this bound to the four regimes in Corollary 2.

(i) On $\mathcal{P}_1$ away from its one-source interfaces, both supporting weights are positive. Substituting the slopes (232)–(233) into (178) leads to

$$L_0 + \frac{1 - ru}{1 - r^2} L_1 + \frac{1 - r/u}{1 - r^2} L_2 \ge \sqrt{V}\, \mathrm{Q}^{-1}(\varepsilon), \tag{179}$$

where V is the first branch of (57). This proves the necessary condition (62). At the $(\mathcal{P}_1, \mathcal{P}_2)$ interface, $r = 0$, so both slopes equal one and $V = 1$; Lemma 8 still applies there.

(ii) On $\mathcal{P}_2$, the weights are $(\lambda_1^*, \lambda_2^*) = (1, 1)$ and $V = 1$. Thus (178) reduces to

$$L_0 + L_1 + L_2 \ge \mathrm{Q}^{-1}(\varepsilon), \tag{180}$$

which is (63), including tuples with a zero private first-order rate. A one-source face or its adjoining interface has a zero supporting weight and requires the scalar converse below.

(iii) On $\mathcal{P}_3$ and its adjoining $\mathcal{P}_1$ interface, $R_0^* + R_1^* = R_X(P_X, D_1)$, $(\lambda_1^*, \lambda_2^*) = (1, 0)$, and $V = 1/2$. Decoder 1 has at most $M_{0,n} M_{1,n}$ reproductions. Its successful set lies in their distortion-ball union regardless of the encoder's observation of Y^n. For each x^n, choose the smallest aggregate index minimizing distortion over all $M_{0,n} M_{1,n}$ decoder reproductions. A finite minimum of continuous distortion functions with this tie rule defines a Borel encoder observing only X^n. Whenever the

original decoder 1 succeeds, this aggregate encoder also succeeds. Its excess-distortion probability is therefore at most the original joint error probability ε_n. The scalar Gaussian converse [6, Theorem 40] applies to the aggregate code. Its normalized rate satisfies

$$\frac{\log(M_{0,n}M_{1,n}) - nR_X(P_X, D_1)}{\sqrt{n}} = \frac{\log M_{0,n} - nR_0^* + \log M_{1,n} - nR_1^*}{\sqrt{n}}, \tag{181}$$

$$\leq L_0 + L_1 + 2\gamma, \tag{182}$$

where Eq. (181) follows from $R_0^* + R_1^* = R_X(P_X, D_1)$, and Eq. (182) follows from (166). The scalar converse and (46) imply $L_0 + L_1 + 2\gamma \geq \mathrm{Q}^{-1}(\varepsilon)/\sqrt{2}$ for every $\gamma > 0$. Consequently,

$$L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 = L_0 + L_1, \tag{183}$$

$$\geq \frac{1}{\sqrt{2}}\mathrm{Q}^{-1}(\varepsilon), \tag{184}$$

$$= \sqrt{V}\,\mathrm{Q}^{-1}(\varepsilon), \tag{185}$$

where Eq. (183) follows from the weights $(1, 0)$, Eq. (184) follows by sending $\gamma \to 0$ in the scalar bound, and Eq. (185) follows from $V = 1/2$. This proves (64).

(iv) On $\mathcal{P}_4$ and its adjoining $\mathcal{P}_1$ interface, $R_0^* + R_2^* = R_Y(P_Y, D_2)$, $(\lambda_1^*, \lambda_2^*) = (0, 1)$, and $V = 1/2$. Apply the same aggregate construction to the $M_{0,n}M_{2,n}$ reconstructions of decoder 2. Its scalar excess-distortion probability is at most ε_n, and (166) bounds its normalized rate by $L_0 + L_2 + 2\gamma$. The scalar converse, followed by $\gamma \to 0$, therefore implies

$$L_0 + \lambda_1^* L_1 + \lambda_2^* L_2 = L_0 + L_2, \tag{186}$$

$$\geq \frac{1}{\sqrt{2}}\mathrm{Q}^{-1}(\varepsilon), \tag{187}$$

$$= \sqrt{V}\,\mathrm{Q}^{-1}(\varepsilon), \tag{188}$$

which proves (65). At the $(\mathcal{P}_1, \mathcal{P}_3)$ interface, $r = u$ makes the curved condition (62) equal to (64); at the $(\mathcal{P}_1, \mathcal{P}_4)$ interface, $r = u^{-1}$ implies (65). Thus the scalar arguments also prove the curved formula at its zero-weight interfaces.

Lemma 2 exhausts the positive-common-rate boundary by these cases. Combining their weighted-offset inequalities with $L_j \geq 0$ for every $j \in \mathcal{Z}(\mathbf{r}^*)$ proves the converse inclusion in (60) and each necessary condition in Corollary 2.

B. Zero Common Rate

At positive common rate, the first-order boundary is differentiable at the operating tuple. One supporting normal combines the three rate offsets, and the converse reduces to the tail of a single centered quadratic fluctuation. At the zero common rate tuple, the local common-rate cost instead has the directional form $\psi_\rho([Z_1 - L_1]^+, [Z_2 - L_2]^+)$. Its optimizing supporting direction depends on the two realized private-rate requirements. Applying the fixed-weight asymptotic converse separately in each direction leads to scalar tail bounds, which do not determine the probability of their union or justify optimizing the direction inside that probability. We therefore establish a finite-blocklength bound on simultaneous successful decoding in each covariance cell, optimize there, and only then take the bivariate Gaussian limit. For $\boldsymbol{\sigma} = [\sigma_{jk}] \in \mathcal{K}$, let $Q^{\boldsymbol{\sigma}} := \mathcal{N}(\mathbf{0}_2, \boldsymbol{\sigma})$ and define

$$\mathcal{C}_n(\boldsymbol{\sigma}) := \left\{\mathbf{x}^n \in \mathbb{R}^{2\times n} : [\hat{\boldsymbol{\sigma}}(\mathbf{x}^n)]_{jk} - \sigma_{jk} \in [-1/(2n), 1/(2n)),\ j, k \in [2],\ j \leq k\right\}. \tag{189}$$

When $\boldsymbol{\sigma}$ ranges over the nearest-grid representatives, these are disjoint cells. The conditional estimate below is valid for every $\boldsymbol{\sigma} \in \mathcal{K}$, not only grid points. For an arbitrary code and each $j \in [2]$, define its marginal information requirement by

$$\delta_{j,n}(\boldsymbol{\sigma}) := \left[\tfrac{1}{2}\log(\sigma_{jj}/D_j) - n^{-1}\log M_{j,n}\right]^+. \tag{190}$$

The successful set $\mathcal{A}_n$ is the set in (13) for that code. Fix $\boldsymbol{\sigma}_{XY} \in \mathbb{S}_{++}^2$, $(D_1, D_2) \in \mathbb{R}_+^2$, and a compact $\mathcal{K} \subset \{\boldsymbol{\sigma} \in \mathbb{S}_{++}^2 : \sigma_{jj} > D_j\ (j \in [2]),\ \sigma_{12} > 0\}$.

Lemma 9. *There are $C \geq 1$, $k \geq 0$, and $n_0 \in \mathbb{N}$, depending only on these fixed parameters, such that every $(n, M_{0,n}, M_{1,n}, M_{2,n})$ code, every $n \geq n_0$, and every $\boldsymbol{\sigma} \in \mathcal{K}$ satisfy*

$$P_{XY}^n(\mathcal{A}_n \mid \mathcal{C}_n(\boldsymbol{\sigma})) \leq \min\left\{1, Cn^k \exp\left[\log M_{0,n} - n\psi_{\rho(\boldsymbol{\sigma})}\big(\delta_{1,n}(\boldsymbol{\sigma}), \delta_{2,n}(\boldsymbol{\sigma})\big)\right]\right\}. \tag{191}$$

The constants do not depend on the code, its rates, its reconstruction centers, or the selected covariance.

Lemma 9 supplies this additional converse step. Its bound holds for every code and reconstruction center, controls the joint success event rather than either marginal success event, and retains both marginal requirements through $\psi_{\rho(\boldsymbol{\sigma})}$. Appendix N proves it from the marginal image estimate, the Gaussian hypercontractive inequality [17], [18], and a change of Gaussian measure. The polynomial cell-mass and likelihood-ratio bounds are uniform before the hypercontractive exponents are optimized.

Thus the direction optimization is performed in a deterministic finite-blocklength bound before the bivariate limit. The lemma does not assert an exact deterministic-shell rate characterization.

Set $a_{j,n}(\boldsymbol{\sigma}) := \sqrt{n}\,\delta_{j,n}(\boldsymbol{\sigma})$. Fix $(\gamma,\zeta) \in \mathbb{R}_+^2$ until after the blocklength limit and suppress these fixed margins in the bad-set notation. Define

$$\mathcal{K}_{\mathrm{bad},n} := \{\boldsymbol{\sigma} \in \mathcal{K} : \psi_{\rho(\boldsymbol{\sigma})}(a_{1,n}(\boldsymbol{\sigma}), a_{2,n}(\boldsymbol{\sigma})) > L_0 + \gamma + \zeta\}. \tag{192}$$

For every $\boldsymbol{\sigma} \in \mathcal{K}_{\mathrm{bad},n}$ and all sufficiently large n,

$$\begin{aligned}
\log M_{0,n} - n\psi_{\rho(\boldsymbol{\sigma})}(\delta_{1,n}(\boldsymbol{\sigma}), \delta_{2,n}(\boldsymbol{\sigma})) &= \log M_{0,n} - \sqrt{n}\,\psi_{\rho(\boldsymbol{\sigma})}(a_{1,n}(\boldsymbol{\sigma}), a_{2,n}(\boldsymbol{\sigma})), && (193)\\
&\le \sqrt{n}[L_0 + \gamma - \psi_{\rho(\boldsymbol{\sigma})}(a_{1,n}(\boldsymbol{\sigma}), a_{2,n}(\boldsymbol{\sigma}))], && (194)\\
&< -\zeta\sqrt{n}, && (195)
\end{aligned}$$

where Eq. (193) follows from positive homogeneity in (277) and the definition of $a_{j,n}$, Eq. (194) follows from (166) with $R_0^* = 0$, and Eq. (195) follows from (192). Define the event

$$\mathcal{E}_{\mathrm{bad},n} := \big\{\mathbf{x}^n \in \mathbb{R}^{2\times n} :\ \hat{\boldsymbol{\sigma}}(\mathbf{x}^n) \in \mathcal{K}_0,\ q_n(\hat{\boldsymbol{\sigma}}(\mathbf{x}^n)) \in \mathcal{K}_{\mathrm{bad},n}\big\}. \tag{196}$$

This event is contained in the union of the full covariance cells with bad representatives. Applying Lemma 9 to these full cells leads to

$$\begin{aligned}
P_{XY}^n(\mathcal{A}_n \cap \mathcal{E}_{\mathrm{bad},n}) &\le \sum_{\substack{q\in[N_n]\\ \boldsymbol{\sigma}_q\in\mathcal{K}_{\mathrm{bad},n}}} P_{XY}^n(\mathcal{A}_n \cap \mathcal{C}_n(\boldsymbol{\sigma}_q)), && (197)\\
&\le Cn^k e^{-\zeta\sqrt{n}} \sum_{\substack{q\in[N_n]\\ \boldsymbol{\sigma}_q\in\mathcal{K}_{\mathrm{bad},n}}} P_{XY}^n(\mathcal{C}_n(\boldsymbol{\sigma}_q)), && (198)\\
&\le Cn^k e^{-\zeta\sqrt{n}}, && (199)
\end{aligned}$$

where Eq. (197) follows from the event inclusion after (196), Eq. (198) follows from (191) and (195), and Eq. (199) follows since the full grid cells are disjoint and their probabilities sum to at most one. Consequently,

$$\begin{aligned}
\varepsilon_n &= P_{XY}^n(\mathcal{A}_n^{\mathrm{c}}), && (200)\\
&\ge P_{XY}^n(\mathcal{A}_n^{\mathrm{c}} \cap \mathcal{E}_{\mathrm{bad},n}), && (201)\\
&= P_{XY}^n(\mathcal{E}_{\mathrm{bad},n}) - P_{XY}^n(\mathcal{A}_n \cap \mathcal{E}_{\mathrm{bad},n}), && (202)\\
&\ge P_{XY}^n(\mathcal{E}_{\mathrm{bad},n}) - Cn^k e^{-\zeta\sqrt{n}}, && (203)
\end{aligned}$$

where Eq. (200) follows from (13) and (14), Eq. (201) follows from restricting the error event, Eq. (202) follows from partitioning $\mathcal{E}_{\mathrm{bad},n}$ into successful and unsuccessful blocks, and Eq. (203) follows from (199). The enlargement to full cells accommodates $\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) \in \mathcal{K}_0$ without applying the conditional bound to a truncated cell.

Recall the safe covariance representative $\tilde{\boldsymbol{\Sigma}}_n = \bar{q}_n(\hat{\boldsymbol{\Sigma}}_n)$ from (102) and the logarithmic power fluctuations $Z_{j,n}$ from (139). The safe representative makes the converse marginal requirements globally defined. The quantization bound (73) and the positive diagonal entries on $\mathcal{K}$ imply a constant $C_{\mathrm{q}} > 0$ such that, for every source-block realization $\mathbf{x}^n$ with $\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) \in \mathcal{K}_0$ and all sufficiently large n,

$$\max_{j\in[2]} \frac{\sqrt{n}}{2} \left|\log \frac{[\bar{q}_n(\hat{\boldsymbol{\sigma}}(\mathbf{x}^n))]_{jj}}{[\hat{\boldsymbol{\sigma}}(\mathbf{x}^n)]_{jj}}\right| \le C_{\mathrm{q}} n^{-1/2}. \tag{204}$$

Thus this pointwise estimate applies to the random source block whenever $\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0$. The probability of the exceptional event is at most $P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\} \to 0$ by (119). Fix this constant throughout the remaining argument. For each $j \in [2]$ and fixed $\gamma > 0$, define the shifted positive marginal requirements by

$$\begin{aligned}
\mathsf{B}_{j,n}^{(\gamma)} &:= [Z_{j,n} - L_j - \gamma - C_{\mathrm{q}} n^{-1/2}]^+, && (205)\\
\mathsf{B}_{j}^{(\gamma)} &:= [Z_j - L_j - \gamma]^+. && (206)
\end{aligned}$$

On $\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0\}$, for every $j \in [2]$,

$$\begin{aligned}
a_{j,n}(\tilde{\boldsymbol{\Sigma}}_n) &= \left[\frac{\sqrt{n}}{2}\log\frac{[\tilde{\boldsymbol{\Sigma}}_n]_{jj}}{D_j} - \frac{\log M_{j,n}}{\sqrt{n}}\right]^+, && (207)\\
&\ge \left[\frac{\sqrt{n}}{2}\log\frac{[\tilde{\boldsymbol{\Sigma}}_n]_{jj}}{[\boldsymbol{\sigma}_{XY}]_{jj}} - L_j - \gamma\right]^+, && (208)\\
&\ge \mathsf{B}_{j,n}^{(\gamma)}, && (209)
\end{aligned}$$

where Eq. (207) follows from (190), Eq. (208) follows from (166) and $R_j^* = \frac{1}{2}\log([\boldsymbol{\sigma}_{XY}]_{jj}/D_j)$ at the corner, and Eq. (209) follows from (139) and the uniform quantization bound (73) inside the logarithm. The fixed constant C_{q} covers both coordinates, and the last term is the shorthand in (205). On $\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0\}$, coordinatewise monotonicity of $\psi_{\rho(\tilde{\boldsymbol{\Sigma}}_n)}$ shows that a value of $\psi_{\rho(\tilde{\boldsymbol{\Sigma}}_n)}(\mathsf{B}_{1,n}^{(\gamma)}, \mathsf{B}_{2,n}^{(\gamma)})$ exceeding $L_0+\gamma+\zeta$ places the representative in $\mathcal{K}_{\mathrm{bad},n}$. Consequently, for each fixed $(\gamma,\zeta) \in \mathbb{R}_+^2$ and all sufficiently large n,

$$\varepsilon_n \geq P_{XY}^n(\mathcal{E}_{\mathrm{bad},n}) - Cn^k e^{-\zeta\sqrt{n}}, \tag{210}$$

$$\geq P_{XY}^n\big\{\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0,\ \psi_{\rho(\tilde{\boldsymbol{\Sigma}}_n)}(\mathsf{B}_{1,n}^{(\gamma)}, \mathsf{B}_{2,n}^{(\gamma)}) > L_0+\gamma+\zeta\big\} - Cn^k e^{-\zeta\sqrt{n}}, \tag{211}$$

$$\geq P_{XY}^n\big\{\psi_{\rho(\tilde{\boldsymbol{\Sigma}}_n)}(\mathsf{B}_{1,n}^{(\gamma)}, \mathsf{B}_{2,n}^{(\gamma)}) > L_0+\gamma+\zeta\big\} - P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\} - Cn^k e^{-\zeta\sqrt{n}}, \tag{212}$$

where Eq. (210) follows from (203), Eq. (211) follows from (209), coordinatewise monotonicity in (277), and the definition (196), and Eq. (212) follows from subtracting the probability of the complementary covariance event. The covariance probability tends to zero by the law of large numbers, and the exponential remainder tends to zero since $\zeta > 0$ is fixed.

We now evaluate the probability in (212) at fixed (γ,ζ). Recall the localization event $\mathcal{V}_n(B)$ from (147). At fixed B, its power and correlation bounds ensure $\hat{\boldsymbol{\Sigma}}_n \in \mathcal{K}_0$ for all sufficiently large n. Quantization then gives $\rho(\tilde{\boldsymbol{\Sigma}}_n) - \rho = O_B(n^{-1/2})$ there. Define the exact statistical perturbation by

$$E_{\psi,n}^{(\gamma)} := \psi_{\rho(\tilde{\boldsymbol{\Sigma}}_n)}(\mathsf{B}_{1,n}^{(\gamma)}, \mathsf{B}_{2,n}^{(\gamma)}) - \psi_\rho([Z_{1,n}-L_1-\gamma]^+, [Z_{2,n}-L_2-\gamma]^+). \tag{213}$$

On the explicitly specified event $\mathcal{V}_n(B)$, the arguments lie in a fixed compact set and the shifts $C_{\mathrm{q}}n^{-1/2}$ vanish. Joint uniform continuity of (58) therefore bounds $|E_{\psi,n}^{(\gamma)}|$ by a deterministic number tending to zero. For every $\eta > 0$, this gives

$$\limsup_{n\to\infty} P_{XY}^n\{|E_{\psi,n}^{(\gamma)}| > \eta\} \leq \limsup_{n\to\infty} P_{XY}^n(\mathcal{V}_n(B)^{\mathrm{c}}) \longrightarrow 0 \quad \text{as } B \to \infty. \tag{214}$$

By (144), continuity of ψ_ρ, and this vanishing probabilistic remainder, the statistic in (212) converges in distribution to $\psi_\rho(\mathsf{B}_1^{(\gamma)}, \mathsf{B}_2^{(\gamma)})$. The open-set weak-convergence bound yields

$$\liminf_{n\to\infty} \varepsilon_n \geq P_{\mathbf{Z}}\big\{\psi_\rho(\mathsf{B}_1^{(\gamma)}, \mathsf{B}_2^{(\gamma)}) > L_0+\gamma+\zeta\big\}, \tag{215}$$

where Eq. (215) follows from (212) and the open-set bound for weak convergence. This bound requires no continuity assumption at the threshold. Finally, send $(\gamma,\zeta) \in \mathbb{R}_+^2$ to zero after the blocklength limit to obtain

$$\liminf_{n\to\infty} \varepsilon_n \geq \lim_{\gamma\to 0}\lim_{\zeta\to 0} P_{\mathbf{Z}}\{\psi_\rho(\mathsf{B}_1^{(\gamma)}, \mathsf{B}_2^{(\gamma)}) > L_0+\gamma+\zeta\}, \tag{216}$$

$$= P_{\mathbf{Z}}\{\psi_\rho([Z_1-L_1]^+, [Z_2-L_2]^+) > L_0\}, \tag{217}$$

where Eq. (216) follows from (215) with $(\gamma,\zeta) \in \mathbb{R}_+^2$ approaching zero after $n \to \infty$, and Eq. (217) follows since monotonicity and continuity of ψ make the events increase to the strict limiting event. Thus the possible atom at zero causes no difficulty. Combining (217) with (46) leads to

$$P_{\mathbf{Z}}\big\{\psi_\rho([Z_1-L_1]^+, [Z_2-L_2]^+) \leq L_0\big\} \geq 1-\varepsilon, \tag{218}$$

where Eq. (218) follows from taking the complement of the strict event in (217). Together with $L_0 \geq 0$, this is the converse inclusion in (61) and matches the two direct bounds. Restricting to $M_{0,n} = 1$ and $L_0 = 0$ also proves the converse inclusion in (348).

The converse proof of Theorem 1 is now completed.

VI. Conclusion

In this paper, we derived exact second-order asymptotics for the lossy Gaussian Gray–Wyner problem under the quadratic distortion measure and a joint excess-distortion probability criterion. We classified the boundary rate tuples into two cases: with a nonzero common rate and with a zero common rate. For the nonzero common rate case, our main contribution was an achievability bound that matches the existing converse bounds [6], [12]. We obtained this bound using a covariance-shell covering lemma analogous to the type covering lemma for discrete sources. For the zero common rate case, we derived new achievability and converse bounds. For achievability, we used the same covering lemma when the second-order common rate is positive and two P2P coverings when it is zero. For the matching converse, we used Gaussian hypercontractivity and a bivariate central limit theorem.

We next discuss future directions. Firstly, in this paper, we derived second-order asymptotics. However, to accurately characterize the performance of optimal codes at small blocklengths, further refined asymptotics including the third-order asymptotics [6, Theorem 27] or the refined moderate deviations [29] are necessary. Secondly, one can complement the previous second-order asymptotics in [11], [10] by studying the boundary rate tuples with a zero common rate and deriving corresponding second-order asymptotics. Thirdly, one can derive second-order asymptotics for other multiterminal lossy source coding problems concerning bivariate Gaussian sources, e.g., the Wyner–Ziv problem [30] or the Heegard–Berger problem [31].

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APPENDIX A
PROOF OF LEMMA 1

We prove the two region inclusions before identifying their common-rate functions. We first recall Yu's Gaussian evaluation of the expected-distortion region in (223). We then prove $\mathcal{R} \subseteq \mathcal{R}_{\mathrm{E}}$ by adding a zero reconstruction on unsuccessful blocks and prove $\mathcal{R}_{\mathrm{E}} \subseteq \mathcal{R}$ by assembling the determinant-based shell codes of Lemma 4 with a fixed rate margin. Only after these inclusions do we identify the operational common rate with the covariance minimization in (231).

For the appendix derivations, (I_Q, h_Q) denote mutual information and differential entropy under a specified distribution Q, $\mathbb{1}_{\mathcal{A}}$ is the indicator of an event $\mathcal{A}$, and $(\mathrm{diag}, \mathrm{rank})$ denote the diagonal-matrix and rank operators. The discrete benchmark uses base-two logarithms. Dividing rates and information quantities by $\log 2$, and variances by $(\log 2)^2$, converts the quantities here to bits. The supporting slopes are unchanged.

We recall the unrestricted auxiliary characterization to identify the residual covariance used by the code. For a source distribution $\tilde{P}_{XY}$ on $\mathbb{R}^2$ and private budgets $(r_1, r_2) \in \mathbb{R}^2_{+,0}$, let $\mathcal{Q}(\tilde{P}_{XY}; r_1, r_2)$ consist of the joint distributions of $(X, Y, W, \hat{X}, \hat{Y})$ of the form

$$Q = \tilde{P}_{XY} Q_{W|XY} Q_{\hat{X}|XW} Q_{\hat{Y}|YW},$$

where W ranges over standard Borel alphabets. Require $\mathsf{E}_Q[d(X, \hat{X})] \le D_1$, $\mathsf{E}_Q[d(Y, \hat{Y})] \le D_2$, $I_Q(X; \hat{X} \mid W) \le r_1$, and $I_Q(Y; \hat{Y} \mid W) \le r_2$. The expected-distortion characterization [9], [13], in the notation of [11, Theorem 1 and Eqs. (11)–(13)], implies

$$\inf\left\{r_0 \in \mathbb{R}_{+,0} : (r_0, r_1, r_2) \in \mathcal{R}_{\mathrm{E}}(D_1, D_2 \mid \tilde{P}_{XY})\right\} = \inf_{Q \in \mathcal{Q}(\tilde{P}_{XY}; r_1, r_2)} I_Q(X, Y; W). \tag{219}$$

For a fixed joint distribution Q_{XW}, $R_{X|W}(Q_{XW}, D_1)$ is the infimum of $I_Q(X; \hat{X} \mid W)$ over reconstruction kernels meeting the first distortion constraint. Define $R_{Y|W}(Q_{YW}, D_2)$ analogously. No Gaussian restriction is imposed in (219).

For $(\boldsymbol{\sigma}, \mathbf{k}) \in (\mathbb{S}^2_{++})^2$ with $\mathbf{k} \preceq \boldsymbol{\sigma}$, consider the Gaussian design $\mathbf{X} = \mathbf{W} + \mathbf{N}$, where $(\mathbf{W}, \mathbf{N}) \in (\mathbb{R}^2)^2$ are independent centered Gaussian vectors with covariances $(\boldsymbol{\sigma} - \mathbf{k}, \mathbf{k})$. Let Q denote the joint distribution of this design. The source marginal is $Q^{\boldsymbol{\sigma}}$ from Section II-B. The conditional covariance given the common auxiliary $\mathbf{W}$ is $\mathbf{k}$, and the design rates are

$$I_Q(X, Y; \mathbf{W}) = \frac{1}{2} \log \frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|}, \tag{220}$$

$$R_{X|\mathbf{W}}(Q_{X\mathbf{W}}, D_1) = \frac{1}{2}[\log(k_{11}/D_1)]^+, \tag{221}$$

$$R_{Y|\mathbf{W}}(Q_{Y\mathbf{W}}, D_2) = \frac{1}{2}[\log(k_{22}/D_2)]^+. \tag{222}$$

The common covariance may be singular, while the residual covariance remains positive definite. The private constraints specify the feasible residual matrices

$$\mathcal{F}(\boldsymbol{\sigma}; r_1, r_2) := \{\mathbf{k} \in \mathbb{S}^2_{++} : \mathbf{k} \preceq \boldsymbol{\sigma},\ k_{jj} \le D_j e^{2r_j},\ j \in [2]\}.$$

Yu's Gaussian optimality evaluates the expected-distortion graph as

$$\inf\left\{r_0 \in \mathbb{R}_{+,0} : (r_0, r_1, r_2) \in \mathcal{R}_{\mathrm{E}}(D_1, D_2 \mid Q^{\boldsymbol{\sigma}})\right\} = \min_{\mathbf{k} \in \mathcal{F}(\boldsymbol{\sigma}; r_1, r_2)} \frac{1}{2} \log \frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|}. \tag{223}$$

The determinant program and its eigenvalue bounds in Lemma 4 are independent of either operational distortion criterion. The covering assertion of Lemma 4 likewise applies to every admissible pair $(\boldsymbol{\sigma}, \mathbf{k})$, with alphabet sizes given directly by its determinant ratio and residual diagonal entries.

The unrestricted expected-distortion expression is (219). Its Gaussian evaluation is the existing result of Yu [13, Theorem 3 and Corollary 2], with scaling to the marginal variances in $\boldsymbol{\sigma}$. To record the connection without repeating that proof, fix $Q \in \mathcal{Q}(Q^{\boldsymbol{\sigma}}; r_1, r_2)$. Then

$$\begin{aligned} I_Q(X; W) &= I_Q(X; W, \hat{X}) - I_Q(X; \hat{X} \mid W), \\ &\ge I_Q(X; \hat{X}) - r_1, \\ &\ge R_X(Q_X^{\boldsymbol{\sigma}}, D_1) - r_1, \end{aligned} \tag{224}$$

where Eq. (224) follows from the chain rule, data processing, the private-rate constraint, and the scalar rate-distortion bound. The same inequality holds for Y. Nonnegativity implies the marginal requirements $(\alpha(\boldsymbol{\sigma}; r_1), \beta(\boldsymbol{\sigma}; r_2))$, so Yu's unrestricted envelope lower-bounds the common rate by $\Upsilon_{\rho(\boldsymbol{\sigma})}(\alpha, \beta)$. His attaining Gaussian auxiliary has a constant conditional covariance $\mathbf{k}^\circ \succ 0$ whose marginal information constraints imply $k^\circ_{jj} \le D_j e^{2r_j}$. Hence it is feasible for the covariance minimization in (223) and attains the same common value.

For clarity about a zero private rate, the standard scalar Gaussian reconstruction can be written explicitly. For each $j \in [2]$, let $\tilde{D}_j := \min\{D_j, k_{jj}\}$ and choose independent Gaussian V_j, independent of $(\mathbf{W}, \mathbf{N})$, with variance $\tilde{D}_j(1 - \tilde{D}_j/k_{jj})$. With $\mathbf{X} = \mathbf{W} + \mathbf{N}$ and $\hat{\mathbf{X}} := (\hat{X}, \hat{Y})^{\mathsf{T}} \in \mathbb{R}^2$, set

$$[\hat{\mathbf{X}}]_j := [\mathbf{W}]_j + \left(1 - \frac{\tilde{D}_j}{k_{jj}}\right)[\mathbf{N}]_j + V_j,$$

$$\mathsf{E}_Q[([\mathbf{X}]_j - [\hat{\mathbf{X}}]_j)^2] = \tilde{D}_j \le D_j, \tag{225}$$

$$I_Q([\mathbf{X}]_j; [\hat{\mathbf{X}}]_j \mid \mathbf{W}) = \tfrac{1}{2}[\log(k_{jj}/D_j)]^+, \tag{226}$$

where Eqs. (225) and (226) follow from independent Gaussian errors. If $D_j \ge k_{jj}$, the reproduction is simply $[\mathbf{W}]_j$. Thus Gaussian substitution establishes the reverse bound and

$$\min_{\mathbf{k} \in \mathcal{F}(\boldsymbol{\sigma}; r_1, r_2)} \frac{1}{2} \log \frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|} = \Upsilon_{\rho(\boldsymbol{\sigma})}(\alpha(\boldsymbol{\sigma}; r_1), \beta(\boldsymbol{\sigma}; r_2)). \tag{227}$$

Together with (223), this evaluates the expected-distortion graph. Its covariance value is continuous by Yu's formula. The shell covering used below has the determinant-based statement of Lemma 4.

We now prove the two inclusions in Lemma 1. For $\mathcal{R} \subseteq \mathcal{R}_{\mathrm{E}}$, fix a code sequence satisfying Definition 2. Its encoder can determine the joint successful set $\mathcal{A}_n$ in (13) from the observed block and the known code maps. At blocklength n, add one common symbol. For each $j \in [2]$ and $m_j \in [M_{j,n}]$, define the modified maps by

$$\tilde{f}(\mathbf{x}^n) := \begin{cases} f(\mathbf{x}^n), & \mathbf{x}^n \in \mathcal{A}_n, \\ (M_{0,n}+1, 1, 1), & \mathbf{x}^n \in \mathcal{E}_n \end{cases},$$

$$\tilde{\phi}_j(m_0, m_j) := \begin{cases} \phi_j(m_0, m_j), & m_0 \in [M_{0,n}], \\ \mathbf{0}_n, & m_0 = M_{0,n}+1 \end{cases}.$$

The private alphabets are unchanged, and the common length increases by $\log(M_{0,n}+1) - \log M_{0,n} \le \log 2$. All new maps are Borel. Write $(\tilde{X}^n, \tilde{Y}^n)$ for the modified reconstructions. Their first expected distortion obeys

$$\begin{aligned} \mathsf{E}_{P^n_{XY}}[d(X^n, \tilde{X}^n)] &\le D_1 + \mathsf{E}_{P^n_{XY}}[n^{-1} \|X^n\|^2 \mathbb{1}_{\mathcal{E}_n}], \\ &\le D_1 + \sqrt{\mathsf{E}_{P^n_{XY}}[(n^{-1}\|X^n\|^2)^2]\, \varepsilon_n}, \\ &= D_1 + \sigma_X^2 \sqrt{(1 + 2/n)\varepsilon_n}, \\ &= D_1 + o(1), \end{aligned} \tag{228}$$

where Eq. (228) follows from the zero reconstruction on $\mathcal{E}_n$, Cauchy–Schwarz, and the Gaussian fourth-moment identity. The final term vanishes because $\varepsilon_n \to 0$. The same argument establishes a limiting upper bound D_2 for Y. Thus every excess-distortion-achievable tuple belongs to $\mathcal{R}_{\mathrm{E}}$. Taking closures proves $\mathcal{R} \subseteq \mathcal{R}_{\mathrm{E}}$.

For the reverse inclusion, fix $(r_1, r_2) \in \mathbb{R}^2_{+,0}$ and $\delta > 0$. Let $\mathbf{k}^*$ maximize the determinant under the source covariance and private caps, and set

$$r_0 := \frac{1}{2} \log \frac{|\boldsymbol{\sigma}_{XY}|}{|\mathbf{k}^*|} + \delta.$$

For each quantized center $\boldsymbol{\sigma}_q$, choose $\mathbf{k}_q$ by (78) and retain its cell precisely when $\frac{1}{2}\log(|\boldsymbol{\sigma}_q|/|\mathbf{k}_q|) \le r_0$. Apply the local shell code of Lemma 4 to every retained pair $(\boldsymbol{\sigma}_q, \mathbf{k}_q)$. The optimizer bounds provide the same compact family $\mathcal{F}_\kappa$ for all centers. The diagonal caps give $\frac{1}{2}[\log([\mathbf{k}_q]_{jj}/D_j)]^+ \le r_j$. Use the cell index and the local common index as the common message, pad each private alphabet to the largest local size, and reserve one common symbol for the zero-reconstruction fallback. Since $N_n \le Cn^3$, the alphabet-size bounds of Lemma 4 yield

$$\log M_j \le nr_j + C(s_n + \log n), \qquad j \in [0:2]. \tag{229}$$

The finite codebooks and the cell partition give Borel code maps, as in Section IV-B. Every retained cell is contained in its shell and therefore has both distortions at most their prescribed levels. The remainder in (229) is $o(n)$ because $s_n = \sqrt{n}/\log n$. The weak law and (73) imply $\tilde{\boldsymbol{\Sigma}}_n \to \boldsymbol{\sigma}_{XY}$ in probability. Continuity of the covariance minimum in (227) now gives

$$\begin{aligned}\limsup_{n\to\infty} \varepsilon_n &\le \limsup_{n\to\infty} P_{XY}^n\{\hat{\boldsymbol{\Sigma}}_n \notin \mathcal{K}_0\} \\ &\quad + \limsup_{n\to\infty} P_{XY}^n \left\{ \min_{\mathbf{k}\in\mathcal{F}(\tilde{\boldsymbol{\Sigma}}_n; r_1, r_2)} \frac{1}{2}\log\frac{|\tilde{\boldsymbol{\Sigma}}_n|}{|\mathbf{k}|} > r_0 \right\}, \\ &= 0. \end{aligned} \tag{230}$$

The first term vanishes by the weak law. The second vanishes because the covariance minimum converges in probability to $\frac{1}{2}\log(|\boldsymbol{\sigma}_{XY}|/|\mathbf{k}^*|)$, which is strictly below r_0. Thus the tuple $(\frac{1}{2}\log(|\boldsymbol{\sigma}_{XY}|/|\mathbf{k}^*|) + \delta, r_1, r_2)$ belongs to $\mathcal{R}$. Sending $\delta \to 0$ after constructing these fixed-margin code sequences and using closure places the covariance graph in $\mathcal{R}$. Equation (223) and upward closure imply $\mathcal{R}_{\mathrm{E}} \subseteq \mathcal{R}$. Together with the reverse inclusion proved above, this establishes Lemma 1. The same argument applies to every nearby nonsingular Gaussian source under consideration. Consequently, the operational common-rate function has the covariance representation

$$\mathsf{R}_0(\boldsymbol{\sigma}; r_1, r_2) = \min_{\mathbf{k}\in\mathcal{F}(\boldsymbol{\sigma}; r_1, r_2)} \frac{1}{2}\log\frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|}. \tag{231}$$

Appendix B
Proof of Proposition 2

For the fixed positive-common-rate tuple, define

$$\eta := \partial_\rho \Upsilon_\rho(\alpha^*, \beta^*).$$

In $\mathcal{P}_1$, recall $r = r_{\alpha^*,\beta^*}$ and u from (54). The derivatives used in the dispersion calculation are

$$\lambda_1^* = \frac{1 - ru}{1 - r^2}, \tag{232}$$

$$\lambda_2^* = \frac{1 - r/u}{1 - r^2}, \tag{233}$$

$$\eta = \frac{re^{\alpha^*+\beta^*}}{1 - r^2} - \frac{\rho}{1 - \rho^2}.$$

On the affine branches and their adjoining interfaces,

$$(\lambda_1^*, \lambda_2^*, \eta) = \begin{cases} (1, 1, -\rho/(1-\rho^2)), & \text{if } (\alpha^*, \beta^*) \in \mathcal{P}_2, \\ (1, 0, 0), & \text{if } (\alpha^*, \beta^*) \in \mathcal{P}_3, \\ (0, 1, 0), & \text{if } (\alpha^*, \beta^*) \in \mathcal{P}_4 \end{cases}. \tag{234}$$

For $(\alpha^*, \beta^*) \in \mathcal{P}_1$, Eqs. (232) and (233) imply

$$\lambda_1^* + \lambda_2^* - 1 = \frac{1 + r^2 - r(u + u^{-1})}{1 - r^2} = \frac{(u - r)(1 - ru)}{u(1 - r^2)}, \tag{235}$$

where the last identity follows by factoring the numerator. Substituting (235) and (234) into (56) proves (57). We verify the derivatives and their agreement at the interfaces as part of the regularity proof.

We now verify the regularity in Proposition 2. For positive (α, β), recall (ν_α, ν_β) from (27) and write $r := r_{\alpha,\beta}$ and $u := (\nu_\beta/\nu_\alpha)e^{\beta-\alpha}$. Let F_k be the kth expression in (33). Differentiation of the curved branch leads to

$$\partial_\alpha r = r - u, \tag{236}$$

$$\partial_\beta r = r - u^{-1},$$

$$\partial_\rho r = e^{\alpha+\beta},$$

$$\partial_\alpha F_1 = 1 + \frac{r\partial_\alpha r}{1 - r^2} = \frac{1 - ru}{1 - r^2}, \tag{237}$$

$$\partial_\beta F_1 = \frac{1 - r/u}{1 - r^2},$$

$$\partial_\rho F_1 = \frac{re^{\alpha+\beta}}{1 - r^2} - \frac{\rho}{1 - \rho^2}, \tag{238}$$

where Eqs. (236)–(238) follow from $\nu'_\alpha = e^{-2\alpha}/\nu_\alpha$, $\nu'_\beta = e^{-2\beta}/\nu_\beta$, and the chain rule. In the strict curved interior, $r > 0$, $r - u = e^{\alpha+\beta}(\rho - \nu_\beta/\nu_\alpha) < 0$, and $r - u^{-1} = e^{\alpha+\beta}(\rho - \nu_\alpha/\nu_\beta) < 0$. Hence $0 < r < \min\{u, u^{-1}\}$, which implies $0 < \lambda_j^* < 1$.

With gradients in (α, β, ρ), substitution at the three interfaces yields

$$\nabla F_1|_{r=0} = (1, 1, -\rho/(1-\rho^2)) = \nabla F_2,$$
$$\nabla F_1|_{\rho=\nu_\beta/\nu_\alpha} = (1, 0, 0) = \nabla F_3, \tag{239}$$
$$\nabla F_1|_{\rho=\nu_\alpha/\nu_\beta} = (0, 1, 0) = \nabla F_4, \tag{240}$$

where Eq. (239) follows from substituting $r = u$, and Eq. (240) follows from interchanging the coordinates. The branch values agree as well. The finitely many adjoining analytic pieces therefore have a common first-order expansion and continuous first derivatives.

Near $\alpha = 0$, $\beta > 0$, the active expression is $F_4 = \beta$, independent of α. The other positive axis is symmetric. This handles the positive-part composition. If $R_j^* = 0$, its marginal requirement is positive because $D_j < [\boldsymbol{\sigma}_{XY}]_{jj}$, so the same branch expressions extend to small negative r_j for differentiation. Operational rates remain nonnegative. Only the marginal requirement origin is excluded. This proves the stated continuously differentiable extension, including all finite interfaces and zero private rates.

For the zero-slope interpretation in Section III-A, fix a positive interface pair (α, β) with $\nu_\beta = \rho\nu_\alpha$. Keep α fixed and let $h > 0$ increase the second marginal requirement into $\mathcal{P}_1$. At the interface $r = u$, while $\partial_\beta r = r - u^{-1}$ and $\partial_\beta \log u = \nu_\beta^{-2}$. Differentiating the curved branch from the right and taking $h \to 0$ leads to

$$\begin{aligned}\partial^2_{\beta,+}\Upsilon_\rho(\alpha,\beta) &= u^{-2} + \frac{1}{\nu_\beta^2(1-u^2)},\\ &= \frac{(1-\rho^2\nu_\alpha^2)^2}{\rho^2\nu_\alpha^2(1-\nu_\alpha^2)(1-\rho^2)} > 0,\end{aligned} \tag{241}$$

$$\Upsilon_\rho(\alpha, \beta+h) = \alpha + \frac{(1-\rho^2\nu_\alpha^2)^2}{2\rho^2\nu_\alpha^2(1-\nu_\alpha^2)(1-\rho^2)}h^2 + o(h^2), \tag{242}$$

where Eq. (241) follows from substituting $\nu_\beta^2 = \rho^2\nu_\alpha^2$ and $u^2 = \rho^2(1-\nu_\alpha^2)/(1-\rho^2\nu_\alpha^2)$. Eq. (242) follows from the zero first derivative. For sufficiently small negative h, the value remains α on $\mathcal{P}_3$. This expansion fixes a positive interface pair and is not asserted uniformly as that pair approaches the corner.

Appendix C
Proof of Lemma 2

We first identify the lower-boundary graph and then check which private coordinates are necessary. If a private rate exceeds its marginal rate-distortion value, reducing it to that value leaves the marginal requirements in (25)–(26) unchanged. If a common rate strictly exceeds (36), it can also be reduced. Thus every boundary rate tuple satisfies (39) and lies on the graph (41).

The marginal requirement bounds in (224) imply $\Upsilon_\rho(\alpha, \beta) \geq \max\{\alpha, \beta\}$. Consequently, zero common rate requires $(\alpha, \beta) = (0, 0)$ and yields $\mathbf{r}_{\mathrm{c}}^*$. Decreasing either positive private coordinate at this tuple creates a positive marginal requirement, so it is a boundary rate tuple. A tuple on a coordinate axis away from the origin is not a boundary rate tuple. For example, fix $\alpha > 0$ and $\beta = 0$. Its second private rate is $R_Y(P_Y, D_2) > 0$. A sufficiently small reduction $t > 0$ of that rate ensures $\nu_t/\nu_\alpha < \rho$, hence $(\alpha, t) \in \mathcal{P}_3$ and $\Upsilon_\rho(\alpha, t) = \alpha = \Upsilon_\rho(\alpha, 0)$. The other axis is symmetric. This proves the stated zero-common-rate case and the positivity of both marginal requirements in the remaining case.

For positive marginal requirements, the rate-coordinate expressions follow directly from Proposition 1. On $\mathcal{P}_1 \cup \mathcal{P}_2$,

$$\begin{aligned}R_0 + R_1 + R_2 &= R_X(P_X, D_1) + R_Y(P_Y, D_2) + \Upsilon_\rho(\alpha,\beta) - \alpha - \beta,\\ &= \begin{cases} R_X(P_X, D_1) + R_Y(P_Y, D_2) - \dfrac{1}{2}\log\dfrac{1-r^2_{\alpha,\beta}}{1-\rho^2}, & \text{if } (\alpha,\beta) \in \mathcal{P}_1,\\ R_X(P_X, D_1) + R_Y(P_Y, D_2) + \dfrac{1}{2}\log(1-\rho^2), & \text{if } (\alpha,\beta) \in \mathcal{P}_2 \end{cases},\end{aligned} \tag{243}$$

where the first equality uses (41) and the second uses the first two branches of (33). The first expression depends on the two marginal requirements through $r_{\alpha,\beta}$; the second is the constant sum-rate face. On $\mathcal{P}_3$, the branch $\Upsilon_\rho(\alpha, \beta) = \alpha$ implies

$$R_0 + R_1 = R_X(P_X, D_1). \tag{244}$$

On $\mathcal{P}_4$, the branch $\Upsilon_\rho(\alpha, \beta) = \beta$ similarly implies

$$R_0 + R_2 = R_Y(P_Y, D_2). \tag{245}$$

The common-rate requirement is nondecreasing in each marginal requirement, since increasing a marginal requirement tightens a marginal-information constraint. In the strict curved interior both derivatives in (237) are positive; on $\mathcal{P}_2$ they equal

one. At the curved/sum-rate interface the derivatives still equal one. Thus every positive private coordinate is necessary at these points. At a $(\mathcal{P}_1, \mathcal{P}_3)$ interface, increasing the first marginal requirement strictly raises the common rate, while increasing the second does so quadratically by (242). The other one-source interface is symmetric. Hence every tuple on $\mathcal{P}_1 \cup \mathcal{P}_2$ is a boundary rate tuple, including the finite interfaces. Monotonicity extends the strict local increase to every larger admissible reduction of a private rate.

Inside the strict $\mathcal{P}_3$ face, a sufficiently small increase of β leaves Υ_ρ equal to α. Therefore $R_2 > 0$ makes that coordinate redundant. Conversely, if $R_2 = 0$, it cannot be reduced. The positive common coordinate is minimal, and reducing a positive R_1 increases α and hence increases the common requirement by the bound $\Upsilon_\rho \geq \alpha$. Such a tuple is a boundary rate tuple. The $\mathcal{P}_4$ case follows by interchanging the two coordinates. These arguments also cover zero private rates at the edges of the feasible rectangle, including the all-common tuple. The two cases exhaust all boundary rate tuples.

Appendix D
Proof of Lemma 3

We prove Lemma 3. For $Q^{\boldsymbol{\sigma}} = \mathcal{N}(\mathbf{0}_2, \boldsymbol{\sigma})$, the mean of a quadratic form gives

$$\mathsf{E}_{Q^\sigma}[\jmath_{XY}(X,Y)] = R_0^* + \operatorname{tr}(\mathbf{g}\boldsymbol{\sigma}) - \operatorname{tr}(\mathbf{g}\boldsymbol{\sigma}_{XY}) = R_0^* + \langle \mathbf{g}, \boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY} \rangle \,, \tag{246}$$

where Eq. (246) follows from (51). Setting $\boldsymbol{\sigma} = \boldsymbol{\sigma}_{XY}$ proves claim (i).

For a symmetric perturbation $d\boldsymbol{\sigma}$ at $\boldsymbol{\sigma}_{XY}$, the chain rule implies

$$\begin{aligned} d\rho(\boldsymbol{\sigma}) &= \frac{d\sigma_{12}}{\sigma_X \sigma_Y} - \frac{\rho}{2\sigma_X^2} d\sigma_{11} - \frac{\rho}{2\sigma_Y^2} d\sigma_{22}, \\ d\mathsf{R}_0 &= \lambda_1^* \frac{d\sigma_{11}}{2\sigma_X^2} + \lambda_2^* \frac{d\sigma_{22}}{2\sigma_Y^2} + \eta \, d\rho(\boldsymbol{\sigma}) = \langle \mathbf{g}, d\boldsymbol{\sigma} \rangle \,, \end{aligned} \tag{247}$$

where Eq. (247) follows from the marginal logarithmic derivatives and counts the symmetric off-diagonal entry twice, which establishes (248). At a positive axis the vanishing marginal requirement has zero boundary derivative, as proved in Appendix B.

Differentiating (24) and (22) leads to the explicit matrix

$$\mathbf{g} = \frac{1}{2} \begin{bmatrix} (\lambda_1^* - \rho\eta)/\sigma_X^2 & \eta/(\sigma_X \sigma_Y) \\ \eta/(\sigma_X \sigma_Y) & (\lambda_2^* - \rho\eta)/\sigma_Y^2 \end{bmatrix}. \tag{248}$$

Let $\boldsymbol{\omega} := \boldsymbol{\sigma}_{XY}^{1/2} \mathbf{g} \boldsymbol{\sigma}_{XY}^{1/2} \in \mathbb{S}^2$ have eigenvalues (ω_1, ω_2), and let (N_1, N_2) be independent standard Gaussians. We use $\overset{\mathrm{d}}{=}$ to denote equality in distribution. Orthogonal diagonalization yields

$$\begin{aligned} \jmath_{XY}(X,Y) - R_0^* &\overset{\mathrm{d}}{=} \omega_1(N_1^2 - 1) + \omega_2(N_2^2 - 1), \\ \mathsf{Var}_{P_{XY}}[\jmath_{XY}(X,Y)] &= 2(\omega_1^2 + \omega_2^2) = 2\operatorname{tr}(\boldsymbol{\omega}^2) = 2\operatorname{tr}[(\mathbf{g}\boldsymbol{\sigma}_{XY})^2], \end{aligned} \tag{249}$$

where Eq. (249) follows from independence and $\mathsf{Var}(N_j^2) = 2$. This representation also ensures a finite third absolute moment. The following argument determines the eigenvalues, making a second expanded variance formula unnecessary.

To evaluate the dispersion through its spectrum, fix a boundary rate tuple $\mathbf{r}^*$ with $R_0^* > 0$, the weights $(\lambda_1^*, \lambda_2^*)$ from (48)–(49), and the derivative $\mathbf{g}$ from (50). Let $\mathbf{k}^*$ be the optimizer of (231) at $(\boldsymbol{\sigma}_{XY}, R_1^*, R_2^*)$, whose existence is proved in Lemma 4. Define the positive caps $\tau_j := D_j e^{2R_j^*}$ for $j \in [2]$.

After dropping the constant $\frac{1}{2}\log|\boldsymbol{\sigma}_{XY}|$, the primal problem minimizes $-\frac{1}{2}\log|\mathbf{k}|$ under the affine diagonal caps and covariance order. A sufficiently small positive multiple of $\mathbf{1}_2$ is strictly feasible. Convex duality therefore ensures the existence of multipliers $(\vartheta_1, \vartheta_2) \in \mathbb{R}^2_{+,0}$ and $\boldsymbol{\lambda}_{\mathrm{cov}} \in \mathbb{S}^2_+$ satisfying

$$\begin{aligned} -\tfrac{1}{2}(\mathbf{k}^*)^{-1} + \operatorname{diag}(\vartheta_1, \vartheta_2) + \boldsymbol{\lambda}_{\mathrm{cov}} &= \mathbf{0}_{2\times 2}, \\ \langle \boldsymbol{\lambda}_{\mathrm{cov}}, \boldsymbol{\sigma}_{XY} - \mathbf{k}^* \rangle &= 0. \end{aligned} \tag{250}$$

For a nearby positive-definite $\boldsymbol{\sigma}$, the same dual multipliers are feasible. Their dual objective changes only by $-\langle \boldsymbol{\lambda}_{\mathrm{cov}}, \boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY} \rangle$. Consequently,

$$\mathsf{R}_0(\boldsymbol{\sigma}; R_1^*, R_2^*) - \tfrac{1}{2}\log|\boldsymbol{\sigma}| \geq R_0^* - \tfrac{1}{2}\log|\boldsymbol{\sigma}_{XY}| - \langle \boldsymbol{\lambda}_{\mathrm{cov}}, \boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY} \rangle \,, \tag{251}$$

$$\mathbf{g} = \tfrac{1}{2}\boldsymbol{\sigma}_{XY}^{-1} - \boldsymbol{\lambda}_{\mathrm{cov}}, \tag{252}$$

where Eq. (251) follows from weak duality with equality at $\boldsymbol{\sigma} = \boldsymbol{\sigma}_{XY}$, and Eq. (252) follows from differentiating the touching inequality in both signs of every symmetric direction, using Proposition 2. No differentiability of the optimizer or of its active set is needed.

Let $\boldsymbol{\delta}_* := \boldsymbol{\sigma}_{XY} - \mathbf{k}^* \in \mathbb{S}_+^2$. Since $R_0^* > 0$, the covariance $\mathbf{k}^*$ cannot equal $\boldsymbol{\sigma}_{XY}$, so $\boldsymbol{\delta}_* \neq \mathbf{0}_{2\times 2}$. For two positive-semidefinite matrices, zero trace product implies zero product: indeed $\operatorname{tr}(\boldsymbol{\lambda}_{\mathrm{cov}}\boldsymbol{\delta}_*) = \left\|\boldsymbol{\lambda}_{\mathrm{cov}}^{1/2}\boldsymbol{\delta}_*^{1/2}\right\|^2 = 0$, so $\boldsymbol{\lambda}_{\mathrm{cov}}^{1/2}\boldsymbol{\delta}_*^{1/2} = \mathbf{0}_{2\times 2}$ and then $\boldsymbol{\lambda}_{\mathrm{cov}}\boldsymbol{\delta}_* = \mathbf{0}_{2\times 2}$. Equation (250) therefore implies

$$\begin{aligned}
\boldsymbol{\lambda}_{\mathrm{cov}}\boldsymbol{\delta}_* &= \mathbf{0}_{2\times 2}, \\
\operatorname{rank}(\boldsymbol{\lambda}_{\mathrm{cov}}) &\leq 1, \qquad (253)\\
\boldsymbol{\omega} &= \boldsymbol{\sigma}_{XY}^{1/2}\mathbf{g}\boldsymbol{\sigma}_{XY}^{1/2} = \tfrac{1}{2}\mathbf{1}_2 - \boldsymbol{\sigma}_{XY}^{1/2}\boldsymbol{\lambda}_{\mathrm{cov}}\boldsymbol{\sigma}_{XY}^{1/2}, \qquad (254)\\
\operatorname{tr}(\boldsymbol{\omega}) &= \operatorname{tr}(\mathbf{g}\boldsymbol{\sigma}_{XY}) = \frac{\lambda_1^* + \lambda_2^*}{2}, \qquad (255)
\end{aligned}$$

where Eq. (253) follows from the nonzero range of $\boldsymbol{\delta}_*$, Eq. (254) follows from substituting (252), and Eq. (255) follows from evaluating (248). The rank bound forces one eigenvalue of $\boldsymbol{\omega}$ to equal $1/2$. The trace determines the other as $(\lambda_1^* + \lambda_2^* - 1)/2$.

We next identify (51) with the distortion-tilted information density in [11, Definition 4]. The same dual touching argument, applied to the cap $\tau_j = D_j e^{2R_j^*}$, gives $\lambda_j^* = 2\vartheta_j\tau_j$. Complementary slackness gives $\vartheta_j(k_{jj}^* - \tau_j) = 0$. Thus, on defining the diagonal matrix $\boldsymbol{\theta}_* := \operatorname{diag}(\lambda_1^*/k_{11}^*, \lambda_2^*/k_{22}^*)$, the stationarity and complementarity relations imply

$$\boldsymbol{\theta}_* = 2\operatorname{diag}(\vartheta_1, \vartheta_2) = (\mathbf{k}^*)^{-1} - 2\boldsymbol{\lambda}_{\mathrm{cov}}. \qquad (256)$$

Moreover,

$$\boldsymbol{\theta}_*\boldsymbol{\delta}_* = (\mathbf{k}^*)^{-1}\boldsymbol{\delta}_*, \qquad (257)$$

where Eq. (257) uses $\boldsymbol{\lambda}_{\mathrm{cov}}\boldsymbol{\delta}_* = \mathbf{0}_{2\times 2}$. This argument also covers a vanishing multiplier.

Let $\mathbf{W}_* \sim \mathcal{N}(\mathbf{0}_2, \boldsymbol{\delta}_*)$ and $\mathbf{N}_* \sim \mathcal{N}(\mathbf{0}_2, \mathbf{k}^*)$ be independent and set $(X, Y)^{\mathsf{T}} = \mathbf{W}_* + \mathbf{N}_*$. Then $W = \mathbf{W}_*$ is an attaining Gaussian common auxiliary. For deterministic $\mathbf{w} \in \mathbb{R}^2$, write $w_j := [\mathbf{w}]_j$. For each $j \in [2]$ with $\lambda_j^* > 0$, the cap is active and $k_{jj}^* = D_j e^{2R_j^*} \geq D_j$. The conditional distortion-tilted information density [11, Eqs. (24)–(25)] is therefore

$$\jmath_j(x_j, D_j \mid \mathbf{w}) = R_j^* + \frac{(x_j - w_j)^2}{2k_{jj}^*} - \frac{1}{2}. \qquad (258)$$

This formula follows by evaluating the Gaussian reconstruction expectation with distortion multiplier $1/(2D_j)$, and remains valid at $R_j^* = 0$, when the reconstruction is the constant w_j. For $\lambda_j^* = 0$, set $\jmath_j(x_j, D_j \mid \mathbf{w}) := 0$; the corresponding zero-weight term does not affect the density below. The continuous version of [11, Eq. (26)] becomes

$$\begin{aligned}
-\log \mathsf{E}\Big[\exp\Big(\sum_{j\in[2]} \lambda_j^*\big(R_j^* - \jmath_j(x_j, D_j \mid \mathbf{W}_*)\big)\Big)\Big] &= -\frac{\lambda_1^* + \lambda_2^*}{2} - \log \mathsf{E}\Big[\exp\Big(-\frac{1}{2}(\mathbf{x} - \mathbf{W}_*)^{\mathsf{T}}\boldsymbol{\theta}_*(\mathbf{x} - \mathbf{W}_*)\Big)\Big], \\
&= \frac{1}{2}\log|\mathbf{1}_2 + \boldsymbol{\delta}_*\boldsymbol{\theta}_*| - \frac{\lambda_1^* + \lambda_2^*}{2} + \frac{1}{2}\mathbf{x}^{\mathsf{T}}(\mathbf{1}_2 + \boldsymbol{\theta}_*\boldsymbol{\delta}_*)^{-1}\boldsymbol{\theta}_*\mathbf{x}. \qquad (259)
\end{aligned}$$

Here the Gaussian integration is valid for positive-semidefinite $(\boldsymbol{\delta}_*, \boldsymbol{\theta}_*)$, so it requires neither a nonsingular common covariance nor positive private supporting weights. Equation (257), its transpose, and (252) imply

$$|\mathbf{1}_2 + \boldsymbol{\delta}_*\boldsymbol{\theta}_*| = |\mathbf{1}_2 + \boldsymbol{\delta}_*(\mathbf{k}^*)^{-1}| = \frac{|\boldsymbol{\sigma}_{XY}|}{|\mathbf{k}^*|} = e^{2R_0^*}. \qquad (260)$$

Furthermore,

$$\begin{aligned}
(\mathbf{1}_2 + \boldsymbol{\theta}_*\boldsymbol{\delta}_*)^{-1}\boldsymbol{\theta}_* &= \boldsymbol{\sigma}_{XY}^{-1}\mathbf{k}^*\big((\mathbf{k}^*)^{-1} - 2\boldsymbol{\lambda}_{\mathrm{cov}}\big), \\
&= \boldsymbol{\sigma}_{XY}^{-1} - 2\boldsymbol{\lambda}_{\mathrm{cov}} = 2\mathbf{g}, \qquad (261)
\end{aligned}$$

where Eq. (260) uses $\boldsymbol{\sigma}_{XY} = \mathbf{k}^* + \boldsymbol{\delta}_*$ and Eq. (261) uses $\mathbf{k}^*\boldsymbol{\lambda}_{\mathrm{cov}} = \boldsymbol{\sigma}_{XY}\boldsymbol{\lambda}_{\mathrm{cov}}$. Substituting these identities and (255) into (259) yields $R_0^* + \mathbf{x}^{\mathsf{T}}\mathbf{g}\mathbf{x} - \operatorname{tr}(\mathbf{g}\boldsymbol{\sigma}_{XY})$, which is exactly (51). Thus the covariance representation and the distortion-tilted definition identify the same density.

For completeness, the slope sum lies in $[1, 2]$. The upper bound follows from $\lambda_j^* \leq 1$. In the strict curved region, the parameters (r, u) in (54)–(233) satisfy

$$\lambda_1^* + \lambda_2^* - 1 = \frac{1 + r^2 - r(u + u^{-1})}{1 - r^2} = \frac{(u - r)(1 - ru)}{u(1 - r^2)} > 0, \qquad (262)$$

where Eq. (262) follows from substitution and factorization, and positivity uses $0 < r < \min\{u, u^{-1}\}$. The affine values and all finite interfaces follow from (234) and continuity. Thus

$$\begin{aligned}\mathsf{Var}_{P_{XY}}[\jmath_{XY}(X,Y)] &= 2\,\mathrm{tr}(\boldsymbol{\omega}^2),\\ &= 2\left[\left(\frac{1}{2}\right)^2 + \left(\frac{\lambda_1^* + \lambda_2^* - 1}{2}\right)^2\right],\\ &= \frac{1 + (\lambda_1^* + \lambda_2^* - 1)^2}{2} = V, \end{aligned} \tag{263}$$

where Eq. (263) follows from the two identified eigenvalues. Together with the branch derivatives in (232)–(234), this proves (56) and (57), completing the proof of Lemma 3. The preceding slope calculation also implies

$$1 \le \lambda_1^* + \lambda_2^* \le 2. \tag{264}$$

Consequently, the dispersion satisfies

$$\frac{1}{2} \le V \le 1. \tag{265}$$

These identities include zero private rates, because the diagonal caps remain strictly positive and Slater's condition remains valid there.

The same diagonalization also identifies the exact finite-blocklength distribution of the centered source fluctuation. Let $(\Xi_{1,n}, \Xi_{2,n})$ be independent chi-squared variables with n degrees of freedom. Independence across source letters and the two whitened coordinates implies

$$\begin{aligned}\frac{1}{\sqrt{n}}\sum_{i\in[n]}[\jmath_{XY}(X_i,Y_i) - R_0^*] &\overset{\mathrm{d}}{=} \frac{1}{2\sqrt{n}}\sum_{i\in[n]}(N_{1,i}^2 - 1) + \frac{\lambda_1^* + \lambda_2^* - 1}{2\sqrt{n}}\sum_{i\in[n]}(N_{2,i}^2 - 1),\\ &\overset{\mathrm{d}}{=} \frac{\Xi_{1,n} - n}{2\sqrt{n}} + \frac{\lambda_1^* + \lambda_2^* - 1}{2\sqrt{n}}(\Xi_{2,n} - n), \end{aligned} \tag{266}$$

where Eq. (266) follows from summing independent squared standard Gaussian variables $N_{j,i}$. This is an exact distributional identity for the linearized source statistic, not for the error probability of the covering code.

Appendix E
Proof of Lemma 7

Fix the positive supporting weights in Lemma 7. Let $Q_{W|XY}$ be a conditional distribution on a standard Borel auxiliary alphabet, and write $Q := P_{XY}Q_{W|XY}$. For every $(\alpha, \beta) \in \mathbb{R}_{+,0}^2$, Yu's unrestricted mutual-information envelope [13, Theorem 3] is

$$\Upsilon_\rho(\alpha,\beta) = \inf_{Q_{W|XY}:\ I_Q(X;W)\ge\alpha,\ I_Q(Y;W)\ge\beta} I_Q(X,Y;W).$$

The marginal requirements are not restricted to $\mathcal{I}_{\boldsymbol{\sigma}_{XY}}$. Independent time sharing between auxiliary kernels makes this envelope convex on the entire nonnegative quadrant. The selected marginal requirements (α^*, β^*) are positive at every boundary rate tuple with $R_0^* > 0$, by the classification in Section II-B. Proposition 2 therefore identifies $(\lambda_1^*, \lambda_2^*)$ as its gradient and global supporting slope, including when a private operational rate equals zero. For $\mathbf{0}_{2\times2} \prec \mathbf{k} \preceq \boldsymbol{\sigma}_{XY}$, let $\alpha_{\mathbf{k}} := \frac{1}{2}\log(\sigma_X^2/k_{11})$, $\beta_{\mathbf{k}} := \frac{1}{2}\log(\sigma_Y^2/k_{22})$, and $t_{\mathbf{k}} := \frac{1}{2}\log(|\boldsymbol{\sigma}_{XY}|/|\mathbf{k}|)$. The supporting plane at (α^*, β^*) implies

$$\begin{aligned} t_{\mathbf{k}} &\ge \Upsilon_\rho(\alpha_{\mathbf{k}}, \beta_{\mathbf{k}}),\\ &\ge R_0^* + \lambda_1^*(\alpha_{\mathbf{k}} - \alpha^*) + \lambda_2^*(\beta_{\mathbf{k}} - \beta^*), \end{aligned} \tag{267}$$

and

$$\begin{aligned}\frac{1}{2}\log\frac{|\boldsymbol{\sigma}_{XY}|}{|\mathbf{k}|} + \sum_{j\in[2]}\frac{\lambda_j^*}{2}\log\frac{k_{jj}}{D_j} &= t_{\mathbf{k}} - \lambda_1^*\alpha_{\mathbf{k}} - \lambda_2^*\beta_{\mathbf{k}} + \sum_{j\in[2]}\frac{\lambda_j^*}{2}\log\frac{[\boldsymbol{\sigma}_{XY}]_{jj}}{D_j},\\ &\ge R_0^* + \lambda_1^* R_1^* + \lambda_2^* R_2^*, \end{aligned} \tag{268}$$

where Eq. (267) follows from the unrestricted envelope and convexity, and Eq. (268) follows from substituting the boundary marginal requirements. At a common-rate optimizer $\mathbf{k}^*$, feasibility implies $\alpha_{\mathbf{k}^*} \ge \alpha^*$ and $\beta_{\mathbf{k}^*} \ge \beta^*$. Since $t_{\mathbf{k}^*} = R_0^*$ and both weights are positive, Eq. (267) forces both inequalities to be equalities. Thus $\mathbf{k}^*$ attains the last bound, proving (173), including $R_j^* = 0$ where $k_{jj}^* = D_j$.

For nearby $\boldsymbol{\sigma}$, evaluating the entropy support at a common-rate optimizer $\mathbf{k}^*_{\boldsymbol{\sigma}}$ yields

$$\begin{aligned}\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma}) &\le \frac{1}{2}\log\frac{|\boldsymbol{\sigma}|}{|\mathbf{k}^*_{\boldsymbol{\sigma}}|} + \sum_{j\in[2]}\frac{\lambda^*_j}{2}\log\frac{[\mathbf{k}^*_{\boldsymbol{\sigma}}]_{jj}}{D_j},\\ &\le \mathsf{R}_0(\boldsymbol{\sigma}; R^*_1, R^*_2) + \lambda^*_1 R^*_1 + \lambda^*_2 R^*_2, \end{aligned} \tag{269}$$

where Eq. (269) follows from the private covariance caps. Equality holds at $\boldsymbol{\sigma}_{XY}$, not necessarily on a neighborhood.

Recall the concave function $\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}$, its matrix supergradient $\mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*}$, and centered quadratic function $g_{\boldsymbol{\lambda}^*}$ defined in Section V-A. The covariance/functional-supergradient correspondence is the imported result [12, Proposition 5]. We only identify its matrix here. Since $\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}$ is finite and concave on the open positive-definite cone, it has a supergradient at $\boldsymbol{\sigma}_{XY}$. Equation (269) provides a differentiable lower touching function for $\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}$. Its derivative at $\boldsymbol{\sigma}_{XY}$ is $-\mathbf{g}$, since the derivatives of the additional log-determinant and reference-linear terms cancel. For $\mathbf{h}\in\mathbb{S}^2$ and sufficiently small $t>0$,

$$\begin{aligned}-\langle \mathbf{g},\mathbf{h}\rangle + o(1) &\le \frac{\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}(\boldsymbol{\sigma}_{XY}+t\mathbf{h}) - \varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}(\boldsymbol{\sigma}_{XY})}{t},\\ &\le \langle \mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*},\mathbf{h}\rangle, \end{aligned} \tag{270}$$

where Eq. (270) follows from combining the touching bound with (171). For every $\mathbf{h}\in\mathbb{S}^2$, sending $t\to 0$ in the directions $\mathbf{h}$ and $-\mathbf{h}$ leads to

$$\begin{aligned}\langle \mathbf{g}_{\mathrm{ent},\boldsymbol{\lambda}^*},\mathbf{h}\rangle &= -\langle \mathbf{g},\mathbf{h}\rangle,\\ g_{\boldsymbol{\lambda}^*}(\mathbf{x}) &= -[\jmath_{XY}(x,y) - R^*_0], \end{aligned} \tag{271}$$

where Eq. (271) follows from centering the quadratic form with matrix $-\mathbf{g}$. The unique supergradient makes $\varphi_{\boldsymbol{\lambda}^*,\boldsymbol{\sigma}_{XY}}$ differentiable at this interior point, and (170) then implies (174). The variance is V by (52). This establishes all parts of Lemma 7 without differentiating a clipped private logarithm.

Appendix F
Proof of Lemma 8

We specialize the positive-weight Gaussian converse of Liu et al. to the present rates and distortion criterion. First we identify its supporting value and centered fluctuation using Lemma 7. We then apply its asymptotic statement when the normalized weighted displacement is finite. Its finite-threshold bound handles a displacement tending to minus infinity and completes the proof under the one-sided rate envelopes. The imported converse is [12, Theorem 17, with Proposition 5 and Eq. (190)]. In its notation, take $m=2$, $Q_{Y^m}=P_{XY}$, $c_0=1$, $c_j=\lambda^*_j$ for $j\in[2]$, and each reference measure to be Lebesgue measure. Let d^* be the supporting value in that theorem. Gaussian optimality [12, Proposition 16 and Eq. (190)] and (169) imply

$$d^* + \sum_{j\in[2]}\frac{\lambda^*_j}{2}\log(2\pi e D_j) = -\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma}_{XY}). \tag{272}$$

The Gaussian auxiliary attaining the support in Lemma 7 also attains d^*. The published functional subgradient is normalized to have mean d^* [12, Eq. (52)]. Proposition 5 of that paper and Lemma 7 consequently identify it as $d^* + g_{\boldsymbol{\lambda}^*}(\mathbf{x})$. Its centered version is $g_{\boldsymbol{\lambda}^*}(\mathbf{x}) = -[\jmath_{XY}(x,y) - R^*_0]$, so its third absolute moment is the same finite β_3 used in (126). Applying that Berry–Esseen bound to its centered sum gives, for every $a\in\mathbb{R}$,

$$P^n_{XY}\left\{\frac{1}{\sqrt{n}}\sum_{i\in[n]} g_{\boldsymbol{\lambda}^*}(\mathbf{X}_i) \le -a\right\} \ge \mathrm{Q}(a/\sqrt{V}) - \frac{C_{\mathrm{BE}}\beta_3}{V^{3/2}\sqrt{n}}.$$

Thus the weighted displacement in the imported theorem equals

$$A := \limsup_{n\to\infty}\frac{\log M_{0,n} + \lambda^*_1\log M_{1,n} + \lambda^*_2\log M_{2,n} - n\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma}_{XY})}{\sqrt{n}}. \tag{273}$$

Lemma 7 identifies the variance as V. The source covariance is positive definite, the distortions and weights are positive, and the support is finite and attained. The imported theorem permits positive weights equal to one. The rate envelopes make A bounded above by a finite number. Suppose first that $A\in\mathbb{R}$. The cited theorem implies

$$\liminf_{n\to\infty}\varepsilon_n \ge \mathrm{Q}(A/\sqrt{V}).$$

For every $\gamma>0$, the rate envelopes and (173) yield

$$A \le L_0 + \lambda^*_1 L_1 + \lambda^*_2 L_2 + (1+\lambda^*_1+\lambda^*_2)\gamma.$$

Combining the last two displays, using that Q is decreasing, and then sending $\gamma\to 0$ proves (177).

It remains to consider $A = -\infty$. Fix any $a \in \mathbb{R}$. By (273), the normalized weighted displacement is eventually at most $a - 1$. In the finite-threshold bound of [12, Theorem 17, Eqs. (193)–(197)], choose the threshold $nd^* - \sqrt{n}a$. The normalized subgradient identified above makes its probability term the event $\sum_{i\in[n]} g_{\boldsymbol{\lambda}^*}(\mathbf{X}_i) \le -\sqrt{n}a$. For each $j \in [2]$, writing vol for Lebesgue volume, the Gaussian distortion-ball volume satisfies

$$\log \mathrm{vol}\left(\mathcal{B}_n(\mathbf{0}_n, \sqrt{nD_j})\right) = \frac{n}{2}\log(2\pi e D_j) + O(\log n).$$

Together with (272), this bounds the logarithm of the exponential remainder by

$$\log M_{0,n} + \sum_{j\in[2]} \lambda_j^* \log M_{j,n} - n\underline{F}_{\boldsymbol{\lambda}^*}(\boldsymbol{\sigma}_{XY}) - \sqrt{n}a + O(\log n) \le -\sqrt{n} + O(\log n).$$

The finite-threshold bound therefore implies

$$\varepsilon_n \ge P_{XY}^n \left\{ \frac{1}{\sqrt{n}} \sum_{i\in[n]} g_{\boldsymbol{\lambda}^*}(\mathbf{X}_i) \le -a \right\} - \exp\{-\sqrt{n} + O(\log n)\}, \tag{274}$$

where Eq. (274) follows from the finite-threshold converse of [12, Theorem 17, Eqs. (193)–(197)] and the distortion-ball volume estimate above. Its logarithmic remainder depends only on the fixed distortions and weights. Taking the blocklength limit yields

$$\liminf_{n\to\infty} \varepsilon_n \ge \mathrm{Q}(a/\sqrt{V}), \tag{275}$$

where Eq. (275) follows from the Berry–Esseen bound above and (274): both the $n^{-1/2}$ normal-approximation term and the exponential remainder vanish at fixed a. Sending $a \to -\infty$ after the blocklength limit implies $\liminf_{n\to\infty} \varepsilon_n \ge 1$, proving (177) in this remaining case.

Appendix G
Proof of Lemma 6

We prove Lemma 6, including the moving branch interfaces and bounded correlation perturbations. Recall the allocation function ψ_ρ in (58). For fixed $\rho \in (0,1)$, define the hypercontractive exponent set by

$$\mathcal{H}_\rho := \left\{(p,q) \in (1,\infty)^2 : (p-1)(q-1) \ge \rho^2\right\}. \tag{276}$$

For $(\delta_1, \delta_2) \in \mathbb{R}^2_{+,0}$, its equivalent dual representation is

$$\psi_\rho(\delta_1,\delta_2) = \sup_{(p,q)\in\mathcal{H}_\rho} \left(\frac{\delta_1}{p} + \frac{\delta_2}{q}\right). \tag{277}$$

We verify this representation first. The supremum includes the limits $(p,q) \to (1,\infty)$ and $(\infty,1)$; every finite feasible exponent remains strictly greater than one.

For $(\delta_1,\delta_2) \in \mathbb{R}^2_{+,0}$, monotonicity in the hypercontractive exponents places the optimization on $(p-1)(q-1) = \rho^2$. With $s := p - 1 > 0$, its objective and derivative are

$$\psi_\rho(\delta_1,\delta_2) = \sup_{s>0} \left\{ \frac{\delta_1}{1+s} + \frac{\delta_2 s}{s+\rho^2} \right\}, \tag{278}$$

$$\frac{d}{ds}\left(\frac{\delta_1}{1+s} + \frac{\delta_2 s}{s+\rho^2}\right) = -\frac{\delta_1}{(1+s)^2} + \frac{\delta_2\rho^2}{(s+\rho^2)^2},$$

where Eq. (278) follows from substituting $q-1 = \rho^2/s$. For positive (δ_1,δ_2), the stationary point is $s = (\rho\sqrt{\delta_2} - \rho^2\sqrt{\delta_1})/(\sqrt{\delta_1} - \rho\sqrt{\delta_2})$, positive exactly when $\rho^2\delta_1 < \delta_2 < \delta_1/\rho^2$. The derivative changes from positive to negative there. Substitution leads to the curved expression in (58). Otherwise the supremum is δ_1 as $s \to 0$ or δ_2 as $s \uparrow \infty$. These limits also determine the values on the axes and interfaces.

Define the compact exponent image by

$$\mathcal{C}_\rho := \{(u,v) \in [0,1]^2 : (1-u)(1-v) \ge \rho^2 uv\}.$$

The compact form of the same optimization is

$$\psi_\rho(\delta_1,\delta_2) = \max_{(u,v)\in\mathcal{C}_\rho} (u\delta_1 + v\delta_2), \tag{279}$$

$$\max\{\delta_1,\delta_2\} \le \psi_\rho(\delta_1,\delta_2) \le \delta_1 + \delta_2, \tag{280}$$

$$\begin{aligned} |\psi_\rho(\delta_1,\delta_2) - \psi_\rho(\delta_1',\delta_2')| &\le \max_{(u,v)\in\mathcal{C}_\rho} |u(\delta_1-\delta_1') + v(\delta_2-\delta_2')|, \\ &\le |\delta_1-\delta_1'| + |\delta_2-\delta_2'|, \end{aligned} \tag{281}$$

where Eq. (279) follows from taking the closure of $(1/p, 1/q)$, Eq. (280) follows from $(1,0),(0,1) \in \mathcal{C}_\rho \subset [0,1]^2$, and Eq. (281) follows from comparing the two maxima. This also proves positive homogeneity and coordinatewise monotonicity. On bounded argument sets and compact subintervals of $\rho \in (0,1)$, the curved formula has a bounded ρ derivative. The constant one-source pieces join continuously. Thus ψ_ρ is locally Lipschitz in ρ, uniformly on those sets.

To prove the uniform expansion, fix $(\delta_1, \delta_2) \in [0, M]^2$. Set $A_t(\delta_1) := (1 - e^{-2t\delta_1})/(2t)$ for $t > 0$. Uniformly on this square,

$$A_t(\delta_1) = \delta_1 \int_0^1 e^{-2t\delta_1 u}\, du = \delta_1 + O_M(t), \tag{282}$$

$$\sqrt{(1 - e^{-2t\delta_1})(1 - e^{-2t\delta_2})} = 2t\sqrt{\delta_1\delta_2} + O_M(t^2), \tag{283}$$

$$r_{t\delta_1, t\delta_2} = \rho + t[\rho(\delta_1 + \delta_2) - 2\sqrt{\delta_1\delta_2}] + O_M(t^2), \tag{284}$$

where Eq. (283) follows from expanding $\sqrt{(1 - e^{-2x})/(2x)}$ continuously at zero, and Eq. (284) follows from expanding the exponential in (28). No division by a vanishing argument occurs. The product in (283) is smaller than ρ for small t, so $\mathcal{P}_2$ is excluded uniformly. Write $\psi_1(\delta_1, \delta_2) := (\delta_1 + \delta_2 - 2\rho\sqrt{\delta_1\delta_2})/(1 - \rho^2)$. On the finite-$t$ curved branch,

$$\begin{aligned} \Upsilon_\rho(t\delta_1, t\delta_2) &= t(\delta_1 + \delta_2) - \frac{1}{2}\log\frac{1 - r^2_{t\delta_1, t\delta_2}}{1 - \rho^2}, \\ &= t\psi_1(\delta_1, \delta_2) + O_M(t^2), \end{aligned} \tag{285}$$

where Eq. (285) follows from applying the logarithmic Taylor expansion to (284). $r_{t\delta_1, t\delta_2}$ remains uniformly bounded away from one, so the logarithm's argument remains bounded away from zero. The finite-t one-source expressions are exactly $t\delta_1$ and $t\delta_2$.

The exact interface with $\mathcal{P}_3$ is $A_t(\delta_2) = \rho^2 A_t(\delta_1)$. A disagreement with the limiting classifier $\delta_2 = \rho^2\delta_1$ can occur only in the strip

$$|\delta_2 - \rho^2\delta_1| \le |\delta_2 - A_t(\delta_2)| + \rho^2|\delta_1 - A_t(\delta_1)| \le C_M t, \tag{286}$$

$$0 \le \psi_1(\delta_1, \delta_2) - \delta_1 = \frac{(\sqrt{\delta_2} - \rho\sqrt{\delta_1})^2}{1 - \rho^2} \le \frac{|\delta_2 - \rho^2\delta_1|}{1 - \rho^2}, \tag{287}$$

where Eq. (286) follows from the opposite signs of the two classifiers and (282), and Eq. (287) follows from $(\sqrt{x} - \sqrt{y})^2 \le |x - y|$. Interchanging (δ_1, δ_2) leads to the other strip and gap. If both one-source classifiers switch, both estimates apply. Thus the coefficient error is $O_M(t)$ wherever the branches disagree, and

$$\sup_{0 \le \delta_1, \delta_2 \le M} |\Upsilon_\rho(t\delta_1, t\delta_2) - t\psi_\rho(\delta_1, \delta_2)| \le C_M t^2 + tC_M t \le C'_M t^2, \tag{288}$$

where Eq. (288) follows from combining the fixed-branch remainder with the strip bounds, including the origin.

For $\rho_t = \rho + th$, $|h| \le H$, all correlations lie in one compact subinterval of $(0,1)$ for small t. The same constants can therefore be used, and

$$\begin{aligned} |\Upsilon_{\rho_t}(t\delta_1, t\delta_2) - t\psi_\rho(\delta_1, \delta_2)| &\le |\Upsilon_{\rho_t}(t\delta_1, t\delta_2) - t\psi_{\rho_t}(\delta_1, \delta_2)| + t|\psi_{\rho_t}(\delta_1, \delta_2) - \psi_\rho(\delta_1, \delta_2)|, \\ &\le C_M t^2 + tC_{M,H} t, \end{aligned} \tag{289}$$

where Eq. (289) follows from (288) uniformly in correlation and the Lipschitz property proved above. This is the uniform estimate in Lemma 6.

For a positive level ℓ, homogeneity and (280) make each ray in $\mathbb{R}^2_+$ meet $\{\psi_\rho = \ell\}$ in a single point. By polar integration, this level set has zero area. Where one positive-part argument vanishes, its preimage at a positive level is a horizontal or vertical line. Since $P_{\mathbf{Z}}$ has a density, $T(\mathbf{Z})$ has no positive atom. Finally, (280) implies $\psi_\rho(\delta_1, \delta_2) = 0$ exactly at the origin, proving (136).

APPENDIX H
PROOF OF LEMMA 10

We first establish the Euclidean covering bound used for the private descriptions. Fix $T \ge 1$.

Lemma 10. *There is $C_T \ge 1$ such that, for every integer $n \ge 3$, $r > 0$, and $R \in [0, Tr]$, a finite set $\mathcal{V} \subset \mathbb{R}^n$ satisfies*

$$\begin{aligned} \mathcal{B}_n(\mathbf{0}_n, R) &\subseteq \bigcup_{\mathbf{v} \in \mathcal{V}} \mathcal{B}_n(\mathbf{v}, r), \\ |\mathcal{V}| &\le C_T n^4 \max\{1, (R/r)^n\}. \end{aligned}$$

When $R \le r$, one may take $\mathcal{V} = \{\mathbf{0}_n\}$. The constant and the threshold $n \ge 3$ do not depend on R/r.

Fix $T \geq 1$, $n \geq 3$, $r > 0$, and $0 \leq R \leq Tr$. If $R \leq r$, the origin suffices. Suppose $R > r$, and define $h := r/(4n)$, $r_- := r - 2h$, and $m := \lceil (R-r)/h \rceil$. For each $k \in [m]$, set

$$R_k := \min\{r + kh, R\}.$$

The radii satisfy $r < R_k \leq R$. Every point with radius in $(r, R]$ is within distance h of a point with the same direction and one of the radii R_k. The inner ball of radius r will be covered by the origin.

For each sphere $\mathcal{S}_{n,k} := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = R_k\}$, choose an h-net $\mathcal{N}_{n,k} \subset \mathcal{S}_{n,k}$ by taking a maximal h-separated set. The disjoint radius-$h/2$ balls around its points imply

$$|\mathcal{N}_{n,k}| \leq \left(1 + \frac{2R_k}{h}\right)^n \leq (1 + 8Tn)^n, \tag{290}$$

where Eq. (290) follows from the ambient-volume comparison and $R_k/r \leq T$.

Choose a center uniformly on the sphere of radius $\sqrt{R_k^2 - r_-^2}$. For a fixed net point, the radius-r_- ball about this center covers it when the cosine of the angle is at least $z_k := \sqrt{1 - r_-^2/R_k^2}$. Write $p_{n,k}$ for this probability and $\omega_n := \Gamma(n/2)/(\sqrt{\pi}\,\Gamma((n-1)/2))$. The coordinate density of a uniform sphere point implies

$$\begin{aligned} p_{n,k} &= \omega_n \int_{z_k}^1 (1-u^2)^{(n-3)/2}\, du, \\ &= \frac{\omega_n}{2} \int_0^{1-z_k^2} w^{(n-3)/2}(1-w)^{-1/2}\, dw, \\ &\geq \frac{1}{2(n-1)} \left(\frac{r_-}{R_k}\right)^{n-1}, \end{aligned} \tag{291}$$

where Eq. (291) follows from the spherical-coordinate density, the substitution $w = 1 - u^2$, $\omega_n \geq 1/2$ for $n \geq 3$, and $(1-w)^{-1/2} \geq 1$. Importantly, this lower bound remains valid as $z_k \to 0$. No positive lower bound on the cap cutoff is used.

Take the following number of independent centers of that radius:

$$J_{n,k} := \left\lceil 2(n-1)\left(\frac{R_k}{r_-}\right)^{n-1} \big[n \log(1 + 8Tn) + 1\big] \right\rceil. \tag{292}$$

Let $\mathcal{F}_{n,k}$ be the event that at least one point of $\mathcal{N}_{n,k}$ is uncovered by their radius-r_- balls. Then

$$\begin{aligned} \Pr(\mathcal{F}_{n,k}) &\leq |\mathcal{N}_{n,k}|(1 - p_{n,k})^{J_{n,k}}, \\ &\leq |\mathcal{N}_{n,k}| \exp\big(-J_{n,k} p_{n,k}\big), \\ &\leq e^{-1} < 1, \end{aligned} \tag{293}$$

where Eq. (293) follows from the union bound, independence, $1 - x \leq e^{-x}$, and (290)–(292). Fix a covering realization for each of the finitely many radii. For an arbitrary point of the outer ball, radial rounding costs at most h and the sphere net costs at most another h. Its distance to a selected center is therefore at most $r_- + 2h = r$. Together with the origin, these centers cover $\mathcal{B}_n(\mathbf{0}_n, R)$.

Finally, $m \leq 4nT + 1$, and

$$\left(\frac{r}{r_-}\right)^{n-1} = \left(1 - \frac{1}{2n}\right)^{-(n-1)} \leq \exp\left(\frac{n-1}{2n-1}\right) \leq e^{1/2}, \tag{294}$$

where Eq. (294) follows from $-\log(1-x) \leq x/(1-x)$. Since $R_k \leq R$ and $R/r > 1$, the total number of centers obeys

$$\begin{aligned} |\mathcal{V}| &\leq 1 + \sum_{k \in [m]} J_{n,k}, \\ &\leq C_T n^3 \big[1 + \log n\big](R/r)^n, \\ &\leq C_T' n^4 (R/r)^n, \end{aligned} \tag{295}$$

where Eq. (295) follows from (292), $m \leq 4nT + 1$, (294), and $\log(1 + 8Tn) \leq \log(1 + 8T) + \log n$. The constants depend only on T, and the proof works for every $n \geq 3$. Absorbing C_T' into C_T completes Lemma 10. This polynomial is not claimed to be optimal. Its logarithm is all that the second-order proof uses.

APPENDIX I
PROOF OF LEMMA 11

We prepare the conditional Gaussian statistics used in the covariance-shell covering proof. Fix a pair satisfying (96); all bounds in this appendix and Appendix J are uniform over this family and its shell blocks. Lemma 4 ensures that all residual optimizers used by the code lie in such a family, with a common κ. Recall $\mathbf{b} = (\boldsymbol{\sigma} - \mathbf{k})^{1/2}$. Under the analysis distribution $Q^{\boldsymbol{\sigma},\mathbf{k}}$, let $\mathbf{U} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{1}_2)$ and $\mathbf{N} \sim \mathcal{N}(\mathbf{0}_2, \mathbf{k})$ be independent, and set $\mathbf{X} = \mathbf{b}\mathbf{U} + \mathbf{N}$. Write $Q := Q^{\boldsymbol{\sigma},\mathbf{k}}$ for this joint distribution. Its source marginal is $Q^{\boldsymbol{\sigma}}$. $Q_{\mathbf{U}}$ is nonsingular even when $\mathbf{W} = \mathbf{b}\mathbf{U}$ is singular. Since $\mathbf{X}$ depends on $\mathbf{U}$ only through $\mathbf{W}$, their mutual informations with $\mathbf{X}$ are equal. For deterministic $(\mathbf{u}, \mathbf{x}) \in (\mathbb{R}^2)^2$, let $\mathbf{c} \in \mathbb{S}^2_{++}$ be the posterior covariance of $\mathbf{U}$ given $\mathbf{X} = \mathbf{x}$. Gaussian conditioning implies

$$
\begin{aligned}
I_{\boldsymbol{\sigma},\mathbf{k}} &:= I_Q(\mathbf{X}; \mathbf{U}) = \frac{1}{2} \log \frac{|\boldsymbol{\sigma}|}{|\mathbf{k}|}, \\
\imath_{\boldsymbol{\sigma},\mathbf{k}}(\mathbf{u}; \mathbf{x}) &:= \log \frac{dQ_{\mathbf{X}|\mathbf{U}=\mathbf{u}}}{dQ^{\boldsymbol{\sigma}}}(\mathbf{x}), \\
&= I_{\boldsymbol{\sigma},\mathbf{k}} - \tfrac{1}{2}(\mathbf{x} - \mathbf{b}\mathbf{u})^{\mathsf{T}} \mathbf{k}^{-1} (\mathbf{x} - \mathbf{b}\mathbf{u}) + \tfrac{1}{2} \mathbf{x}^{\mathsf{T}} \boldsymbol{\sigma}^{-1} \mathbf{x}, \qquad (296) \\
\mathbf{c} &:= \mathbf{1}_2 - \mathbf{b}^{\mathsf{T}} \boldsymbol{\sigma}^{-1} \mathbf{b} = (\mathbf{1}_2 + \mathbf{b}^{\mathsf{T}} \mathbf{k}^{-1} \mathbf{b})^{-1} \succ 0, \qquad (297) \\
\mathbf{U} \mid \mathbf{X} = \mathbf{x} &\sim \mathcal{N}(\mathbf{b}^{\mathsf{T}} \boldsymbol{\sigma}^{-1} \mathbf{x}, \mathbf{c}),
\end{aligned}
$$

where Eq. (296) follows from dividing the Gaussian densities and Eq. (297) follows from the Woodbury identity. All matrix coefficients are uniformly bounded under (96).

For a fixed shell block $\mathbf{x}^n$, draw $\mathbf{U}_i \in \mathbb{R}^2$, $i \in [n]$, independently from these posterior kernels, form $\mathbf{U}^n := [\mathbf{U}_1 \ \cdots \ \mathbf{U}_n] \in \mathbb{R}^{2\times n}$, and write $Q(\cdot \mid \mathbf{x}^n)$ for their product. For each $j \in [2]$ and $\mathbf{u}^n \in \mathbb{R}^{2\times n}$, define

$$
\begin{aligned}
j_n(\mathbf{u}^n; \mathbf{x}^n) &:= \sum_{i\in[n]} \imath_{\boldsymbol{\sigma},\mathbf{k}}(\mathbf{u}_i; \mathbf{x}_i), \\
e_{j,n}(\mathbf{u}^n; \mathbf{x}^n) &:= \sum_{i\in[n]} ([\mathbf{x}_i]_j - (\mathbf{b}\mathbf{u}_i)_j)^2, \\
J_n &:= j_n(\mathbf{U}^n; \mathbf{x}^n), \\
E_{j,n} &:= e_{j,n}(\mathbf{U}^n; \mathbf{x}^n).
\end{aligned}
$$

The residual $\mathbf{N}_i = \mathbf{x}_i - \mathbf{b}\mathbf{U}_i$ has posterior mean $\mathbf{k}\boldsymbol{\sigma}^{-1}\mathbf{x}_i$ and covariance $\mathbf{k} - \mathbf{k}\boldsymbol{\sigma}^{-1}\mathbf{k}$. Summing its second moments and substituting (296) yields

$$
\begin{aligned}
\mathsf{E}_Q[E_{j,n} \mid \mathbf{x}^n] &= nk_{jj} + n\mathbf{e}_j^{\mathsf{T}} \mathbf{k}\boldsymbol{\sigma}^{-1}(\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) - \boldsymbol{\sigma})\boldsymbol{\sigma}^{-1}\mathbf{k}\mathbf{e}_j, \qquad (298) \\
\mathsf{E}_Q[J_n \mid \mathbf{x}^n] &= nI_{\boldsymbol{\sigma},\mathbf{k}} + \frac{n}{2} \operatorname{tr}\left[\boldsymbol{\sigma}^{-1}(\boldsymbol{\sigma} - \mathbf{k})\boldsymbol{\sigma}^{-1}(\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) - \boldsymbol{\sigma})\right], \qquad (299) \\
|\mathsf{E}_Q[J_n \mid \mathbf{x}^n] - nI_{\boldsymbol{\sigma},\mathbf{k}}| + \sum_{j\in[2]} |\mathsf{E}_Q[E_{j,n} \mid \mathbf{x}^n] - nk_{jj}| &\le Cn \|\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) - \boldsymbol{\sigma}\| \le C', \qquad (300)
\end{aligned}
$$

where Eqs. (298) and (299) follow from the empirical second-moment matrix and cyclicity of trace, and Eq. (300) follows from the uniform matrix bounds and shell width C_0/n. These are the conditional centers needed by the common covering argument. Fix $(\boldsymbol{\sigma}, \mathbf{k})$ in the family (96) and retain the posterior statistics $(J_n, E_{1,n}, E_{2,n})$ defined above.

Lemma 11. *For every $\mathbf{x}^n \in \mathbb{R}^{2\times n}$ and every orthogonal $\mathbf{o} \in \mathbb{R}^{n\times n}$, the posterior distribution of $(J_n, E_{1,n}, E_{2,n})$ is the same for $\mathbf{x}^n$ and $\mathbf{x}^n\mathbf{o}$. Hence that distribution depends on the block only through $\mathbf{x}^n(\mathbf{x}^n)^{\mathsf{T}}$.*

Let $\mathbf{Z}^n \in \mathbb{R}^{2\times n}$ have independent standard Gaussian columns. The posterior can be represented as

$$
\begin{aligned}
\mathbf{U}^n(\mathbf{x}^n) &= \mathbf{b}^{\mathsf{T}} \boldsymbol{\sigma}^{-1} \mathbf{x}^n + \mathbf{c}^{1/2} \mathbf{Z}^n, \\
\mathbf{U}^n(\mathbf{x}^n\mathbf{o}) &\overset{\mathrm{d}}{=} \mathbf{b}^{\mathsf{T}} \boldsymbol{\sigma}^{-1} \mathbf{x}^n\mathbf{o} + \mathbf{c}^{1/2} \mathbf{Z}^n\mathbf{o} = \mathbf{U}^n(\mathbf{x}^n)\mathbf{o}, \qquad (301)
\end{aligned}
$$

where Eq. (301) follows from $\mathbf{Z}^n\mathbf{o} \overset{\mathrm{d}}{=} \mathbf{Z}^n$. Thus both the source and residual row Gram matrices are jointly invariant. Their residual diagonal entries are $(E_{1,n}, E_{2,n})$, while

$$
J_n = nI_{\boldsymbol{\sigma},\mathbf{k}} - \tfrac{1}{2} \operatorname{tr}[\mathbf{k}^{-1}(\mathbf{x}^n - \mathbf{b}\mathbf{U}^n)(\mathbf{x}^n - \mathbf{b}\mathbf{U}^n)^{\mathsf{T}}] + \tfrac{1}{2} \operatorname{tr}[\boldsymbol{\sigma}^{-1}\mathbf{x}^n(\mathbf{x}^n)^{\mathsf{T}}]. \qquad (302)
$$

Equation (302) follows from summing (296), proving joint invariance of all three statistics. Blocks with the same row Gram matrix are related by a right orthogonal transformation: the Gram-preserving isometry of their row spaces extends to $\mathbb{R}^n$.

For $n \geq 3$, define $\mathbf{o}_{\mathrm{frame},n} \in \mathbb{R}^{2\times n}$ to have ith column $\sqrt{2/n}(\cos(2\pi i/n), \sin(2\pi i/n))^{\mathsf{T}}$. The trigonometric sums imply $\mathbf{o}_{\mathrm{frame},n}\mathbf{o}_{\mathrm{frame},n}^{\mathsf{T}} = \mathbf{1}_2$. Define $\boldsymbol{\sigma}_{\mathrm{emp}} := \hat{\boldsymbol{\sigma}}(\mathbf{x}^n)$. The bounded representative satisfies

$$\begin{aligned} \mathbf{x}_\circ^n &:= \sqrt{n}\, \boldsymbol{\sigma}_{\mathrm{emp}}^{1/2}\mathbf{o}_{\mathrm{frame},n}, \\ \hat{\boldsymbol{\sigma}}(\mathbf{x}_\circ^n) &= \boldsymbol{\sigma}_{\mathrm{emp}}^{1/2}\mathbf{o}_{\mathrm{frame},n}\mathbf{o}_{\mathrm{frame},n}^{\mathsf{T}}\boldsymbol{\sigma}_{\mathrm{emp}}^{1/2} = \boldsymbol{\sigma}_{\mathrm{emp}}, \\ \max_{i\in[n]} \|\mathbf{x}_{\circ,i}\| &\leq \sqrt{2\lambda_{\max}(\boldsymbol{\sigma}_{\mathrm{emp}})} \leq C, \end{aligned} \tag{303}$$

where Eq. (303) follows from the compact shell family. Lemma 11 therefore permits conditional aggregate probabilities to be evaluated at this representative even if the original block has large individual coordinates. Replace the original block by $\mathbf{x}_\circ^n$ and relabel this representative as $\mathbf{x}^n$ for the following conditional calculation. In particular, its columns satisfy (303).

Appendix J
Proof of Lemma 12

Fix a pair $(\boldsymbol{\sigma}, \mathbf{k})$ in (96) and a block $\mathbf{x}^n \in \mathcal{T}_n(\boldsymbol{\sigma})$. Use the bounded representative from Appendix I, and retain the posterior distribution $Q(\cdot \mid \mathbf{x}^n)$ and its statistics defined there.

For each $i \in [n]$, define the centered vector $\mathbf{T}_i \in \mathbb{R}^3$ by

$$\mathbf{T}_i := \begin{bmatrix} \imath_{\boldsymbol{\sigma},\mathbf{k}}(\mathbf{U}_i; \mathbf{x}_i) \\ ([\mathbf{x}_i]_1 - (\mathbf{b}\mathbf{U}_i)_1)^2 \\ ([\mathbf{x}_i]_2 - (\mathbf{b}\mathbf{U}_i)_2)^2 \end{bmatrix} - \mathsf{E}_Q\left[\begin{bmatrix} \imath_{\boldsymbol{\sigma},\mathbf{k}}(\mathbf{U}_i; \mathbf{x}_i) \\ ([\mathbf{x}_i]_1 - (\mathbf{b}\mathbf{U}_i)_1)^2 \\ ([\mathbf{x}_i]_2 - (\mathbf{b}\mathbf{U}_i)_2)^2 \end{bmatrix} \middle| \mathbf{x}_i \right].$$

These vectors are independent, not necessarily identically distributed. At the bounded representative, they are quadratic polynomials of standard Gaussian innovations with uniformly bounded coefficients. Consequently,

$$\sup_{i\in[n]} \mathsf{E}_Q[\|\mathbf{T}_i\|^3 \mid \mathbf{x}_i] \leq C. \tag{304}$$

Lemma 12. *There are $c > 0$ and $n_0 \in \mathbb{N}$ such that, for all $n \geq n_0$,*

$$Q\Big\{ \Big\| \sum\nolimits_{i\in[n]} \mathbf{T}_i \Big\| \leq \frac{\sqrt{n}}{64 \log n} \;\Big|\; \mathbf{x}^n \Big\} \geq \frac{c}{(\log n)^3},$$

uniformly over (96) *and $\mathbf{x}^n \in \mathcal{T}_n(\boldsymbol{\sigma})$.*

The argument separates variance directions: after whitening, the large-variance subspace is handled by a multivariate Berry–Esseen bound, whereas the small-variance subspace stays inside the target window with high probability by a second-moment estimate. By Lemma 11, use the bounded representative. Let $\boldsymbol{\gamma}_n := \sum_{i\in[n]} \mathsf{Cov}_Q(\mathbf{T}_i \mid \mathbf{x}_i) \in \mathbb{S}_+^3$, the covariance of the centered sum. Equation (304) implies $\lambda_{\max}(\boldsymbol{\gamma}_n) \leq \mathrm{tr}(\boldsymbol{\gamma}_n) \leq Cn$. Set $\tau_n := n/(\log n)^8$ and $a_n := \sqrt{n}/(64 \log n)$. Let $(\mathbf{p}_{\mathrm{L}}, \mathbf{p}_{\mathrm{S}}) \in (\mathbb{S}_+^3)^2$ be the orthogonal projections onto the eigenspaces of $\boldsymbol{\gamma}_n$ with eigenvalues at least τ_n and below it, respectively. Markov's inequality implies

$$\begin{aligned} Q\Big\{ \Big\| \mathbf{p}_{\mathrm{S}} \sum\nolimits_{i\in[n]} \mathbf{T}_i \Big\| > a_n/2 \;\Big|\; \mathbf{x}^n \Big\} &\leq \frac{4\,\mathrm{tr}(\mathbf{p}_{\mathrm{S}}\boldsymbol{\gamma}_n)}{a_n^2}, \\ &\leq \frac{12\tau_n}{a_n^2} \leq \frac{C}{(\log n)^6}, \end{aligned} \tag{305}$$

where Eq. (305) follows since the trace contains at most three eigenvalues smaller than τ_n. Let $q := \mathrm{rank}(\mathbf{p}_{\mathrm{L}})$. If $q = 0$, this proves the lemma.

Otherwise $q \in [3]$. Choose an orthonormal basis matrix $\mathbf{o}_{\mathrm{L}} \in \mathbb{R}^{3\times q}$ for the large-variance subspace and define $\tilde{\boldsymbol{\gamma}}_{\mathrm{L}} := \mathbf{o}_{\mathrm{L}}^{\mathsf{T}}\boldsymbol{\gamma}_n\mathbf{o}_{\mathrm{L}} \in \mathbb{S}_{++}^q$ and $\mathbf{Z}_i' := \tilde{\boldsymbol{\gamma}}_{\mathrm{L}}^{-1/2}\mathbf{o}_{\mathrm{L}}^{\mathsf{T}}\mathbf{T}_i \in \mathbb{R}^q$, $i \in [n]$. Then

$$\mathsf{Cov}_Q\Big(\sum\nolimits_{i\in[n]} \mathbf{Z}_i' \;\Big|\; \mathbf{x}^n \Big) = \mathbf{1}_q, \tag{306}$$

$$\begin{aligned} \sum_{i\in[n]} \mathsf{E}_Q[\|\mathbf{Z}_i'\|^3 \mid \mathbf{x}^n] &\leq \tau_n^{-3/2} \sum_{i\in[n]} \mathsf{E}_Q[\|\mathbf{T}_i\|^3 \mid \mathbf{x}_i], \\ &\leq Cn\tau_n^{-3/2} = C(\log n)^{12}/\sqrt{n}, \end{aligned} \tag{307}$$

where Eq. (306) follows from whitening, and Eq. (307) follows since $\lambda_{\min}(\tilde{\boldsymbol{\gamma}}_{\mathrm{L}}) \geq \tau_n$. Bentkus' theorem for independent summands [16] therefore bounds the error on every convex Borel set by the last expression, uniformly since $q \leq 3$.

Choose $d_0 > 0$ independent of the parameters so that the inverse whitening map sends a ball of radius $d_0/\log n$ into one of radius $a_n/2$. This is possible because $\lambda_{\max}(\tilde{\gamma}_{\mathrm{L}}) \le Cn$. A standard Gaussian vector $\mathbf{N}_q \in \mathbb{R}^q$ with distribution $\mathcal{N}(\mathbf{0}_q, \mathbf{1}_q)$ assigns that ball probability at least $c(\log n)^{-q} \ge c(\log n)^{-3}$, by a lower bound on its density in the unit ball. Hence, we have

$$\begin{aligned} Q\Big\{ \Big\| \mathbf{p}_{\mathrm{L}} \sum\nolimits_{i\in[n]} \mathbf{T}_i \Big\| \le a_n/2 \;\Big|\; \mathbf{x}^n \Big\} &\ge \Pr\{\|\mathbf{N}_q\| \le d_0/\log n\} - C(\log n)^{12}/\sqrt{n}, \\ &\ge c'(\log n)^{-3}, \end{aligned} \tag{308}$$

and

$$\begin{aligned} Q\Big\{ \Big\| \sum\nolimits_{i\in[n]} \mathbf{T}_i \Big\| \le a_n \;\Big|\; \mathbf{x}^n \Big\} &\ge c'(\log n)^{-3} - C(\log n)^{-6}, \\ &\ge c''(\log n)^{-3}, \end{aligned} \tag{309}$$

where Eq. (308) follows from applying the convex-set Berry–Esseen estimate and $(\log n)^{12}/\sqrt{n} = o((\log n)^{-3})$. Eq. (309) follows from the triangle inequality and subtracting the small-component failure bound (305). No independence between the projections is required.

Appendix K
Proof of Lemma 4

We first verify the covariance margins used by the optimizer and covering bounds. Fix $\boldsymbol{\sigma} = [\sigma_{jk}] \in \mathcal{K}$. The Frobenius norm gives $|\sigma_{jk} - [\boldsymbol{\sigma}_{XY}]_{jk}| \le \|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\| \le 2\eta_{\mathrm{cov}}$ for every entry, and the operator norm satisfies $\|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\|_{\mathrm{op}} \le \|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\|$. Weyl's perturbation bound [32, Chapter III] therefore gives $\lambda_{\min}(\boldsymbol{\sigma}) \ge \lambda_{\min}(\boldsymbol{\sigma}_{XY}) - 2\eta_{\mathrm{cov}}$. This bound also follows directly by minimizing $\mathbf{u}^{\mathsf{T}}\boldsymbol{\sigma}\mathbf{u} \ge \mathbf{u}^{\mathsf{T}}\boldsymbol{\sigma}_{XY}\mathbf{u} - \|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\|_{\mathrm{op}}$ over unit vectors $\mathbf{u} \in \mathbb{R}^2$. By (68), $2\eta_{\mathrm{cov}} \le \lambda_{\min}(\boldsymbol{\sigma}_{XY})/4$, so

$$\lambda_{\min}(\boldsymbol{\sigma}) \ge \lambda_{\min}(\boldsymbol{\sigma}_{XY}) - 2\eta_{\mathrm{cov}} \ge \frac{3}{4}\lambda_{\min}(\boldsymbol{\sigma}_{XY}) > 0. \tag{310}$$

For each $j \in [2]$, the entry bound and $2\eta_{\mathrm{cov}} \le ([\boldsymbol{\sigma}_{XY}]_{jj} - D_j)/4$ similarly yield

$$\sigma_{jj} - D_j \ge [\boldsymbol{\sigma}_{XY}]_{jj} - D_j - 2\eta_{\mathrm{cov}} \ge \frac{3}{4}([\boldsymbol{\sigma}_{XY}]_{jj} - D_j) > 0. \tag{311}$$

Thus the factor $3/4$ in both bounds is $1 - 1/4$ from the radius choice. The spectral margin and compactness supply uniform constants (m, M) for the source covariance and permit a strictly feasible residual $\delta\mathbf{1}_2$. The diagonal margin keeps the Gaussian marginal rate-distortion formulas (22)–(23) valid throughout $\mathcal{K}$.

Every covariance on the segment from $\boldsymbol{\sigma}_{XY}$ to $\boldsymbol{\sigma}$ is positive definite with diagonal entries at least $(\sigma_X^2/2, \sigma_Y^2/2)$. On this segment, the absolute derivatives of $\rho(\boldsymbol{\sigma}) = \sigma_{12}/\sqrt{\sigma_{11}\sigma_{22}}$ with respect to the two diagonal entries are bounded by $(\sigma_X^{-2}, \sigma_Y^{-2})$, and the derivative with respect to σ_{12} is bounded by $2/(\sigma_X\sigma_Y)$. Applying the mean-value bound to the three independent entries gives

$$|\rho(\boldsymbol{\sigma}) - \rho| \le c_{\mathrm{cor}} \|\boldsymbol{\sigma} - \boldsymbol{\sigma}_{XY}\| \le 2c_{\mathrm{cor}}\eta_{\mathrm{cov}} \le \frac{1}{4}\min\{\rho, 1-\rho\}. \tag{312}$$

Consequently,

$$\rho/2 \le \rho(\boldsymbol{\sigma}) \le (1+\rho)/2. \tag{313}$$

This compact correlation interval keeps every center in the domain of (35) and supplies the uniform correlation bounds used in the zero-common-rate expansion and the conditional cell converse.

The covariance $\delta\mathbf{1}_2$ is feasible for every nonnegative private-rate pair. Maximizing the determinant over the compact feasible set provides an optimizer $\mathbf{k}^*$ satisfying

$$|\mathbf{k}^*| \ge \delta^2 > 0. \tag{314}$$

Since $\mathbf{k}^* \preceq \boldsymbol{\sigma} \preceq M\mathbf{1}_2$, it follows that

$$\lambda_{\min}(\mathbf{k}^*) = \frac{|\mathbf{k}^*|}{\lambda_{\max}(\mathbf{k}^*)} \ge \frac{\delta^2}{M}. \tag{315}$$

Our choice of κ proves (97). Since the source determinant is fixed, maximizing $|\mathbf{k}|$ also minimizes the determinant ratio $\frac{1}{2}\log(|\boldsymbol{\sigma}|/|\mathbf{k}|)$ over the covariance constraints. It remains to prove the simultaneous covering assertion for every pair in $\mathcal{F}_\kappa$. We use the uniform conditional small-ball probability from Lemma 12, a finite net of the entire shell, and the private coverings (85) to establish (98)–(100).

We also evaluate and verify this optimizer explicitly. Fix a covariance $\boldsymbol{\sigma} = [\sigma_{jk}] \in \mathcal{K}$ and private rates $(r_1, r_2) \in \mathbb{R}^2_{+,0}$. Write $\rho_{\boldsymbol{\sigma}} := \rho(\boldsymbol{\sigma})$, $\alpha := \alpha(\boldsymbol{\sigma}; r_1)$, and $\beta := \beta(\boldsymbol{\sigma}; r_2)$. Define the diagonal scaling matrix $\mathbf{t}_{\boldsymbol{\sigma}} \in \mathbb{S}^2_{++}$ and the normalized marginal

variances by $\mathbf{t}_{\boldsymbol{\sigma}} := \operatorname{diag}(\sqrt{\sigma_{11}}, \sqrt{\sigma_{22}})$, $v_1 := e^{-2\alpha} = \min\{1, D_1 e^{2r_1}/\sigma_{11}\}$, and $v_2 := e^{-2\beta} = \min\{1, D_2 e^{2r_2}/\sigma_{22}\}$. Here $\nu_\alpha = \sqrt{1-v_1}$ and $\nu_\beta = \sqrt{1-v_2}$ retain their definitions in (27). Let $\tilde{\mathbf{k}}^* \in \mathbb{S}^2_{++}$ be the normalized residual matrix specified below. The unique optimizer of the determinant maximization (78), with the covariance and caps fixed above, is

$$\mathbf{k}^*(\boldsymbol{\sigma}; r_1, r_2) = \mathbf{t}_{\boldsymbol{\sigma}} \tilde{\mathbf{k}}^* \mathbf{t}_{\boldsymbol{\sigma}}, \tag{316}$$

where the branches below use correlation $\rho_{\boldsymbol{\sigma}}$:

$$\tilde{\mathbf{k}}^* = \begin{cases} \begin{bmatrix} v_1 & \rho_{\boldsymbol{\sigma}} - \nu_\alpha \nu_\beta \\ \rho_{\boldsymbol{\sigma}} - \nu_\alpha \nu_\beta & v_2 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_1, \\ \begin{bmatrix} v_1 & 0 \\ 0 & v_2 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_2, \\ \begin{bmatrix} v_1 & \rho_{\boldsymbol{\sigma}} v_1 \\ \rho_{\boldsymbol{\sigma}} v_1 & 1 - \rho_{\boldsymbol{\sigma}}^2 + \rho_{\boldsymbol{\sigma}}^2 v_1 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_3, \\ \begin{bmatrix} 1 - \rho_{\boldsymbol{\sigma}}^2 + \rho_{\boldsymbol{\sigma}}^2 v_2 & \rho_{\boldsymbol{\sigma}} v_2 \\ \rho_{\boldsymbol{\sigma}} v_2 & v_2 \end{bmatrix}, & \text{if } (\alpha, \beta) \in \mathcal{P}_4 \end{cases}. \tag{317}$$

The expressions agree on shared interfaces and extend continuously to the axes. At $(\alpha, \beta) = (0, 0)$, take $\mathbf{k}^* = \boldsymbol{\sigma}$. To verify the formulas, let $\tilde{\boldsymbol{\sigma}} := \mathbf{t}_{\boldsymbol{\sigma}}^{-1} \boldsymbol{\sigma} \mathbf{t}_{\boldsymbol{\sigma}}^{-1}$. For the candidates in (317), the covariance difference on $\mathcal{P}_1$ is

$$\tilde{\boldsymbol{\sigma}} - \tilde{\mathbf{k}}^* = \begin{bmatrix} \nu_\alpha \\ \nu_\beta \end{bmatrix} \begin{bmatrix} \nu_\alpha & \nu_\beta \end{bmatrix} \succeq 0.$$

On $\mathcal{P}_3$, it is

$$\tilde{\boldsymbol{\sigma}} - \tilde{\mathbf{k}}^* = (1 - v_1) \begin{bmatrix} 1 \\ \rho_{\boldsymbol{\sigma}} \end{bmatrix} \begin{bmatrix} 1 & \rho_{\boldsymbol{\sigma}} \end{bmatrix} \succeq 0,$$

where both identities follow by subtraction. On $\mathcal{P}_2$, the difference has nonnegative diagonal entries and determinant $\nu_\alpha^2 \nu_\beta^2 - \rho_{\boldsymbol{\sigma}}^2 \geq 0$. On $\mathcal{P}_3$, its second private cap follows from $1 - \rho_{\boldsymbol{\sigma}}^2 (1 - v_1) \leq v_2$. The fourth branch is symmetric. The residual determinants are

$$|\tilde{\mathbf{k}}^*| = \begin{cases} v_1 v_2 (1 - r_{\alpha,\beta}^2), & \text{if } (\alpha, \beta) \in \mathcal{P}_1, \\ v_1 v_2, & \text{if } (\alpha, \beta) \in \mathcal{P}_2, \\ v_1 (1 - \rho_{\boldsymbol{\sigma}}^2), & \text{if } (\alpha, \beta) \in \mathcal{P}_3, \\ v_2 (1 - \rho_{\boldsymbol{\sigma}}^2), & \text{if } (\alpha, \beta) \in \mathcal{P}_4 \end{cases}. \tag{318}$$

These are positive, verifying positive definiteness and all caps. Finally,

$$\begin{aligned} \frac{1}{2} \log \frac{|\boldsymbol{\sigma}|}{|\mathbf{k}^*|} &= \frac{1}{2} \log \frac{1 - \rho_{\boldsymbol{\sigma}}^2}{|\tilde{\mathbf{k}}^*|}, \\ &= \Upsilon_{\rho_{\boldsymbol{\sigma}}}(\alpha, \beta). \end{aligned} \tag{319}$$

Equation (319) follows from determinant scaling and (318). For any feasible $\mathbf{k}$, take independent centered Gaussian vectors $(\mathbf{W}, \mathbf{N})$ with covariances $(\boldsymbol{\sigma} - \mathbf{k}, \mathbf{k})$ and set $(X, Y)^{\mathsf{T}} = \mathbf{W} + \mathbf{N}$. Their common mutual information is $\frac{1}{2} \log(|\boldsymbol{\sigma}|/|\mathbf{k}|)$. Moreover, $I(X; \mathbf{W}) = \frac{1}{2} \log(\sigma_{11}/k_{11}) \geq \alpha$ and $I(Y; \mathbf{W}) = \frac{1}{2} \log(\sigma_{22}/k_{22}) \geq \beta$ by the private caps and $\mathbf{k} \preceq \boldsymbol{\sigma}$. Yu's unrestricted envelope [13, Theorem 3] therefore gives $\frac{1}{2} \log(|\boldsymbol{\sigma}|/|\mathbf{k}|) \geq \Upsilon_{\rho_{\boldsymbol{\sigma}}}(\alpha, \beta)$. The displayed candidates attain this lower bound and therefore maximize the determinant. Strict concavity of $\log |\mathbf{k}|$ on the convex positive-definite feasible set proves uniqueness. The candidates agree on interfaces and extend to the axes. At zero marginal requirements, $\mathbf{k}^* = \boldsymbol{\sigma}$.

Fix the distortions, shell width, and compact covariance bounds in Lemma 4. All constants in this proof may depend on those fixed quantities, but not on n, the selected pair $(\boldsymbol{\sigma}, \mathbf{k})$, or the sequence in its shell. We analyze the one-shell code defined in Section IV-B, with common size M_0 specified by (82). Define its normalized common-codebook distribution by

$$Q_{0,n} := \left(\mathcal{N}(\mathbf{0}_2, \mathbf{1}_2)^{\otimes n} \right)^{\otimes M_0}. \tag{320}$$

The private codebooks and decoding functions are fixed as in that construction. The auxiliary posterior from Appendix I is an analysis distribution, not an extra random operation performed by the encoder. Probabilities under P_{XY} refer only to the actual source.

Recall the fixed codebooks in Section IV-B, the local maps in (92) and (94), and the global maps in (93) and (95). For one covariance pair, suppress q and write $(\mathcal{A}_{0,n}(\mathbf{x}^n), \mathcal{A}_{j,n}(\mathbf{x}^n, m_0), \mathcal{H}_n)$ for the candidate sets and eligible domain in Section IV-B. Their finite norm tests and smallest-index rules define the Borel encoder (92). The whole-shell argument in this appendix proves $\mathcal{T}_n(\boldsymbol{\sigma}) \subseteq \mathcal{H}_n$ for the selected common-codebook realization.

The private covering constant in (86) is uniform because, for every $n \geq 3$ and $j \in [2]$,

$$\frac{\xi_{j,n}}{\sqrt{nD_j}} \leq \sqrt{\frac{\kappa^{-1} + \chi_\kappa}{\min\{D_1, D_2\}}}.$$

Thus Lemma 10 applies with one radius-ratio bound, including when a private first-order rate is zero. Translating the same offsets by different common codewords preserves their cardinalities.

For the global construction, the cells (75) satisfy $\mathcal{T}_{n,q} \subseteq \mathcal{T}_n(\boldsymbol{\sigma}_q)$ by (73). The common alphabet (89) carries the pair (q, m_0), so neither private alphabet needs a cell index. The encoder (93) and the fallback convention following (95) specify the code on every other input and every unused padded private index. The finite cell partition and Borel local maps therefore define a Borel global encoder.

For a fixed shell block, define the analysis-only set

$$\mathcal{G}_n(\mathbf{x}^n) := \big\{\mathbf{u}^n \in \mathbb{R}^{2\times n} :\ j_n(\mathbf{u}^n; \mathbf{x}^n) \leq nI_{\boldsymbol{\sigma},\mathbf{k}} + s_n/16,\ e_{j,n}(\mathbf{u}^n; \mathbf{x}^n) \leq nk_{jj} + s_n/16,\ j \in [2]\big\}.$$

The bounded mean shifts in (300) and Lemma 12 imply $Q\{\mathcal{G}_n(\mathbf{x}^n) \mid \mathbf{x}^n\} \geq c/(\log n)^3$ for large n. The posterior and $Q^n_{\mathbf{U}}$ are mutually absolutely continuous by (297). Bayes' formula therefore yields

$$\begin{aligned} Q^n_{\mathbf{U}}\{\mathcal{G}_n(\mathbf{x}^n)\} &= \mathsf{E}_Q[e^{-J_n}\mathbb{1}_{\mathcal{G}_n(\mathbf{x}^n)} \mid \mathbf{x}^n], \\ &\geq \exp\big(-nI_{\boldsymbol{\sigma},\mathbf{k}} - s_n/16\big) Q\{\mathcal{G}_n(\mathbf{x}^n) \mid \mathbf{x}^n\}, \\ &\geq \frac{c}{(\log n)^3}\exp\big(-nI_{\boldsymbol{\sigma},\mathbf{k}} - s_n/16\big) =: p_n, \end{aligned} \tag{321}$$

where Eq. (321) follows from $dQ^n_{\mathbf{U}}/dQ_{\mathbf{U}^n|\mathbf{x}^n} = e^{-J_n}$, the information-density cutoff, and the conditional small-ball bound.

For the independent codebook draws in (320), let $\mathcal{E}^{(0)}_n(\mathbf{x}^n) := \bigcap_{m_0\in[M_0]}\{\mathbf{U}^n(m_0) \notin \mathcal{G}_n(\mathbf{x}^n)\}$. Take a maximal δ_n-separated subset $\mathcal{N}_n$ of the shell, with $\delta_n := 1/\log n$. It is a finite net with centers in the shell. Since the shell lies in a radius-$C\sqrt{n}$ ball in $\mathbb{R}^{2n}$, a disjoint-ball volume comparison implies

$$|\mathcal{N}_n| \leq (1 + C\sqrt{n}\log n)^{2n}, \tag{322}$$

$$Q_{0,n}(\mathcal{E}^{(0)}_n(\mathbf{x}^n)) \leq (1-p_n)^{M_0} \leq \exp\big(-M_0 p_n\big) \leq \exp\{-c(\log n)^{-3}e^{s_n/16}\}, \tag{323}$$

where Eq. (323) follows from independence and the common codebook size (82). For $\mathcal{E}^{(1)}_n := \bigcup_{\mathbf{x}^n\in\mathcal{N}_n}\mathcal{E}^{(0)}_n(\mathbf{x}^n)$,

$$\begin{aligned} Q_{0,n}(\mathcal{E}^{(1)}_n) &\leq |\mathcal{N}_n|\exp\{-c(\log n)^{-3}e^{s_n/16}\}, \\ &\leq \exp\{Cn\log n - c(\log n)^{-3}e^{s_n/16}\}, \end{aligned} \tag{324}$$

where Eq. (324) follows from the union bound and (322). Since $e^{s_n/16}/((\log n)^3 n\log n) \to \infty$, the right-hand side tends to zero. Fix a common codebook good for every net point.

For an arbitrary shell block $\mathbf{x}^n$, take a nearest net point $\bar{\mathbf{x}}^n$ and a codeword good for it. Then, for $j \in [2]$,

$$\|(\mathbf{x}^n)_j - (\mathbf{b}\mathbf{u}^n)_j\|^2 \leq \left(\sqrt{nk_{jj} + s_n/16} + \delta_n\right)^2 \leq nk_{jj} + \chi_\kappa s_n = \xi^2_{j,n}, \tag{325}$$

where Eq. (325) follows from the triangle inequality, $\delta_n\sqrt{n} = s_n$, $k_{jj} \leq \kappa^{-1}$, and the fixed margin (81). Only the residual test, not the information-density test, is needed away from net points. The previously specified private coverings therefore ensure both distortion constraints for the encoder and decoders of (92) and (94).

Their cardinalities from Lemma 10 obey

$$\begin{aligned} \log M_j &\leq \log C_{\mathrm{cov}} + 4\log n + \frac{n}{2}\left[\log\frac{k_{jj} + \chi_\kappa s_n/n}{D_j}\right]^+, \\ &\leq \frac{n}{2}[\log(k_{jj}/D_j)]^+ + \frac{n}{2}\log\left(1 + \frac{\chi_\kappa s_n}{nk_{jj}}\right) + C\log n, \\ &\leq \frac{n}{2}[\log(k_{jj}/D_j)]^+ + Cs_n + C\log n, \end{aligned} \tag{326}$$

where Eq. (326) follows from the positive-part inequality, $\log(1+x) \leq x$, and $k_{jj} \geq \kappa$. Together with (82), this proves Lemma 4. All constants are uniform, including when $\boldsymbol{\sigma} - \mathbf{k}$ loses rank.

APPENDIX L
PROOF OF LEMMA 13

Fix a compact $\mathcal{K} \subset \mathbb{S}^2_{++}$ for the following cell estimates.

Lemma 13. *There are constants $(c, C) \in \mathbb{R}^2_+$ with $\min\{c, C\} > 0$ and $n_0 \in \mathbb{N}$ such that, for $n \geq n_0$ and every $\boldsymbol{\sigma} \in \mathcal{K}$,*

$$(Q^{\boldsymbol{\sigma}})^n\{\mathcal{C}_n(\boldsymbol{\sigma})\} \geq cn^{-3/2}, \tag{327}$$

$$\sup_{(\mathbf{x}^n,\mathbf{y}^n)\in\mathcal{C}_n(\boldsymbol{\sigma})^2} \left|\log \frac{dP^n_{XY}}{d(Q^{\boldsymbol{\sigma}})^n}(\mathbf{x}^n) - \log \frac{dP^n_{XY}}{d(Q^{\boldsymbol{\sigma}})^n}(\mathbf{y}^n)\right| \leq C. \tag{328}$$

Let Γ be Euler's gamma function and $\Gamma_2(a) := \sqrt{\pi}\Gamma(a)\Gamma(a-1/2)$ for $a > 1/2$. Under $(Q^{\boldsymbol{\sigma}})^n$, the random matrix $n\hat{\boldsymbol{\Sigma}}_n$ has the following Wishart density, evaluated at $\mathbf{a} \in \mathbb{S}^2_{++}$, with respect to its three independent entries:

$$f_{n,\boldsymbol{\sigma}}(\mathbf{a}) := \frac{|\mathbf{a}|^{(n-3)/2}\exp\bigl(-\operatorname{tr}(\boldsymbol{\sigma}^{-1}\mathbf{a})/2\bigr)}{2^n|\boldsymbol{\sigma}|^{n/2}\Gamma_2(n/2)}.$$

For bounded $\mathbf{h} \in \mathbb{S}^2$ and $\boldsymbol{\sigma} \in \mathcal{K}$,

$$\log|n\boldsymbol{\sigma} + \mathbf{h}| = 2\log n + \log|\boldsymbol{\sigma}| + n^{-1}\operatorname{tr}(\boldsymbol{\sigma}^{-1}\mathbf{h}) + O(n^{-2}), \tag{329}$$

$$\log f_{n,\boldsymbol{\sigma}}(n\boldsymbol{\sigma} + \mathbf{h}) = -\tfrac{3}{2}\log n + O(1), \tag{330}$$

where Eq. (329) follows from the uniform log-determinant expansion and Eq. (330) follows from substituting it into the density and applying Stirling's formula. Compact positive definiteness makes the constants uniform and keeps $n\boldsymbol{\sigma} + \mathbf{h}$ positive definite for large n. In these coordinates the cell contains a fixed positive-volume box of $\mathbf{h}$ values. Integrating the density lower bound over this box establishes (327). Half-open boundaries have zero mass.

For the true source, division of Gaussian densities leads to

$$\log \frac{dP^n_{XY}}{d(Q^{\boldsymbol{\sigma}})^n}(\mathbf{x}^n) = \frac{n}{2}\log\frac{|\boldsymbol{\sigma}|}{|\boldsymbol{\sigma}_{XY}|} - \frac{n}{2}\operatorname{tr}[(\boldsymbol{\sigma}_{XY}^{-1} - \boldsymbol{\sigma}^{-1})\hat{\boldsymbol{\sigma}}(\mathbf{x}^n)],$$

$$\left|\log \frac{dP^n_{XY}}{d(Q^{\boldsymbol{\sigma}})^n}(\mathbf{x}^n) - \log \frac{dP^n_{XY}}{d(Q^{\boldsymbol{\sigma}})^n}(\mathbf{y}^n)\right| \leq \frac{n}{2}\left\|\boldsymbol{\sigma}_{XY}^{-1} - \boldsymbol{\sigma}^{-1}\right\|\left\|\hat{\boldsymbol{\sigma}}(\mathbf{x}^n) - \hat{\boldsymbol{\sigma}}(\mathbf{y}^n)\right\| \leq C, \tag{331}$$

where Eq. (331) follows from the $O(1/n)$ cell diameter and compactness for $(\mathbf{x}^n, \mathbf{y}^n) \in \mathcal{C}_n(\boldsymbol{\sigma})^2$. This proves (328).

APPENDIX M
PROOF OF LEMMA 14

Fix $D > 0$ and a compact interval $\mathcal{I} \subset (D, \infty)$ for the marginal image estimate.

Lemma 14. *For some $C < \infty$ and all sufficiently large n,*

$$\mathcal{N}(0, s\mathbf{1}_n)\bigl(\mathcal{B}_n(\hat{x}^n, \sqrt{nD}) \cap \mathcal{S}_n(s)\bigr) \leq n^C \exp\{-\tfrac{n}{2}\log(s/D)\}, \tag{332}$$

uniformly for $s \in \mathcal{I}$ and every center $\hat{x}^n \in \mathbb{R}^n$.

Write a vector under $\mathcal{N}(0, s\mathbf{1}_n)$ as $X^n = R\boldsymbol{\Theta}$, where $R \in \mathbb{R}_{+,0}$ and $\boldsymbol{\Theta} \in \{\mathbf{v} \in \mathbb{R}^n : \|\mathbf{v}\| = 1\}$ are independent and $\boldsymbol{\Theta}$ is uniform on the unit sphere. For a center of norm $\sqrt{n}\,t$ with $t > 0$ and conditional power $u := R^2/n > D$, the distortion ball restricts the angle by

$$\cos\theta \geq \frac{u + t^2 - D}{2t\sqrt{u}},$$

$$\min_{t>0}\frac{u + t^2 - D}{2t\sqrt{u}} = \sqrt{1 - D/u}, \tag{333}$$

where Eq. (333) follows since the minimum is attained at $t^2 = u - D$. For $t = 0$ or a cutoff exceeding one, the intersection is empty. The first coordinate of $\boldsymbol{\Theta}$ has density $\omega_n(1-v^2)^{(n-3)/2}$, with $\omega_n = \Gamma(n/2)/(\sqrt{\pi}\Gamma((n-1)/2)) \leq C\sqrt{n}$. For $z > 0$,

$$\int_z^1 (1-v^2)^{(n-3)/2}\,dv = \frac{1}{2}\int_0^{1-z^2} w^{(n-3)/2}(1-w)^{-1/2}\,dw,$$

$$\leq \frac{(1-z^2)^{(n-1)/2}}{z(n-1)}, \tag{334}$$

where Eq. (334) follows from substituting $w = 1 - v^2$ and using $\sqrt{1-w} \geq z$. On the power shell, $u = s + O(1/n)$ uniformly. Compactness of $\mathcal{I} \subset (D, \infty)$ keeps $z = \sqrt{1 - D/u}$ bounded away from zero. Hence

$$\begin{aligned} \Pr\{X^n \in \mathcal{B}_n(\hat{x}^n, \sqrt{nD}) \mid R\} &\leq Cn^{-1/2}(D/u)^{(n-1)/2}, \\ &\leq C'(D/s)^{n/2}, \end{aligned} \tag{335}$$

where Eq. (335) follows from the cap bound and $(s/u)^{(n-1)/2} = O(1)$ on the thin shell. Integrating over the shell radii, whose probability is at most one, proves (332) uniformly in s and the center.

Appendix N
Proof of Lemma 9

We prove Lemma 9. Lemmas 13 and 14 bound the covariance-cell mass, likelihood-ratio oscillation, and private decoding images. We combine these estimates using Gaussian hypercontractivity. Finally, summing over common messages and changing the source distribution leads to the conditional joint-success bound. For a fixed code, $m_0 \in [M_{0,n}]$, each $j \in [2]$, and $s \in \mathbb{R}$, define

$$\begin{aligned} \mathcal{A}_{n,m_0} &:= \mathcal{A}_n \cap f_0^{-1}(\{m_0\}), \\ \mathcal{A}_{j,m_0} &:= \bigcup_{m_j \in [M_{j,n}]} \mathcal{B}_n(\phi_j(m_0, m_j), \sqrt{nD_j}), \\ \mathcal{S}_n(s) &:= \{x^n \in \mathbb{R}^n : |n^{-1}\|x^n\|^2 - s| \leq 1/(2n)\}. \end{aligned}$$

The first set fixes the common index of successful blocks. The second is the private decoding image. Membership in $\mathcal{C}_n(\boldsymbol{\sigma})$ requires the jth row to lie in $\mathcal{S}_n(\sigma_{jj})$.

We now combine the two auxiliary estimates. Fix a code, $\boldsymbol{\sigma} \in \mathcal{K}$, and n sufficiently large. For each $j \in [2]$, put $Q_j^{\boldsymbol{\sigma}} := \mathcal{N}(0, \sigma_{jj})$. By the union bound and Lemma 14,

$$\begin{aligned} (Q_j^{\boldsymbol{\sigma}})^n(\mathcal{A}_{j,m_0} \cap \mathcal{S}_n(\sigma_{jj})) &\leq \min\{1, M_{j,n} n^C e^{-\frac{n}{2}\log(\sigma_{jj}/D_j)}\}, \\ &\leq \exp\big(-n\delta_{j,n}(\boldsymbol{\sigma}) + C\log n\big), \end{aligned} \tag{336}$$

where Eq. (336) follows from the definition (190), including zero marginal requirements.

Let $\rho_{\boldsymbol{\sigma}} = \rho(\boldsymbol{\sigma})$ and let $P_{\mathrm{G},n} := \mathcal{N}(\mathbf{0}_n, \mathbf{1}_n)$ be the standard Gaussian distribution on $\mathbb{R}^n$. For a measurable function $h : \mathbb{R}^n \to \mathbb{R}$ and $p > 1$, write $\|h\|_p := (\int |h|^p \, dP_{\mathrm{G},n})^{1/p}$. For $\mathbf{x} \in \mathbb{R}^n$, define the Ornstein–Uhlenbeck operator by

$$(T_{\rho_{\boldsymbol{\sigma}}} h)(\mathbf{x}) := \int h\big(\rho_{\boldsymbol{\sigma}}\mathbf{x} + \sqrt{1-\rho_{\boldsymbol{\sigma}}^2}\,\mathbf{z}\big)\, P_{\mathrm{G},n}(d\mathbf{z}).$$

Recall the exponent set $\mathcal{H}_{\rho_{\boldsymbol{\sigma}}}$ from (276). For a pair of Borel subsets $(\mathcal{A}_1, \mathcal{A}_2)$ of $\mathbb{R}^n$, set $h_j(\mathbf{x}) := \mathbb{1}_{\mathcal{A}_j}(\sqrt{\sigma_{jj}}\,\mathbf{x})$, $j \in [2]$. Gaussian conditioning identifies $T_{\rho_{\boldsymbol{\sigma}}} h_2$ with the conditional expectation of the second standardized indicator given the first standardized source. For $(p, q) \in \mathcal{H}_{\rho_{\boldsymbol{\sigma}}}$, define $p' := p/(p-1)$. If $q \leq p'$, then $1 < q \leq p' < \infty$ and $\rho_{\boldsymbol{\sigma}}^2 \leq (q-1)/(p'-1)$. Nelson's Gaussian hypercontractive inequality [17, Theorem 3], with the dimension extension described in [18, Corollary 4.1 and the proof of Corollary 4.2], therefore implies

$$\|T_{\rho_{\boldsymbol{\sigma}}} h\|_{p'} \leq \|h\|_q . \tag{337}$$

The exponent condition is precisely $(p-1)(q-1) \geq \rho_{\boldsymbol{\sigma}}^2$, and the constant is one in every dimension n. When $q > p'$, the same norm inequality follows from contraction in $L^q(P_{\mathrm{G},n})$ and the probability-space embedding into $L^{p'}(P_{\mathrm{G},n})$. Consequently,

$$(Q^{\boldsymbol{\sigma}})^n(\mathcal{A}_1 \times \mathcal{A}_2) = \mathsf{E}_{P_{\mathrm{G},n}}[h_1 T_{\rho_{\boldsymbol{\sigma}}} h_2] \leq \|h_1\|_p \, \|T_{\rho_{\boldsymbol{\sigma}}} h_2\|_{p'} , \tag{338}$$

$$\leq \|h_1\|_p \, \|h_2\|_q = (Q_1^{\boldsymbol{\sigma}})^n(\mathcal{A}_1)^{1/p}(Q_2^{\boldsymbol{\sigma}})^n(\mathcal{A}_2)^{1/q}, \tag{339}$$

where Eq. (338) follows from Hölder's inequality, and Eq. (339) follows from (337) or the preceding contraction argument. Endpoint contributions are obtained by limits, never by setting $p = 1$.

Successful decoding with common index m_0 inside a cell implies membership in both marginal images and power shells. Therefore

$$\begin{aligned} (Q^{\boldsymbol{\sigma}})^n(\mathcal{A}_{n,m_0} \cap \mathcal{C}_n(\boldsymbol{\sigma})) &\leq (Q^{\boldsymbol{\sigma}})^n\big((\mathcal{A}_{1,m_0} \cap \mathcal{S}_n(\sigma_{11})) \times (\mathcal{A}_{2,m_0} \cap \mathcal{S}_n(\sigma_{22}))\big), \\ &\leq \inf_{(p,q)\in\mathcal{H}_{\rho_{\boldsymbol{\sigma}}}} \exp\big(-n(\delta_{1,n}(\boldsymbol{\sigma})/p + \delta_{2,n}(\boldsymbol{\sigma})/q) + C\log n\big), \\ &= \exp\big(-n\psi_{\rho_{\boldsymbol{\sigma}}}(\delta_{1,n}(\boldsymbol{\sigma}), \delta_{2,n}(\boldsymbol{\sigma})) + C\log n\big), \end{aligned} \tag{340}$$

where Eq. (340) follows from combining (336) and (339) and applying (277). The polynomial factor is uniform before optimization since $1/p + 1/q \le 2$. Summing this bound over $M_{0,n}$ common messages leads to

$$(Q^{\boldsymbol{\sigma}})^n(\mathcal{A}_n \cap \mathcal{C}_n(\boldsymbol{\sigma})) \le \exp\big(\log M_{0,n} - n\psi_{\rho_{\boldsymbol{\sigma}}}(\delta_{1,n}(\boldsymbol{\sigma}), \delta_{2,n}(\boldsymbol{\sigma})) + C\log n\big). \tag{341}$$

No separation of the common encoder across sources has been assumed.

Set $\mathcal{C} := \mathcal{C}_n(\boldsymbol{\sigma})$ and $\ell := dP_{XY}^n/d(Q^{\boldsymbol{\sigma}})^n$. The cell has positive probability by Lemma 13, so

$$\begin{aligned} P_{XY}^n(\mathcal{A}_n \mid \mathcal{C}) &= \frac{\int_{\mathcal{A}_n \cap \mathcal{C}} \ell \, d(Q^{\boldsymbol{\sigma}})^n}{\int_{\mathcal{C}} \ell \, d(Q^{\boldsymbol{\sigma}})^n}, \\ &\le \frac{\sup_{\mathcal{C}} \ell}{\inf_{\mathcal{C}} \ell} \frac{(Q^{\boldsymbol{\sigma}})^n(\mathcal{A}_n \cap \mathcal{C})}{(Q^{\boldsymbol{\sigma}})^n(\mathcal{C})}, \\ &\le Cn^k \exp\big(\log M_{0,n} - n\psi_{\rho(\boldsymbol{\sigma})}(\delta_{1,n}(\boldsymbol{\sigma}), \delta_{2,n}(\boldsymbol{\sigma}))\big), \end{aligned} \tag{342}$$

where Eq. (342) follows from bounding the conditional likelihood ratio and applying (328), (327), and (341). Taking the minimum with one proves Lemma 9. Its constants were fixed independently of the code, rates, centers, and selected covariance.

Appendix O
Proof of Corollary 2

On $\mathcal{P}_1$, the slope formulas (232)–(233) and the dispersion identity (57) imply (62) by substitution into (60). On $(\mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_4)$, the derivative values in (234) are $(1,1)$, $(1,0)$, and $(0,1)$, respectively, and their dispersions are 1, $1/2$, and $1/2$. This implies (63)–(65). The zero-rate constraints in (60) are retained in every case.

At the curved/sum-rate interface $r = 0$, the curved weights equal $(1,1)$ and $V = 1$. At the curved/first-source interface $r = u$, they equal $(1,0)$ and $V = 1/2$; at the other one-source interface $r = u^{-1}$, they equal $(0,1)$ and $V = 1/2$. These values agree with the adjacent affine conditions. Lemma 2 specifies which strict one-source tuples are admissible boundary rate tuples. This completes the proof.

Appendix P
Additional Boundary Consequences

The following calculations justify the boundary discussions after Theorem 1. To check the case of zero private rates, suppose that $\boldsymbol{\sigma}_{XY} - \operatorname{diag}(D_1, D_2) \succeq 0$. At the all-common first-order tuple $\mathbf{r}^* = (R_0^*, 0, 0)$, the marginal requirements $(R_X(P_X, D_1), R_Y(P_Y, D_2))$ lie in $\mathcal{P}_2$, including its interface with $\mathcal{P}_1$. The residual covariance is $\operatorname{diag}(D_1, D_2)$, both slopes equal one, and

$$\begin{aligned} R_0^* &= \frac{1}{2}\log\frac{|\boldsymbol{\sigma}_{XY}|}{D_1 D_2}, \\ V &= 1. \end{aligned}$$

The slice $L_1 = L_2 = 0$ of (60) therefore requires $L_0 \ge \mathrm{Q}^{-1}(\varepsilon)$. Private descriptions of size $o(\sqrt{n})$ nats can be included in the common message without changing this offset, which establishes the consistency check with joint coding of (X, Y) under the two componentwise quadratic distortions.

A zero slope need not make a private description redundant. At a $(\mathcal{P}_1, \mathcal{P}_3)$ interface with $R_2^* > 0$, one has $\nu_{\beta^*} = \rho\nu_{\alpha^*}$ and $(\lambda_1^*, \lambda_2^*) = (1, 0)$. Reducing R_2^* by a small amount enters the curved region and increases the required common rate quadratically, as shown in (242). Thus reducing this coordinate leaves the rate region. A private-rate perturbation of order $n^{-1/2}$ affects the common rate by only $O(n^{-1})$ in this direction, which explains the absence of L_2 from (60). The other one-source interface is symmetric.

At $R_0^* > 0$, the eigenvalues of the symmetric matrix $\boldsymbol{\sigma}_{XY}^{1/2}\mathbf{g}\boldsymbol{\sigma}_{XY}^{1/2}$ are $1/2$ and $(\lambda_1^* + \lambda_2^* - 1)/2$. Consequently, the dispersion depends on the supporting slopes only through their sum, as in (56). This does not reduce the whole region to that sum: its rate displacement remains $L_0 + \lambda_1^* L_1 + \lambda_2^* L_2$. Nor does it imply correlation independence at a fixed rate tuple, because the slopes themselves depend on the source and the selected tuple. The matrix $\mathbf{g}$ is still needed to identify the empirical fluctuation and match the converse. Appendix D proves the eigenvalue statement, including affine interfaces and zero private rates.

There is no path-independent scalar dispersion at the corner. For sufficiently small $t > 0$, use the two nonredundant tuples from (41) with pairs of marginal requirements (t, t) and (t, β_t), where

$$\beta_t := -\tfrac{1}{2}\log(1 - \rho^2 + \rho^2 e^{-2t}).$$

Along (t,t), symmetry and $r_{t,t}=1-(1-\rho)e^{2t}$ imply that both slopes equal $1/(1+r_{t,t})$. Along the second path, $\nu_{\beta_t}=\rho\nu_t$ places the tuple on the $(\mathcal{P}_1,\mathcal{P}_3)$ interface, with slopes $(1,0)$. If $(V^{(1)}(t),V^{(2)}(t))$ denote their dispersions, then

$$\lim_{t\to 0} V^{(1)}(t)=\frac{1}{2}\left[1+\left(\frac{2}{1+\rho}-1\right)^2\right]=\frac{1+\rho^2}{(1+\rho)^2}, \tag{343}$$

$$\lim_{t\to 0} V^{(2)}(t)=\frac{1}{2}, \tag{344}$$

where Eqs. (343)–(344) follow from substituting the respective slope limits into (56). Both paths converge to $\mathbf{r}_{\mathrm{c}}^*$, but the limits differ for $0<\rho<1$. Theorem 1 fixes its operating tuple before the blocklength limit. It does not assert a uniform transition theorem for an n-dependent tuple approaching the corner.

Under the differentiability and support conditions of [11, Theorem 4], the discrete result has the same weighted-half-space form as (60). To distinguish the two operating points in this comparison only, attach a subscript D to its slopes and dispersion:

$$L_0+\lambda_{1,\mathrm{D}}^* L_1+\lambda_{2,\mathrm{D}}^* L_2\geq\sqrt{V_{\mathrm{D}}}\,\mathrm{Q}^{-1}(\varepsilon).$$

The definition of each slope is the same negative private-rate derivative. The values depend on the source. The discrete source derivative is represented by centered Gray–Wyner tilted information. Here, the covariance derivative satisfies

$$\sqrt{n}\left\langle \mathbf{g},\hat{\boldsymbol{\Sigma}}_n-\boldsymbol{\sigma}_{XY}\right\rangle=\frac{1}{\sqrt{n}}\sum_{i\in[n]}[\jmath_{XY}(X_i,Y_i)-R_0^*].$$

Lemma 4 replaces the generalized type-covering lemma in [11, Lemma 8]. Local expansion and a central limit theorem then complete the direct argument in the same order. Table I compares the centered quantities. The Gaussian auxiliary and empirical descriptor retain the domains specified in Section II-A. At the corner, the directional expansion and Lemma 9 replace the scalar reduction.

TABLE I
THE LOCAL-BOUNDARY REDUCTION AT DIFFERENTIABLE POINTS. THE DISCRETE COLUMN REFERS TO [11, THEOREM 4 AND SECTION IV-A].

Role	Discrete lossy Gray–Wyner	Gaussian lossy Gray–Wyner
Empirical descriptor	Joint type $\hat{T}_{x^n y^n}$, [11, Section II-A]	Empirical matrix $\hat{\boldsymbol{\sigma}}(\mathbf{x}^n)$, (66)
Local benchmark	Source-dependent common rate, [11, Eqs. (19)–(21)]	Covariance-dependent common rate, (36)
Uniform coding result	Generalized type covering, [11, Lemma 8]	Covariance-shell covering, Lemma 4 and (99)–(100)
Centered first variation	Centered tilted information, [11, Lemmas 2–3]	$\jmath_{XY}(x,y)-R_0^*$, (51)
Dispersion	Variance of the first variation, [11, Eq. (30)]	$V=[1+(\lambda_1^*+\lambda_2^*-1)^2]/2$, (56)
Rate offset	$L_0+\lambda_{1,\mathrm{D}}^* L_1+\lambda_{2,\mathrm{D}}^* L_2$, [11, Theorem 4]	$L_0+\lambda_1^* L_1+\lambda_2^* L_2$, (60)

At the corner, for every $(\delta_1,\delta_2)\in\mathbb{R}_{+,0}^2$, Lemma 6 establishes the uniform directional expansion underlying

$$\lim_{t\to 0}\frac{\Upsilon_\rho(t\delta_1,t\delta_2)}{t}=\psi_\rho(\delta_1,\delta_2).$$

Appendix G treats the moving interfaces and covariance perturbations in detail. The values $\psi_\rho(\delta_1,0)=\delta_1$ and $\psi_\rho(\delta_1,\delta_1)=2\delta_1/(1+\rho)$ show how a common description serves balanced and unbalanced marginal requirements differently. The correlation ρ enters this allocation cost, whereas ρ^2 is the correlation of the limiting squared-source fluctuations in (59). Consequently, the corner offset region depends only on (ρ,ε). The marginal variances and distortions determine its first-order location.

The positive-weight converse is imported from [12, Theorem 17]. Lemma 8 states its specialization after Lemma 7 identifies the support and fluctuation. A scalar Gaussian converse handles a zero weight. At the corner, separate fixed-direction bounds cannot be combined by moving a supremum inside a probability. The decisive additional statement is the finite-blocklength conditional-success bound (191) in Lemma 9: it holds uniformly over codes and covariance cells, with the supporting-direction optimization already inside its deterministic exponent. Applying that bound before the bivariate limit preserves the joint success event.

At $\mathbf{r}_{\mathrm{c}}^*$ and $L_0=0$, the condition in (61) is

$$P_{\mathbf{Z}}\{Z_1\leq L_1,Z_2\leq L_2\}=\Phi_2(\sqrt{2}L_1,\sqrt{2}L_2;\rho^2)\geq 1-\varepsilon,$$

where $\Phi_2(\cdot,\cdot;r)$ is the standard bivariate Gaussian CDF with correlation $r\in(-1,1)$. Indeed,

$$\begin{aligned}
&P_{\mathbf{Z}}\{\psi_\rho([Z_1-L_1]^+,[Z_2-L_2]^+)\leq 0\}\\
&\qquad=P_{\mathbf{Z}}\{[Z_1-L_1]^+=[Z_2-L_2]^+=0\}, &(345)\\
&\qquad=P_{\mathbf{Z}}\{Z_1\leq L_1,Z_2\leq L_2\}, &(346)\\
&\qquad=\Phi_2(\sqrt{2}L_1,\sqrt{2}L_2;\rho^2), &(347)
\end{aligned}$$

where Eq. (345) follows from the zero set of ψ_ρ, Eq. (346) follows from the positive parts, and Eq. (347) follows from rescaling (59).

In fact, an absent common message attains this entire slice. Let $\mathcal{L}_\varnothing(D_1, D_2, \varepsilon)$ be the closure of the private offset pairs satisfying Definition 4 at the two marginal first-order rates with $M_{0,n} = 1$ for every n. Then

$$\mathcal{L}_\varnothing(D_1, D_2, \varepsilon) = \{(L_1, L_2) \in \mathbb{R}^2 : (0, L_1, L_2) \in \mathcal{L}(\mathbf{r}_{\mathrm{c}}^*, D_1, D_2, \varepsilon)\}. \tag{348}$$

The private-ball construction and its analysis in Section IV-E prove achievability without a common header. The converse follows from the arbitrary-code bound in Section V-B. Thus the $o(\sqrt{n})$ common nats allowed by $L_0 = 0$ are unnecessary for this second-order slice.

The remaining coupling is probabilistic, not a need for common communication. For $\ell_\varepsilon := \mathrm{Q}^{-1}(\varepsilon)/\sqrt{2}$,

$$P_{\mathbf{Z}}\{Z_1 \le \ell_\varepsilon, Z_2 \le \ell_\varepsilon\} = P_{\mathbf{Z}}\{Z_1 \le \ell_\varepsilon\} - P_{\mathbf{Z}}\{Z_1 \le \ell_\varepsilon, Z_2 > \ell_\varepsilon\}, \tag{349}$$

$$< 1 - \varepsilon, \tag{350}$$

where Eq. (349) follows from partitioning the marginal event, and Eq. (350) follows from the strictly positive joint Gaussian density. Each marginal error may equal ε while the joint target fails. A genuinely second-order common description can improve the private offsets only when $L_0 > 0$.

To justify the closure in Definition 4, consider the regions in Theorem 1 on the one-sided feasible offset domain. The nonzero common rate region is a closed half-space intersected with the coordinate constraints. At the corner, the closed success sets and reverse Fatou imply upper semicontinuity of their probabilities in the offsets, so the probability superlevel set is closed. Thus taking the closure in Definition 4 adds no points.

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